

AegisCare HMS

Complete Documentation Package

v5.16

AegisCare / LabTrack

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Total documents: 11

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AegisCare HMS — Complete System Documentation

System Title: AegisCare Hospital Management System (HMS)
Document Version: 1.0.0
Target Environment: Kenya Ministry of Health (MoH) Level 1–6 Facilities
Classification: Technical & System Architecture Reference

1. System Overview

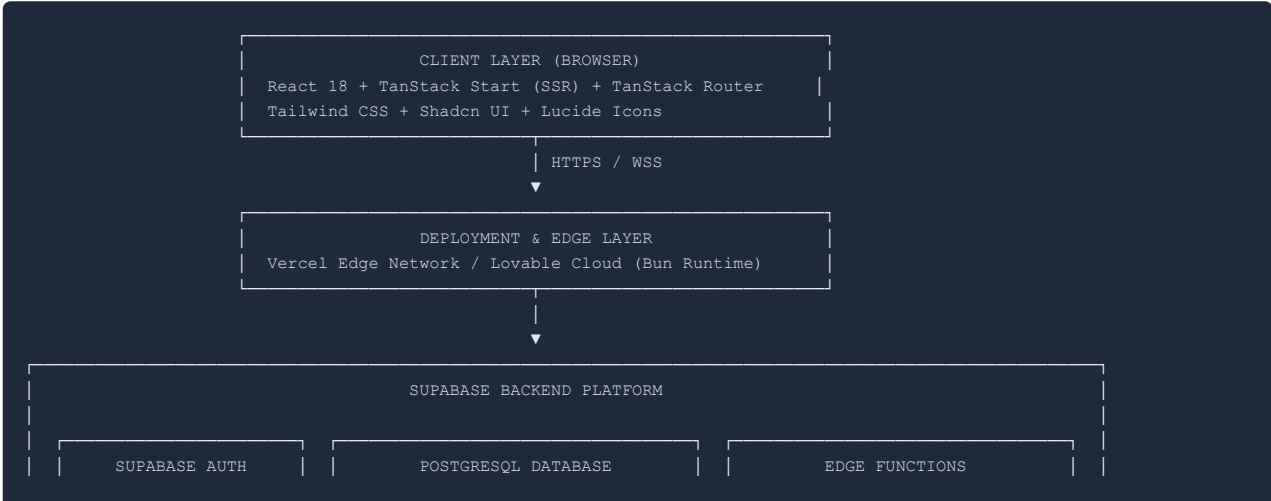
AegisCare HMS is an enterprise-grade, cloud-native Hospital Management System engineered specifically for the Kenyan healthcare ecosystem. It provides comprehensive digital management for healthcare institutions ranging from primary care dispensaries and health centres (MoH Levels 1–3) to comprehensive county referral hospitals and tertiary teaching institutions (MoH Levels 4–6).

Core Capabilities:

- Full Clinical Workflow Management:** Seamless care pathways spanning OPD intake, triage vital scoring, doctor consultations with WHO ICD-11 coding, electronic diagnostic ordering, inpatient ward management, surgical admissions, and discharge reconciliations.
- Specialized Critical Units:** Dedicated clinical subsystems for Intensive Care Units (ICU hourly flow-sheets, ventilator monitoring, RASS sedation scales) and Renal Hemodialysis units (vascular access, dialyzer tracking, ultrafiltration fluid metrics).
- Statutory Kenya MoH Alignment:** Built-in two-layer indicator tagging engine generating automated monthly aggregate reports for MOH 705A (Under-5 Outpatient Morbidity), MOH 705B (Over-5 Outpatient Morbidity), MOH 706 (Laboratory), MOH 707 (Inpatient), MOH 717 (Workload), Family Planning (FP), and Maternal & Child Health (MCH).
- National Health Insurance & Social Health Authority (SHA) Engine:** End-to-end support for the Social Health Insurance Act (SHIA) 2023, handling Primary Healthcare Fund (PHF), Social Health Insurance Fund (SHIF), and Emergency, Chronic and Critical Illness Fund (ECCIF) packages with automated FHIR R4 claim bundling.
- National Digital Health Authority (DHA) Interoperability:** Pre-integrated FHIR R4 interoperability layer for bidirectional synchronization with the national Shared Health Record (SHR) via DHA AfyaLink Health Information Exchange (HIE).
- Automated Communication & Consent:** Real-time SMS notifications for laboratory test completion alerts and Kenya Data Protection Act (ODPC) compliant OTP digital consent verification powered by Africa's Talking.
- Granular Multi-Store Inventory & Ward Dispensing:** Real-time stock ledgers tracking central warehouse deliveries, inter-departmental transfers (Main Store, Central Pharmacy, ICU Store, Dialysis Store), batch numbers, expiry dates, and automated dispensing deductions.

2. Architecture Overview

AegisCare HMS employs a modern, full-stack reactive architecture designed for zero data loss, strict multi-tenant isolation, real-time clinical updates, and low-latency operation over standard Kenyan broadband networks.





Architectural Tenets:

1. **Model A Deployment Architecture:** Each healthcare facility operates on an isolated Supabase project and dedicated frontend deployment. Patient databases are physically partitioned—no cross-facility data leakage is possible.
2. **Database-Driven Business Integrity:** Critical invariants (financial balance computations, inventory deductions, encounter locks, statutory demographic tagging) are executed directly inside PostgreSQL transaction blocks via triggers and RPCs.
3. **Optimistic UI with Realtime Synchronization:** User interfaces leverage TanStack Query with real-time WebSocket subscriptions on PostgreSQL write-ahead logs (`supabase_realtime`), updating triage queues, lab results, and bed states instantaneously.

3. Module Documentation (19 System Modules)

Module 1: Patient Registration

- **Purpose:** Centralized administrative intake capturing patient identity, demographic data, next-of-kin contacts, national identifiers, payment classifications, and opening clinical encounters.
- **Primary Roles:** `receptionist`, `records_officer`, `admin`, `system_admin`.
- **Key Workflows:**
 1. Patient lookup via Master Patient Index (File Number, National ID, Phone, or Name).
 2. If new: Create record in `patients`, generating sequential file number (`P000001`).
 3. Capture emergency vs. standard intake flag.
 4. Select payment mode (`cash`, `sha_shif`, `insurance`, `corporate`, `exemption`).
 5. Select initial destination room (Triage, Insurance Desk, Casualty, or Specialty Clinic).
 6. Automatically create linked row in `encounters` and `encounter_room_visits`.
- **Key DB Tables:** `patients`, `encounters` (view `patient_registrations`), `encounter_room_visits`, `app_settings`.
- **Business Rules Enforced:**
 - Mandatory file number generation via PostgreSQL sequence `patient_file_seq`.
 - Automatic uppercase normalization on names and national identity strings.
 - Emergency encounters bypass cashier clearance and proceed directly to resuscitation/casualty rooms.
- **Edge Cases Handled:**
 - Duplicate registration detection based on National ID and telephone matches.

- Unknown birth date handling: Computes approximate birth date from user-supplied `estimated_age` while flagging `dob_known = false`.

Module 2: Consent & OTP Management

- Purpose:** Secures compliance with the Kenya Data Protection Act (ODPC) 2019 and DHA certification requirements by acquiring verified digital consent before electronic health data capture or national HIE transmission.
- Primary Roles:** `receptionist`, `records_officer`, `triage_nurse`, `doctor`.
- Key Workflows:**
 - Intake staff triggers OTP generation on the patient registration form.
 - System generates a 6-digit numeric token with a 10-minute expiry and saves it in `consent_otps`.
 - Dispatches SMS via Africa's Talking Edge Function (`send-sms`) to the patient's mobile number.
 - Patient provides the code; staff inputs it into the UI.
 - System validates token, sets `patient_consents.consented = true` and `hie_data_sharing_consented = true`, logging timestamp and authenticated user ID.
- Key DB Tables:** `consent_otps`, `patient_consents`, `audit_log`.
- Business Rules Enforced:**
 - OTP tokens expire after 600 seconds (10 minutes).
 - Maximum 3 failed verification attempts before token invalidation.
 - HIE data synchronization in `claims-dispatcher` edge function is strictly suppressed if `hie_data_sharing_consented` is false.
- Edge Cases Handled:**
 - Minors / Incapacitated Patients: Parent/guardian phone number capture with explicit relationship logging.
 - Network Outage / SMS Delay: Authorized manual override with mandatory written paper consent upload/reference.

Module 3: Queue Management

- Purpose:** Real-time orchestrator directing patient flow across physical service rooms, calculating wait times, preventing bottlenecks, and maintaining service transparency.
- Primary Roles:** `receptionist`, `triage_nurse`, `doctor`, `clinical_officer`, `lab_tech`, `radiologist`, `pharmacist`, `accountant`.
- Key Workflows:**
 - Front desk routes an active encounter to a room (`encounters.room_id = target_room_id`).
 - Departmental workbench displays live patient list filtered by room and status (`waiting`, `in_progress`, `done`).
 - Practitioner clicks "Call Patient" → Status updates to `in_progress` and start timestamp recorded.
 - On service completion, practitioner selects next destination (e.g., Doctor → Lab → Doctor → Pharmacy).
 - System records history in `encounter_room_visits` and resets status to `waiting` in destination room.
- Key DB Tables:** `encounters`, `rooms`, `encounter_room_visits`, `user_room_access`.
- Business Rules Enforced:**
 - Practitioners can only access queues for rooms where they possess explicit `user_room_access` or have `admin` role (`can_access_room()` function).
 - Automated return-to-requesting-room routing upon lab or radiology completion (`send_lab_results_to_requesting_room()`).
- Edge Cases Handled:**
 - "Patient Absent / Skipped" state transitions preserve queue sequence without dropping encounter.
 - Simultaneous multi-department orders (e.g., Lab + Ultrasound) allow parallel queue presence.

Module 4: Triage

- Purpose:** Systematic physiological assessment of arriving outpatients, capturing baseline vitals, calculating body mass indexes, screening pediatric malnutrition, and scoring acuity.
- Primary Roles:** `triage_nurse`, `nurse`, `clinical_officer`, `doctor`.
- Key Workflows:**

1. Triage nurse pulls patient from triage room queue.
 2. Captures Systolic/Diastolic Blood Pressure, Pulse Rate, Body Temperature, Respiratory Rate, Oxygen Saturation (SpO2), Random Blood Sugar (RBS), Weight, Height, and MUAC.
 3. System automatically computes BMI and categorizes nutritional status.
 4. Clinical urgency flag assigned (`normal` , `urgent` , `critical_emergency`).
 5. Vitals JSON payload committed to `encounters.vitals` and patient routed to consultation room.
- **Key DB Tables:** `encounters` , `rooms` , `audit_log` .
 - **Business Rules Enforced:**
 - Automatic critical alerts triggered for systolic BP > 180 or < 90 mmHg, SpO2 < 90%, Temperature > 39.0°C.
 - Demographic indicators (`tag_encounter_demographics()`) evaluated upon triage save.
 - **Edge Cases Handled:**
 - Pediatric weight-for-height and MUAC fields dynamically required for patients aged < 5 years.
 - Unresponsive/Emergency triage pathways allow direct casualty transfer with partial vitals capture.
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Module 5: Insurance Desk (SHA + Private + Corporate)

- **Purpose:** Comprehensive policy verification, pre-authorization capture, benefit category mapping, co-payment processing, and contracted tariff application.
 - **Primary Roles:** `insurance_agent` , `accountant` , `admin` .
 - **Key Workflows:**
 1. Front desk routes insured patient to Insurance Desk.
 2. Agent selects underwriter from `insurance_providers` and specific scheme from `insurance_benefit_plans` .
 3. Verifies member eligibility number and captures Pre-Authorization Code (for inpatient/specialized procedures).
 4. Enforces copayment requirements by appending copay charge to open invoice.
 5. Marks insurance status as `cleared` and routes patient to Triage/Consultation.
 - **Key DB Tables:** `insurance_providers` , `insurance_benefit_plans` , `insurance_benefit_categories` , `insurance_contracted_prices` , `encounters` , `sha_claims` .
 - **Business Rules Enforced:**
 - Visit frequency limits enforced via `enforce_insurance_visit_limit()` trigger.
 - Automatic pricing engine overrides baseline cash catalog prices with insurer contracted tariffs (`get_contracted_price()` RPC).
 - SHA fund types (`phf` , `shif` , `eccif`) automatically classified based on facility level and service room via `set_sha_fund_type()` .
 - **Edge Cases Handled:**
 - Expired / Suspended policy: Agent can reject clearance, prompting system to seamlessly convert encounter to cash mode with full invoice recalculation.
 - Co-pay waiver workflows with mandatory supervisor sign-off.
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Module 6: Consultation / Clinical

- **Purpose:** The core clinical diagnostic and therapeutic workstation for outpatient and inpatient clinician workflows.
- **Primary Roles:** `doctor` , `clinical_officer` , `dental_officer` .
- **Key Workflows:**
 1. Clinician reviews patient past medical history, previous visits, allergy alerts, and triage vitals.
 2. Documents Chief Complaints, History of Present Illness, Physical Examination, and Treatment Plans in `clinical_notes` .
 3. Searches live WHO ICD-11 database via `icd11-search` edge function and attaches primary and secondary diagnoses to `encounter_diagnoses` .
 4. Orders diagnostic tests (Lab / Radiology) and prescribes pharmaceutical regimens.
 5. Initiates inpatient ward admission if required.
 6. Finalizes and electronically signs encounter (`sign_encounter` permission), cryptographically locking records.

- **Key DB Tables:** `clinical_notes`, `encounter_diagnoses`, `lab_orders`, `radiology_orders`, `prescriptions`, `admissions`, `encounters`.
 - **Business Rules Enforced:**
 - At least one valid ICD-11 diagnosis code required before signing encounter.
 - Encounter signing irreversibly locks clinical notes, diagnoses, and orders (`enforce_encounter_lock()` trigger).
 - Only authorized clinical roles (`doctor`, `clinical_officer`, `dental_officer`, `admin`) can execute encounter signing.
 - **Edge Cases Handled:**
 - Break-Glass emergency override allows clinicians to view restricted records with mandatory clinical justification (`log_break_glass_access()`).
 - Offline / Uncoded fallback: Assigns provisional `UNCODED` classification with manual description until ICD-11 connection resolves.
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Module 7: Laboratory

- **Purpose:** Diagnostic specimen accessioning, bench testing, analyzer result capture, panic-value flagging, and technologist validation.
 - **Primary Roles:** `lab_tech`, `staff`, `admin`.
 - **Key Workflows:**
 1. Patient arrives at lab; lab tech reviews pending `lab_orders`.
 2. Collects specimen, logs sample collection timestamp, and updates order status to `in progress`.
 3. Performs diagnostic assay and inputs quantitative/qualitative values into `lab_results`.
 4. System automatically validates results against age/sex reference ranges.
 5. Technologist clicks "Verify & Publish" → Result locked, automated SMS sent to patient, and encounter automatically routed back to ordering clinician room.
 - **Key DB Tables:** `lab_orders`, `lab_results`, `lab_test_catalog`, `encounter_room_visits`, `dha_outbound_queue`.
 - **Business Rules Enforced:**
 - Automatic invoice line-item generation upon lab test order (`sync_invoice_line_items_from_tests()`).
 - Critical panic values trigger prominent red visual badges on clinician screens.
 - Results publication creates a FHIR Observation resource queued for DHA SHR transmission (`queue_lab_result_fhir()`).
 - **Edge Cases Handled:**
 - Hemolyzed / Insufficient Specimen: Technologist rejects specimen with structured reason; system alerts ordering doctor for redraw.
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Module 8: Radiology

- **Purpose:** Diagnostic medical imaging order tracking, modality examination management, radiologist reporting, and PACs/film handover.
 - **Primary Roles:** `radiologist`, `doctor`, `admin`.
 - **Key Workflows:**
 1. Imaging requisition received in radiology queue (`radiology_orders`).
 2. Radiographer completes scan (X-Ray, Ultrasound, CT, MRI, ECG).
 3. Radiologist inputs structured radiological interpretation, organ findings, and conclusion in `radiology_results`.
 4. Radiologist signs report → Order status set to `completed`.
 5. Automated SMS alert sent to patient and encounter auto-returned to requesting doctor (`send_radiology_results_to_requesting_room()`).
 - **Key DB Tables:** `radiology_orders`, `radiology_results`, `rooms`, `invoices`, `invoice_line_items`.
 - **Business Rules Enforced:**
 - Automatic billing line item synchronization upon imaging requisition (`sync_invoice_line_item_from_radiology()`).
 - Modification locked once report is signed by radiologist.
 - **Edge Cases Handled:**
 - Repeat exposures or contrast surcharge additions automatically append supplemental line items to invoice.
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Module 9: Pharmacy

- **Purpose:** Outpatient and inpatient prescription validation, contraindication screening, batch-specific drug dispensing, and automated inventory deduction.
- **Primary Roles:** `pharmacist`, `admin`.
- **Key Workflows:**
 1. Pharmacist opens prescription queue filtered by pending orders (`prescriptions.status = 'pending'`).
 2. Verifies drug strength, dosage, route, and interaction profile.
 3. Verifies invoice payment or insurance clearance before dispensing.
 4. Selects physical drug batch and clicks "Dispense".
 5. System marks prescription `dispensed`, logs `dispensed_at` and `dispensed_by`, and automatically decrements inventory from Pharmacy Store.
- **Key DB Tables:** `prescriptions`, `stock_items`, `stock_movements`, `invoices`, `invoice_line_items`.
- **Business Rules Enforced:**
 - Stock Guard: Dispensing blocked if warehouse available quantity < requested quantity (`dispense_prescription_stock()` trigger).
 - ICU prescriptions are segregated: General pharmacy cannot dispense ICU store medications (`sprint13b_dispense_stock_guard.sql`).
 - Dispense event automatically queues a FHIR `MedicationDispense` resource (`trg_medication_dispense_fhir()`).
- **Edge Cases Handled:**
 - Partial dispensing: Pharmacist dispenses available units and leaves remainder pending for subsequent replenishment.
 - Drug substitution / Generic switch with mandatory clinician consultation note.

Module 10: Inpatient (Wards + ICU + Dialysis)

- **Purpose:** Complete inpatient lifecycle management across General Wards, Maternity, Pediatrics, Surgical units, Intensive Care Units, and Hemodialysis units.
- **Primary Roles:** `nurse`, `icu_nurse`, `dialysis_nurse`, `doctor`, `clinical_officer`, `admin`.
- **Key Workflows:**
 1. **Admission:** Doctor issues admission order → Patient assigned ward and bed (`admissions`, `beds.status = 'occupied'`).
 2. **Ward Care:** Nurses record vitals, administer scheduled drugs in Medication Administration Record (`medication_administrations`), and log nursing notes.
 3. **ICU Unit:** ICU nurses maintain hourly charts (`icu_hourly_charts`) tracking arterial pressures, ventilator parameters (PEEP, FiO2, VT), RASS sedation scores, and direct ICU store drug dispensing.
 4. **Dialysis Unit:** Renal nurses log sessions (`dialysis_sessions`), dialyzer models, pre/post weights, BFR, and consumed dialyzer consumables, auto-billed via `process_dialysis_session_billing()`.
 5. **Discharge:** Doctor completes mandatory discharge summary (`require_discharge_summary()`), pharmacy reconciles take-home drugs, accountant clears bill, and bed is reset to `available`.
- **Key DB Tables:** `admissions`, `wards`, `beds`, `icu_hourly_charts`, `dialysis_sessions`, `medication_administrations`, `invoices`.
- **Business Rules Enforced:**
 - Nightly cron job (`accrue_daily_bed_charges()`) accrues daily bed fees at 00:00 EAT.
 - Admitting to ICU automatically applies one-off ICU admission fee (`charge_icu_admission_fee()`).
 - Discharge blocked without comprehensive discharge summary text.
- **Edge Cases Handled:**
 - Inter-ward bed transfers: Automatically updates ward assignment, recalibrates daily bed rates, and logs audit trigger (`audit_ward_transfers`).

Module 11: Mortuary

- **Purpose:** Cold-room storage management, body intake logging (internal ward deaths vs. external police/home cases), preservation tracking, post-mortem records, and body release billing.
- **Primary Roles:** `mortician`, `accountant`, `admin`.

- **Key Workflows:**

1. **Intake:** Mortician admits body, assigns sequential mortuary reference (`assign_mortuary_reference()`), records body source (`internal` vs. `external`), next-of-kin, and cold storage slot.
2. **Preservation & Services:** Logs embalming, post-mortem examination, post-mortem pathology notes, and dressing requests.
3. **Daily Accrual:** Nightly daemon (`accrue_daily_mortuary_charges()`) posts daily refrigeration charges to mortuary invoice.
4. **Handover & Release:** Next-of-kin presents national ID and burial permit, accountant verifies invoice clearance, and mortician executes formal body release.

- **Key DB Tables:** `mortuary_records` , `invoices` , `invoice_line_items` , `rooms` .

- **Business Rules Enforced:**

- External body intake automatically provisions a dedicated mortuary invoice linked to next-of-kin (`create_external_mortuary_invoice()`).
- Daily storage rate applied every 24 hours until `released_at` timestamp is committed.
- Body release strictly locked until invoice balance reaches zero or an authorized waiver is applied.

- **Edge Cases Handled:**

- Unclaimed bodies / State medical examiner cases: Flagged with police OB number and exempted from automated standard billing accruals.

Module 12: Billing & Accounting

- **Purpose:** Core financial engine handling ledger balances, multi-channel payment collections, supervisor waivers, credit note adjustments, and cashier cash drawer closures.

- **Primary Roles:** `accountant` , `admin` , `director` .

- **Key Workflows:**

1. Cashier searches invoice by patient file number, invoice number, or encounter ID.
2. Reviews itemized billable line items (consultations, lab, imaging, drugs, bed days, procedures).
3. Selects payment method (Cash, M-Pesa, Credit Card, Bank Transfer, Insurance Guarantee).
4. Submits payment → System records `invoice_payments` , recalculates invoice balance, issues receipt number, and sends SMS receipt.
5. Performs End-of-Day (EOD) reconciliation reconciling cash collections against system totals.

- **Key DB Tables:** `invoices` , `invoice_line_items` , `invoice_payments` , `encounters` .

- **Business Rules Enforced:**

- Real-time balance integrity maintained via PostgreSQL triggers (`recalc_invoice_payments()` , `recalc_invoice_totals()`).
- Invoice balance = `amount_total` - `amount_paid` - `waiver_amount` .
- Cashier payment reversal strictly locked—requires accounting supervisor credit note transaction.

- **Edge Cases Handled:**

- Overpayment handling: Automatically calculates change or holds overpayment as patient unallocated credit for future visits.

Module 13: Invoices

- **Purpose:** Master fiscal ledger maintaining real-time itemized accounting for every billable clinical and administrative transaction.

- **Primary Roles:** `accountant` , `admin` , `insurance_agent` , `receptionist` .

- **Key Workflows:**

1. Encounter creation automatically provisions linked master invoice (`create_invoice_for_encounter()`).
2. Downstream clinical actions (lab orders, imaging orders, pharmacy dispenses, bed stays) dynamically attach line items via database triggers.
3. Invoices display real-time status: `draft` , `pending` , `partial` , `paid` , `waived` , `claimed` .
4. Printable MoH compliant invoice generated with institutional letterhead, tax registration, and KMHFL codes.

- **Key DB Tables:** `invoices` , `invoice_line_items` , `invoice_payments` , `patients` .

- **Business Rules Enforced:**

- Immutability of line-item amounts once payment is applied against them.
- Strict foreign key constraints binding every line item to its parent invoice and source clinical order.

- **Edge Cases Handled:**

- Split billing: Supports encounters split across insurance coverage caps with residual balances routed to self-pay cash invoices.

Module 14: Stock Management

- **Purpose:** Multi-location inventory warehouse management, supplier delivery receiving, batch tracking, inter-store transfer requisitions, and departmental consumption tracking.
- **Primary Roles:** `store_keeper`, `pharmacist`, `lab_tech`, `admin`.
- **Key Workflows:**
 1. Storekeeper receives commercial vendor shipment in Main Store (`a0000000-0000-0000-0000-000000000001`), logging batch number, expiry date, purchase cost, and quantity in `stock_deliveries`.
 2. Department heads (Pharmacy, ICU, Dialysis, Lab) submit transfer requests.
 3. Storekeeper approves and dispatches items (`transfer_stock_between_locations()` RPC).
 4. Clinical usage logged directly at bedside or during procedures (`record_stock_usage()` RPC).
 5. System generates automated stock valuation ledgers and reorder alerts.
- **Key DB Tables:** `stock_items`, `stock_locations`, `stock_movements`, `stock_transfers`, `stock_deliveries`, `stock_usage`, `user_stock_location_access`.
- **Business Rules Enforced:**
 - Double-entry stock ledger: Every physical quantity change demands a corresponding immutable row in `stock_movements`.
 - Negative inventory strictly prevented via database check constraints.
 - Sub-location access controlled via `grant_stock_location_access()` security matrix.
- **Edge Cases Handled:**
 - Damaged / Expired stock write-offs: Logged under `adjustment` movement type with mandatory supervisory reason notes.

Module 15: Appointments

- **Purpose:** Forward scheduling for outpatient specialist clinics (GOPC, SOPC, MCH/ANC, Dental, Eye Clinic, Physiotherapy, Nutrition), managing clinic capacities and automated patient reminders.
- **Primary Roles:** `receptionist`, `nurse`, `doctor`, `records_officer`.
- **Key Workflows:**
 1. Staff schedules booking specifying patient, target clinician/clinic room, appointment date, time window, and clinical indication.
 2. System records entry in `appointments` and calculates start/end ranges via `set_appointment_time_range()`.
 3. Automated SMS reminder queued for delivery 24 hours prior to appointment.
 4. Patient arrives on scheduled date → Receptionist clicks "Check-In" → System converts booking into active clinical encounter via `create_encounter_from_appointment()`.
- **Key DB Tables:** `appointments`, `encounters`, `rooms`, `patients`.
- **Business Rules Enforced:**
 - Double-booking prevention guards on individual practitioner calendars.
 - Automatic status transitions: `scheduled` → `checked_in` → `completed` / `cancelled` / `no_show`.
- **Edge Cases Handled:**
 - Walk-in patient prioritization during clinic sessions without displacing booked appointments.

Module 16: Reports (MOH + Finance + Clinical)

- **Purpose:** Comprehensive institutional business intelligence and statutory epidemiological compliance reporting engine.
- **Primary Roles:** `records_officer`, `accountant`, `director`, `admin`.
- **Key Workflows:**
 1. **MOH Statutory Reports:** Records officer selects reporting month and executes `get_moh_705_report(start_date, end_date, '705A')` for under-5 morbidity or `'705B'` for over-5 morbidity.

2. **Financial Reports:** Accountant generates Revenue Ledgers, Departmental Sales, Cashier Collections, Insurance Aging, and Waiver Summaries.
3. **Clinical Reports:** Clinicians analyze Disease Incidence Trends, Lab Test Yields, Inpatient Length of Stay, and Mortality Rates.
4. One-click export to CSV and formatted printable PDF.

- **Key DB Tables:** `moh_monthly_aggregates`, `encounter_indicator_tags`, `moh_indicator_definitions`, `invoices`, `encounters`.
- **Business Rules Enforced:**
 - Two-layer tagging architecture ensures 100% automated indicator compilation without manual tally sheets.
 - Demographics (Under 5, Over 5, Gender, New vs. Revisit) tagged universally on every encounter.
- **Edge Cases Handled:**
 - Historical retrospective aggregate recalculation supported via `refresh_moh_aggregates(target_month)`.

Module 17: Admin

- **Purpose:** Centralized facility configuration suite managing hospital metadata, master price tariffs, clinical test templates, rooms, wards, beds, medical machines, and user roles.
- **Primary Roles:** `admin`, `system_admin`.
- **Key Workflows:**
 1. **Facility Settings:** Configures KMHFL code, SHA facility ID, facility level, contact info, and operational parameters.
 2. **User & Access Management:** Invites staff, approves accounts, and assigns roles (`user_roles`) and permissions (`role_permissions`).
 3. **Room & Ward Rosters:** Creates consultation suites, wards, and beds with associated base rates.
 4. **Machine Registers:** Logs biomedical equipment (`machines`) and schedules preventative maintenance (`machine_logs`).
 5. **Tariffs & Pricing:** Sets master prices and insurer-contracted rate schedules.
- **Key DB Tables:** `app_settings`, `profiles`, `user_roles`, `role_permissions`, `rooms`, `wards`, `beds`, `machines`, `machine_logs`, `facility_features`.
- **Business Rules Enforced:**
 - Self-demotion guard: Administrators cannot revoke their own admin permissions.
 - Facility level gating: Modules automatically disabled if facility MoH level does not support them (e.g., ICU locked for Level 2/3).
- **Edge Cases Handled:**
 - Emergency user deactivation instantly terminates active Supabase Auth sessions.

Module 18: Audit Log

- **Purpose:** Immutable forensic logging providing complete accountability, regulatory data protection compliance, and tamper-evident operational trails.
- **Primary Roles:** `admin`, `system_admin`, `director`.
- **Key Workflows:**
 1. System triggers (`audit_trigger_fn()`) capture every database mutation across 15+ core tables.
 2. Writes immutable record containing actor UUID, timestamp, action type (`INSERT`, `UPDATE`, `DELETE`, `BREAK_GLASS`), table name, record ID, and before/after JSON data.
 3. Administrators review filtered security feeds in the UI (`/admin/audit-log`).
 4. Nightly daemon (`archive_old_audit_logs()`) moves records older than 2 years to `audit_log_archive` for 20-year statutory retention.
- **Key DB Tables:** `audit_log`, `audit_log_archive`, `audit_archive_runs`.
- **Business Rules Enforced:**
 - Append-only security: Explicit RLS policies reject all manual `UPDATE` and `DELETE` queries on `audit_log` and `audit_log_archive`.
- **Edge Cases Handled:**
 - Automated tracking and notification of emergency break-glass electronic record access.

Module 19: Dashboard + System Health

- **Purpose:** Real-time executive situational awareness, operational clinical monitoring, financial summary KPIs, and backend system health metrics.
- **Primary Roles:** `director`, `admin`, `accountant`, `doctor`.
- **Key Workflows:**
 1. Dashboard loads live clinical throughput cards (Today's Registrations, Active Inpatients, Lab Queue, Pharmacy Queue).
 2. Visualizes epidemiological disease trends via `dashboard_top_diseases()` RPC.
 3. Displays 7-day OPD vs. Admitted trends via `dashboard_admitted_opd_trend()`.
 4. Monitors system health (Supabase connection, Edge Function response latency, SMS gateway credit balance).
- **Key DB Tables:** `encounters`, `admissions`, `invoices`, `lab_orders`, `prescriptions`, `app_settings`.
- **Business Rules Enforced:**
 - Role-gated dashboard views: Financial revenue cards hidden from clinical-only staff roles.
- **Edge Cases Handled:**
 - Graceful degradation to cached summary snapshots during heavy database workload spikes.

4. Integration Points

AEGISCARE HMS INTEGRATIONS			
Service / Platform	Protocol / Standard	Status	Key Functionality
Africa's Talking SMS	REST HTTP (Form/JSON)	✅ Production	OTP, Lab & Bills
WHO ICD-11 Cloud API	REST OAuth2 (MMS)	✅ Production	Real-time Diagnosis
Kenya DHA AfyaLink HIE	FHIR R4 REST API	⚠️ Stubs Ready	National SHR Sync
SHA Claims API	FHIR R4 / JSON API	⚠️ Stubs Ready	Claim Dispatch
IPRS National Identity	REST HTTPS / DHA Proxy	⚠️ Stubs Ready	Citizen ID Verify
Safaricom M-Pesa Daraja	REST HTTPS (STK Push)	📅 Planned (Ph 3)	Mobile Cashier Pay

1. Africa's Talking SMS Gateway

- **Status:** ✅ Active & Verified in Production.
- **Configuration:** Supabase secrets `AT_USERNAME`, `AT_API_KEY`.
- **Functionality:**
 - Standardizes telephone numbers to Kenyan international format (`+254...`).
 - Dispatches OTP verification tokens during patient intake.
 - Sends automated diagnostic test completion alerts when lab/radiology reports are signed.
 - Employs sandbox endpoint when username is configured as `sandbox`.

2. WHO ICD-11 Cloud API

- **Status:** ✅ Active & Operational.
- **Configuration:** Supabase secrets `ICD_CLIENT_ID`, `ICD_CLIENT_SECRET`.
- **Functionality:**
 - Executes OAuth2 Client Credentials authentication against WHO servers with automated in-memory token caching.
 - Performs flexisearch against ICD-11 MMS (Mortality & Morbidity Statistics) 2024 linearization.
 - Strips HTML markup and normalizes diagnostic titles for clean clinical display and indexing.

3. DHA AfyaLink National HIE (Shared Health Record)

- **Status:** ⚠️ Architectural Stubs Built & Certification Ready (Awaiting National HIE Gateway Credentials).
- **Functionality:**
 - Compiles FHIR R4 Bundle resources containing Patient, Encounter, Condition, and MedicationDispense resources.

- Transmits payloads via `dha_outbound_queue` with automated retry, backoff, and mediator transaction tracking.
- Governed by patient consent verification (`hie_data_sharing_consented = true`).

4. Social Health Authority (SHA) Claims API

- **Status:** 🏗️ Architectural Stubs Built & Database Foundation Complete.
- **Functionality:**
 - Evaluates encounter diagnoses, procedures, and pharmaceutical items against SHA tariff schedules.
 - Automatically compiles standard FHIR `Claim` resources (`build_fhir_claim()`) RPC).
 - Handles SHIF, PHF, and ECCIF benefit package allocations and pre-authorization linkages.

5. Safaricom M-Pesa Daraja API

- **Status:** 📅 Planned for Phase 3 Roadmap.
- **Functionality:**
 - Direct cashier STK Push prompts to patient mobile phones.
 - Real-time C2B payment validation callbacks automatically clearing invoice balances.

5. Security & Compliance

1. Row-Level Security (RLS) Policies

AegisCare enforces PostgreSQL Row-Level Security across all 35+ relational database tables.

- **User Approval Gate:** All access demands `public.is_approved(auth.uid()) = true` . Unapproved user accounts cannot read or write any clinical data.
- **Role-Based Permissions:** Data mutations require explicit roles verified via `public.has_role()` or permissions evaluated via `public.user_has_permission()` .
- **Departmental Room Security:** Room clinical data restricted to assigned practitioners via `public.can_access_room()` .
- **Append-Only Auditing:** Audit tables strictly reject direct client-side `INSERT` , `UPDATE` , and `DELETE` operations.

2. Kenya Data Protection Act (ODPC) Compliance

- **Consent by Default:** Clinical data capture demands documented consent.
- **Two-Factor Digital Verification:** OTP SMS consent mechanism validates citizen authorization.
- **Purpose Specification & Access Minimization:** Practitioners only see clinical fields necessary for their functional department.

3. Break-Glass Emergency Access

- In acute trauma or unconscious emergency scenarios where patient consent cannot be acquired, authorized doctors and clinical officers can invoke **Break Glass Access**.
- The clinician must provide an explicit, mandatory written medical justification.
- The system immediately invokes `log_break_glass_access()` , writing an immutable `BREAK_GLASS` security event in `audit_log` capturing accessor email, timestamp, and justification.

4. Audit Logging & 20-Year Archival Retention

- Every database insert, update, and delete on patient records, diagnoses, prescriptions, lab results, and financial invoices triggers `audit_trigger_fn()` .
- Nightly scheduled daemons (`archive_old_audit_logs()`) move records exceeding two years to `audit_log_archive` to meet Kenya's statutory 20-year medical record retention standard.

6. Cron Jobs (Automated Background Daemons)

AegisCare utilizes `pg_cron` inside PostgreSQL to manage mission-critical background hospital operations:

```
-- 1. Accrue Daily Inpatient Ward Bed Charges (Nightly at 00:01 EAT = 21:01 UTC)
SELECT cron.schedule('accrue-bed-charges-nightly', '1 21 * * *',
  $$ SELECT public.accrue_daily_bed_charges(); $$
);

-- 2. Accrue Daily Mortuary Cold Storage Charges (Nightly at 00:05 EAT = 21:05 UTC)
SELECT cron.schedule('accrue-mortuary-charges-nightly', '5 21 * * *',
  $$ SELECT public.accrue_daily_mortuary_charges(); $$
);

-- 3. Archive Stale Audit Logs to Long-Term Storage (Nightly at 02:00 EAT = 23:00 UTC)
SELECT cron.schedule('archive-audit-logs-nightly', '0 23 * * *',
  $$ SELECT public.archive_old_audit_logs(); $$
);
```

Daemon Functions:

1. `accrue_daily_bed_charges()` : Iterates over all active inpatient admissions (`status = 'admitted'`), queries the ward's daily rate, and appends a `bed_day` line item to the patient's open invoice.
2. `accrue_daily_icu_charges()` : Applies specialized intensive care monitoring daily surcharges for patients admitted to ICU wards.
3. `accrue_daily_mortuary_charges()` : Iterates over non-released mortuary records, calculating daily refrigeration storage fees and appending them to the mortuary invoice.
4. `archive_old_audit_logs()` : Identifies audit entries older than 2 years, copies them into `audit_log_archive` , purges them from the operational `audit_log` table, and logs the batch run metrics in `audit_archive_runs` .

7. Supabase Edge Functions

Function Name	Runtime	Method	Purpose & Integration
<code>send-sms</code>	Deno / TS	POST	Dispatches SMS via Africa's Talking API for OTPs and lab completion alerts. Formats numbers to <code>+254...</code> .
<code>icd11-search</code>	Deno / TS	POST	Proxies live search requests to WHO ICD-11 API with server-side OAuth2 caching and HTML stripping.
<code>claims-dispatcher</code>	Deno / TS	POST	Evaluates completed encounters, builds FHIR claim payloads, and inserts them into <code>dha_outbound_queue</code> .
<code>fhir-patient</code>	Deno / TS	POST	Compiles a FHIR R4 <code>Patient</code> resource from database patient demographics.
<code>fhir-encounter</code>	Deno / TS	POST	Generates a FHIR R4 <code>Encounter</code> resource including class, diagnoses, and service provider.
<code>fhir-condition</code>	Deno / TS	POST	Generates an array of FHIR R4 <code>Condition</code> resources mapped to ICD-11 URLs.
<code>fhir-bundle</code>	Deno / TS	POST	Compiles a full FHIR R4 <code>Bundle</code> (type=collection) for DHA SHR submission.

8. Known Limitations & Pending Items

- **DHA AfyaLink Production Credentials:** Outbound HIE synchronization queues payloads locally in `dha_outbound_queue` awaiting DHA production gateway endpoint access.
- **SHA Biometric Smart Reader Hardware:** Frontend UI components architected; physical USB smart-card / biometric scanner drivers scheduled for Phase 3 field testing.

- **🕒 Safaricom M-Pesa STK Push Gateway:** Cashier payment workflow currently supports manual M-Pesa reference capture; automated STK push API integration planned for Phase 3.
- **🕒 NLMIS Supply Chain Auto-Sync:** Pharmaceutical inventory supports NLMIS commodity codes; automated national KEMSA/NLMIS stock balance reporting currently requires manual CSV export.

9. Future Roadmap

AEGISCARE HMS ROADMAP		
Phase 1 (Completed ✅)	Phase 2 (Current 🕒)	Phase 3 (Planned 📅)
- Core OPD / IPD Clinical	- DHA Certification	- M-Pesa STK Push Integration
- Pharmacy & Multi-Store	- SHA FHIR Claim Testing	- Biometric Hardware Drivers
- Lab & Radiology Flow	- Outbound Queue Workers	- Telemedicine Video Suite
- ICU & Dialysis Units	- Multi-Facility Rollout	- AI Clinical Decision Support
- Mortuary Management	- MOH 705/717 Auto-Sync	- Patient Mobile Portal App
- Africa's Talking SMS		- Offline Sync PWA Node
- WHO ICD-11 Search		

AegisCare HMS — Role-Based User Manual

Document Title: Standard Operating Procedures & Role-Based User Manual

Document Version: 1.0.0

Target Audience: Clinical, Administrative, and Financial Hospital Staff

Applicable Facilities: Kenya MoH Level 1–6 Facilities

Manual Structure & Table of Contents

1. [Receptionist](#)
2. [Insurance Agent](#)
3. [Triage Nurse](#)
4. [Doctor / Clinical Officer](#)
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10. [Accountant / Cashier](#)
11. [Mortician](#)
12. [Store Keeper](#)
13. [System Administrator \(Admin\)](#)
14. [Medical Director / Executive Management](#)

1. Receptionist

Task 1.1: Register a New Patient & Capture Digital Consent OTP

Who: `receptionist`, `records_officer`

Steps:

1. Log into AegisCare HMS and navigate to **Register Patient** (`/register-patient`) on the main navigation bar.
2. In the **Patient Identification** section, enter the patient's National ID, Passport Number, or Birth Certificate Number (for minors).
3. Fill in the demographic details:
 - Full Legal Name (First Name, Middle Name, Family Name).
 - Date of Birth (or check *Unknown DOB* and enter *Estimated Age*).
 - Gender / Biological Sex (`Male` / `Female`).
 - Mobile Phone Number (Must be formatted as `07...` or `01...` or `+254...`).
 - County of Residence and Residential Sub-County / Estate / Town.
 - Next of Kin Full Name, Relationship, and Contact Telephone.
4. In the **Consent Verification (ODPC Compliance)** section:
 - Click **Send Consent OTP SMS**.
 - An automated SMS containing a 6-digit numeric verification code is dispatched immediately via Africa's Talking to the patient's phone.
 - Request the 6-digit code from the patient and type it into the **Enter OTP Code** field.
 - Click **Verify & Accept Consent**. The status badge will change to a green **Verified & Consented** banner.
5. In the **Encounter & Payment Setup** section:

- Select **Payment Mode**: `Cash`, `SHA SHIF / PHF`, `Private Insurance`, `Corporate Scheme`, or `Government Exemption`.
 - If Insurance or SHA: Select the provider and scheme, and input the Member / Policy Number.
 - Check the **Emergency Case** checkbox if the patient presents in acute clinical distress.
6. In the **Destination Room** dropdown, select the next operational station:
- For Cash patients: Select **Triage** (or **Consultation Room** if pre-triaged).
 - For Insurance / SHA patients: Select **Insurance Desk**.
 - For Acute Emergencies: Select **Resuscitation / Casualty Room**.
7. Click **Complete Registration & Open Encounter**.
8. The system generates a permanent **Patient File Number** (e.g., `P000248`), opens an active encounter, generates a blank master invoice, and routes the patient's name to the live queue of the selected room.

Notes:

-  **Emergency Override**: If a patient is unconscious, severely traumatized, or in shock and cannot receive an SMS OTP, check the **Emergency Intake Bypass** toggle. You must provide a brief mandatory justification note. The encounter will proceed immediately, and consent must be regularized upon patient stabilization.
- Ensure the phone number is verified carefully; critical lab panic alerts and digital payment receipts are delivered to this number.

Task 1.2: Check-In a Returning Patient (Revisit)**Who:** `receptionist`, `records_officer`**Steps:**

1. Navigate to **Patients** (`/patients`) or use the global top search bar.
2. Search for the returning patient by **File Number**, **National ID**, **Phone Number**, or **Name**.
3. Click on the patient record to open their master file.
4. Verify demographic details and update phone numbers or residential addresses if changed.
5. Click the **New Encounter** button in the top right corner.
6. Select the **Encounter Type** (`Outpatient`, `Specialist Clinic`, `Emergency`).
7. Confirm the **Payment Mode** and select the **Destination Room** (e.g., Triage, Insurance Desk, or MCH Clinic).
8. Click **Start Encounter**. The encounter is automatically categorized as an `OPD_REVISIT` for statutory MoH epidemiological reporting.

Notes:

- Never create a duplicate master file for an existing patient. Always search the central repository first to preserve longitudinal medical history.

Task 1.3: Check-In an Appointment Booking**Who:** `receptionist`**Steps:**

1. Navigate to **Appointments** (`/appointments`).
2. Review the daily calendar or list view for scheduled patients.
3. Locate the arriving patient and verify their identity.
4. Click the **Check-In** action button next to their booking.
5. Confirm the payment mode and destination room.
6. Click **Confirm Arrival**. The system invokes `create_encounter_from_appointment`, transitions the appointment status to `checked_in`, creates an active clinical encounter, and sends the patient to the assigned clinician's waiting room.

Notes:

- If the patient fails to show up by clinic close, click **Mark No-Show** to release the clinician's calendar slot.

2. Insurance Agent

Task 2.1: Process Insurance Clearance (SHA / Private / Corporate)

Who: `insurance_agent`, `accountant`

Steps:

1. Navigate to **Admin** → **Insurance Clearance** or open the **Insurance Desk** room queue (`/admin/insurance`).
2. Select a patient from the **Waiting for Clearance** queue.
3. Review the patient's insurance details:
 - Underwriter Name (e.g., Social Health Authority, Jubilee, Britam, CIC, APA).
 - Scheme / Benefit Plan (e.g., SHA Outpatient, Corporate Comprehensive, Executive Inpatient).
 - Policy / Member Number.
4. **Eligibility Check:**
 - For SHA: Verify active member status on the SHA portal / NHIF legacy terminal.
 - For Private / Corporate: Verify physical/digital smart card, biometric validity, and active member status on the insurer's portal.
5. **Pre-Authorization (Preauth):**
 - If the visit involves specialized procedures, MRI/CT imaging, or inpatient admission, input the approved **Pre-Authorization Code** provided by the insurer into the preauth field.
6. **Co-Payment:**
 - If the plan requires a fixed co-pay (e.g., KSh 500), verify the co-pay amount. The system automatically appends a co-pay line item to the patient's invoice.
7. Click **Approve Clearance**.
8. Select the destination room (e.g., **Triage** or **Doctor Consultation Room**) and click **Dispatch Patient**.
9. The encounter status changes to `insurance_status = 'cleared'`, unlocking doctor consultations, lab orders, and pharmacy dispenses under contracted insurance tariffs.

Notes:

-  **Visit Limit Warnings:** If a patient has exceeded their monthly or annual insurance visit cap, the system trigger `enforce_insurance_visit_limit` will display an on-screen alert. Do not approve clearance without documented authorization from the underwriter.

Task 2.2: Reject Clearance & Switch Patient to Cash Mode

Who: `insurance_agent`

Steps:

1. If the patient's insurance policy is lapsed, invalid, exhausted, or rejected by the underwriter:
2. Click **Reject Clearance** on the insurance clearance panel.
3. Select a structured rejection reason (e.g., `Policy Inactive / Expired`, `Benefit Limit Exhausted`, `Service Not Covered`, `Dependent Not Registered`).
4. Click **Convert Encounter to Cash**.
5. Inform the patient politely that the insurer has declined coverage.
6. Route the patient to the **Cashier / Triage** for cash-mode processing. The billing engine instantly switches price lookups from contracted insurance tariffs to standard cash prices.

Notes:

- Always document the underwriter's rejection reference code in the notes field for future dispute resolution.

Task 2.3: Track & View SHA / Insurance Claim Status by Date

Who: `insurance_agent`, `accountant`

Steps:

1. Navigate to **Insurance & SHA Claims** (`/admin/insurance`).
2. Use the date range picker to filter claims by admission, discharge, or submission date.
3. Filter claims by status:
 - `draft` : Encounter concluded, claim pending review.
 - `pending` : Queued in `dha_outbound_queue` awaiting transmission.
 - `submitted` : Transmitted to SHA / Insurer clearinghouse.
 - `approved` : Underwriter approved payment.
 - `rejected` : Underwriter query or rejection.
4. Click on any claim ID to inspect itemized diagnostic line items, attached ICD-11 codes, and the compiled FHIR Claim resource JSON bundle.

Notes:

- For queried claims, review the adjudicator notes, attach requested medical reports or lab results, and click **Re-Queue Claim**.

3. Triage Nurse

Task 3.1: Capture Patient Vital Signs & Acuity Assessment

Who: `triage_nurse`, `nurse`

Steps:

1. Open **Queue** (`/queue`) and switch to your assigned **Triage Room**.
2. Select the next patient listed under **Waiting** and click **Call Patient**.
3. Measure and enter physiological parameters:
 - **Blood Pressure (BP):** Systolic and Diastolic in mmHg (e.g., `120` / `80`).
 - **Pulse / Heart Rate:** Beats per minute (e.g., `76` bpm).
 - **Body Temperature:** Degrees Celsius (e.g., `36.8` °C).
 - **Respiratory Rate:** Breaths per minute (e.g., `18` /min).
 - **Oxygen Saturation (SpO2):** Percentage on room air (e.g., `98` %).
 - **Random Blood Sugar (RBS):** In mmol/L (if clinically indicated or diabetic).
 - **Weight (kg) & Height (cm):** System automatically calculates BMI and displays nutritional status.
4. **Pediatric Assessment (For Patients Under 5 Years):**
 - Enter **MUAC (Mid-Upper Arm Circumference)** in centimeters:
 -  Red (<11.5 cm): Severe Acute Malnutrition (SAM).
 -  Yellow (11.5–12.5 cm): Moderate Acute Malnutrition (MAM).
 -  Green (>12.5 cm): Normal.
5. **Clinical Acuity Tagging:**
 - Select **Triage Priority:** `Green (Routine)`, `Yellow (Urgent)`, `Red (Immediate Resuscitation)` .
6. Click **Save Vitals & Route Patient**.
7. Select the target **Consultation Room** (e.g., Doctor Room 1, Pediatric Clinic, MCH/FP Clinic, Dental) from the dropdown and click **Send to Doctor**.

Notes:

-  **Panic Vitals:** If Systolic BP >180 mmHg, Temperature >39.5°C, or SpO2 <90%, the system flags a prominent red **Critical Alert**. Escort the patient immediately to the emergency resuscitation room and notify the casualty medical officer verbally.

4. Doctor / Clinical Officer

Task 4.1: Conduct Clinical Consultation & Review History

Who: `doctor`, `clinical_officer`, `dental_officer`

Steps:

1. Open **Queue** (`/queue`) and select your assigned **Consultation Room** (`/rooms/Sid`).
2. The waiting list shows triaged patients prioritized by acuity (Red/Yellow/Green) and arrival time.
3. Select the patient and click **Start Consultation**.
4. Review the patient banner:
 - Demographics, Age, Gender, Payment Mode, Allergy Alerts.
 - Live Triage Vitals and computed BMI/MUAC.
 - Past medical history, chronic condition flags, previous clinic visits, and previous prescriptions.
5. In the **Clinical Notes** tab, document structured clinical findings:
 - **Chief Complaints (CC):** Presenting symptoms and duration.
 - **History of Presenting Illness (HPI):** Systematic chronological progression.
 - **Review of Systems (ROS) & Physical Examination (PE):** General examination, systemic findings (CVS, RS, PA, CNS).
 - **Treatment Plan:** Intended clinical management strategy.
6. Click **Save Clinical Notes**.

Notes:

-  **Break-Glass Emergency:** If accessing a restricted or non-consented file during an emergency, click the **Break-Glass** shield button. Enter a mandatory medical justification (e.g., "Unconscious road traffic accident victim requiring immediate neurotrauma review"). The file will open instantly and record an immutable security audit event.

Task 4.2: Assign WHO ICD-11 Diagnosis Codes

Who: `doctor`, `clinical_officer`, `dental_officer`

Steps:

1. Scroll to the **Diagnoses** section on the consultation workbench.
2. In the **ICD-11 Search Bar**, begin typing the disease name, symptom, or code (minimum 2 characters, e.g., "Malaria", "Hypertension", "Pneumonia").
3. The system queries WHO's live ICD-11 API via the `icd11-search` edge function and displays standardized international search results.
4. Click on the correct matching condition.
5. Select the **Diagnosis Type**:
 - `Primary`: The main condition responsible for this encounter.
 - `Secondary`: Co-existing medical conditions or chronic diseases.
 - `Working`: Provisional clinical hypothesis pending lab/radiology confirmation.
 - `Differential`: Alternative diagnostic possibilities.
6. Click **Add Diagnosis**. The condition is saved in `encounter_diagnoses` with its official ICD-11 MMS URI and description.

Notes:

-  **MOH Reporting Compliance:** Every encounter must possess at least one primary ICD-11 diagnosis before it can be legally signed.

Task 4.3: Order Diagnostic Investigations (Laboratory & Radiology)

Who: `doctor`, `clinical_officer`

Steps:

1. In the **Diagnostic Orders** tab:
 - **To Order Lab Tests:** Search the `lab_test_catalog` (e.g., *Full Hemogram*, *Malaria BS*, *Urinalysis*, *Lipid Profile*, *LFTs*). Check urgent priority if required. Add specific clinical instructions.

- **To Order Imaging:** Select radiology modality and anatomical site (e.g., *Chest X-Ray PA, Abdominal Ultrasound, Head CT Plain*). Enter clinical indications.
- Click **Submit Diagnostic Orders**.
 - The system instantly generates `lab_orders` and `radiology_orders`, attaches billable line items to the patient's invoice, and routes the patient's name to the respective Laboratory and Radiology queue screens.
 - Route the patient physically to the diagnostic departments.

Notes:

- When laboratory or imaging results are completed and verified, the patient will automatically re-appear in your room queue with a **Results Ready** notification badge.

Task 4.4: Prescribe Medications & Formulations**Who:** `doctor`, `clinical_officer`, `dental_officer`**Steps:**

- Navigate to the **Prescriptions** section on the consultation screen.
- Search for the required drug in the master formulary (e.g., *Amoxicillin/Clavulanic Acid 625mg Tab, Paracetamol 500mg Tab, Ceftriaxone 1g Inj*).
- Specify the prescription parameters:
 - **Dose & Formulation:** e.g., `625 mg Tablet`.
 - **Frequency:** e.g., `TID (Three times daily)`, `BID (Twice daily)`, `OD (Once daily)`, `PRN (As needed)`.
 - **Duration:** Number of days (e.g., `7 days`).
 - **Route:** `Oral`, `Intravenous (IV)`, `Intramuscular (IM)`, `Topical`, `Inhalation`.
 - **Special Instructions:** e.g., *"Take with food", "Complete full course"*.
- Click **Add Prescription**.
- Repeat for all required medications and click **Submit Prescriptions**.
- The prescriptions are transmitted to the Pharmacy workbench, and invoice line items are generated automatically.

Notes:

- Review the patient's allergy alerts on the top header before prescribing penicillins, NSAIDs, or sulfa drugs.

Task 4.5: Admit Patient to an Inpatient Ward / ICU**Who:** `doctor`, `clinical_officer`**Steps:**

- If the patient requires inpatient hospitalization, click the **Admit to Ward** button.
- Select the target **Ward**:
 - *General Male Medical, General Female Surgical, Maternity, Pediatric, HDU, or Intensive Care Unit (ICU).*
- Select an **Available Bed** from the live bed occupancy map.
- Input the **Admission Diagnosis** (ICD-11) and author the initial **Admission Orders** (nursing instructions, IV fluids, monitoring frequency).
- Click **Confirm Admission**.
- The system executes the following:
 - Updates `admissions` table with status `admitted`.
 - Updates bed status to `occupied`.
 - If admitting to ICU: Automatically executes `charge_icu_admission_fee` to post the mandatory one-time ICU admission charge.
 - Routes the electronic chart to the Inpatient Nursing Station.

Notes:

- Ensure the ward selected is compatible with the patient's gender and age (e.g., adults cannot be admitted to Pediatric Ward).

Task 4.6: Finalize & Sign Clinical Encounter

Who: `doctor`, `clinical_officer`, `dental_officer` (Requires `sign_encounter` permission)

Steps:

- Once all clinical documentation, diagnoses, lab reviews, and prescriptions are complete:
- Click the green **Sign & Finalize Encounter** button at the bottom of the clinical screen.
- The system checks validation constraints:
 - Confirms at least one valid ICD-11 diagnosis exists.
 - Confirms clinical notes are not empty.
- A confirmation dialog appears: "Are you sure you want to sign this encounter? This action will cryptographically lock all notes and diagnoses against further changes."
- Click **Sign Encounter**.
- The system:
 - Sets `encounters.status = 'signed'`.
 - Records your digital signature, user ID, and exact timestamp.
 - Irreversibly locks clinical notes, diagnoses, and orders (`enforce_encounter_lock` trigger).
 - Triggers `auto_generate_sha_claim` for SHA-insured patients.
 - Queues FHIR R4 Encounter and Condition resources in `dha_outbound_queue` for national SHR synchronization.
 - Routes patient to **Pharmacy** (for drug collection) or **Cashier** (for final bill settlement).

Notes:

- ⚠ **Immutability:** Once an encounter is signed, it cannot be edited by any staff member. Any supplementary clinical observations must be entered as an addendum in a new encounter.

5. Laboratory Technician

Task 5.1: Accession Specimens & Enter Laboratory Results

Who: `lab_tech`, `staff`

Steps:

- Open the **Laboratory Workbench** (`/laboratory` or `/laboratory/$id`).
- The queue displays pending diagnostic orders grouped by patient and urgency.
- When the patient presents at the lab reception:
 - Verify patient identity against their printed lab requisition or file number.
 - Collect the designated biological sample (e.g., Venous Blood, Urine, Stool, Sputum, Swab).
 - Click **Collect Sample / In Progress**.
- Perform the analytical assay on the designated laboratory machine or manual bench test.
- In the laboratory results entry screen:
 - Input the quantitative values (e.g., Hemoglobin `13.4` g/dL, WBC `6.2` x10⁹/L, Platelets `250` x10⁹/L) or qualitative results (e.g., Malaria BS `No Malaria Parasites Seen`, Urinalysis Protein `Negative`).
 - The system displays standard reference ranges and highlights values outside normal limits with yellow or red flags.
- Enter technologist remarks or specimen quality notes if applicable.
- Click **Save Draft Results** to save progress, or proceed to verification.

Notes:

- For panic/critical results (e.g., Blood Glucose <2.2 mmol/L, Hemoglobin <5.0 g/dL, Potassium >6.5 mmol/L), call the ordering clinician immediately via internal phone in addition to logging the value in the system.

Task 5.2: Verify Results, Trigger Automated SMS & Auto-Return Patient

Who: `lab_tech`

Steps:

1. Review the entered test values for quality control and technical consistency.
2. Click **Verify & Publish Results**.
3. The system executes the following automated workflow atomically:
 - Updates `lab_results.status = 'verified'` and records `verified_by` and `verified_at`.
 - Locks results against further modification.
 - Invokes Edge Function `send-sms` via Africa's Talking to dispatch an automated notification SMS to the patient's phone: "Dear Patient, your laboratory test results at [Hospital Name] are now ready. Please proceed to your consultation room."
 - Executes `send_lab_results_to_requesting_room()`, automatically transferring the patient back into the requesting doctor's room queue.
 - Queues FHIR R4 `Observation` resources in `dha_outbound_queue` for DHA SHR transmission.

Notes:

- If a specimen is hemolyzed, clotted, or insufficient, click **Reject Specimen**, select the rejection reason, and the ordering clinician will be prompted to request a redraw.

6. Radiologist / Radiographer

Task 6.1: Process Radiology Order & Finalize Diagnostic Imaging Report

Who: `radiologist`, `doctor`

Steps:

1. Navigate to **Radiology** (`/radiology` or `/radiology/$id`).
2. Select a patient from the **Pending Imaging Queue**.
3. Position patient and perform requested modality scan (X-Ray, Ultrasound, CT, MRI, ECG, Mammography).
4. In the **Radiology Report Editor**:
 - Select the **Modality** and **Anatomical Site**.
 - Input structured findings:
 - *Clinical Indications:* (Pre-filled from doctor's order).
 - *Technique / Projection:* e.g., "Standard Erect Postero-Anterior (PA) view of the chest".
 - *Detailed Observations:* Systematic organ review (e.g., lung fields, cardiac silhouette, bony thorax, diaphragmatic costophrenic angles).
 - *Conclusion / Impression:* Concise diagnostic summary (e.g., "Right lower lobe consolidation consistent with lobar pneumonia. No pleural effusion.").
5. Click **Finalize & Sign Report**.
6. The system executes:
 - Commits report to `radiology_results`.
 - Sends automated completion SMS to the patient.
 - Invokes `send_radiology_results_to_requesting_room()`, routing the patient back to the ordering doctor's consultation queue.

Notes:

- If additional contrast or specialized extra projections were required during the scan, add the supplementary consumables so they are billed accurately on the invoice.

7. Pharmacist

Task 7.1: Dispense Outpatient Prescriptions & Real-Time Stock Deduction

Who: pharmacist, admin

Steps:

1. Open the **Pharmacy Dispensing Workbench** (/rooms/\$pharmacyRoomId).
2. Select a patient from the **Pending Prescriptions** list.
3. Review the ordered medication list:
 - Drug Name, Strength, Dose, Frequency, Duration, and Special Instructions.
4. **Payment / Clearance Verification:**
 - Verify that the invoice line item for the prescription is marked **paid** (for Cash patients) or **cleared** (for SHA / Insurance patients).
 - If unpaid in cash mode: Direct the patient to the Cashier to settle the bill before dispensing.
5. In the dispensing panel, for each prescribed medication:
 - Select the active **Batch Number** and verify expiry date from the Central Pharmacy Store (a0000000-0000-0000-0000-000000000003).
 - Confirm available warehouse stock balance.
 - Pack the physical drugs, labeling each container with dosage, frequency, and cautionary instructions.
6. Click **Dispense Selected Medications**.
7. The system:
 - Updates `prescriptions.status = 'dispensed'` and stamps `dispensed_at` and `dispensed_by`.
 - Executes database trigger `dispense_prescription_stock()`, creating a negative movement in `stock_movements` and reducing physical stock balance.
 - Triggers `trg_medication_dispense_fhir` to compile a FHIR `MedicationDispense` resource for national health record sync.
8. Verbally counsel the patient on correct drug usage, potential side effects, and storage conditions. Hand over the medications.

Notes:

-  **Stock Guard Violation:** If the pharmacy has zero available stock, the system will block dispensing. You must notify the storekeeper to execute an inter-store transfer from the Main Store or contact the clinician for an alternative therapeutic drug substitution.
-  **ICU Segregation:** General Pharmacy cannot dispense ICU store medications (`sprint13b_dispense_stock_guard`). ICU ward nurses dispense ICU medications directly from the ICU sub-store.

8. ICU Nurse

Task 8.1: Maintain ICU Hourly Flow-Sheet & Critical Monitoring

Who: nurse, icu_nurse

Steps:

1. Open the **Inpatient ICU Module** or select the **ICU Clinical Room** (a839ad3e-8721-4428-a825-6eeef75207b).
2. Select an admitted ICU patient bed.
3. Click **New Hourly Flow-Sheet Entry**.
4. Document the critical monitoring parameters:
 - **Hemodynamics:** Arterial Blood Pressure (Systolic, Diastolic, Mean Arterial Pressure - MAP), Central Venous Pressure (CVP), Heart Rate, Cardiac Rhythm.
 - **Mechanical Ventilation Settings:** Ventilation Mode (e.g., SIMV, PRVC, CPAP), Tidal Volume (VT), PEEP (cmH2O), FiO2 (%), Peak Inspiratory Pressure (PIP), Respiratory Rate.
 - **Neurological & Sedation:** Glasgow Coma Scale (GCS /15), Richmond Agitation-Sedation Scale (RASS: -5 to +4), Pupil size and light reactivity (Left/Right).
 - **Fluid Balance & Renal:** Hourly Urine Output (mL/hr), Gastric Aspirate / NG Drainage, Chest Tube Drainage, Central Line Infusions.
 - **Inotrope / Vasopressor Infusions:** Noradrenaline (mcg/min), Adrenaline, Dopamine, Dobutamine, Midazolam, Fentanyl infusion rates.
 - **Arterial Blood Gas (ABG):** pH, PaO2, PaCO2, HCO3, Base Excess, Lactate, SaO2.

5. Click **Save Hourly Record**. The chart compiles into a longitudinal real-time trend line.

Notes:

- Maintain strict hourly documentation. For MAP <65 mmHg, Urine Output <0.5 mL/kg/hr for 2 consecutive hours, or sudden desaturation, alert the Critical Care Physician immediately.

Task 8.2: Dispense Emergency Medications from ICU Sub-Store

Who: `icu_nurse`

Steps:

- On the ICU Patient Chart, navigate to **ICU Bedside Dispensing**.
- Select the required critical care drug from the dedicated **ICU Store** (`a99583cd-9354-470a-9299-73457734284d`).
- Enter the quantity administered (e.g., *Noradrenaline 4mg Ampoule x 2, Propofol 1% 20mL Vial x 1*).
- Enter clinical indication and administering nurse initials.
- Click **Confirm ICU Administration**.
- The system automatically logs stock consumption in `stock_usage`, updates the ICU stock ledger, and appends the medication line item to the patient's active inpatient invoice.

Notes:

- Emergency ICU medications are administered immediately without waiting for cashier payment clearance.

9. Dialysis Nurse

Task 9.1: Record Hemodialysis Session & Auto-Bill Consumables

Who: `dialysis_nurse`, `nurse`

Steps:

- Open the **Dialysis Unit Room** (`/rooms/${dialysisRoomId}`).
- Select an arriving renal patient or click **Start New Dialysis Session**.
- In the **Pre-Dialysis Assessment** section:
 - Record Pre-Dialysis Weight (kg), Dry Weight (kg), and computed Fluid Overload.
 - Record Pre-Dialysis Blood Pressure, Pulse, Temperature.
 - Assess Vascular Access: Type (`AV Fistula`, `AV Graft`, `Temporary IJ Catheter`, `Permacath`), Access Site, Thrill/Bruit status, and signs of infection.
- In the **Machine & Prescription Parameters** section:
 - Assign Dialysis Machine Number (`machines` table).
 - Select Dialyzer Model / Surface Area (e.g., *High-Flux F60, Low-Flux F6*).
 - Set Blood Flow Rate (BFR in mL/min, e.g., `300`), Dialysate Flow Rate (DFR in mL/min, e.g., `500`).
 - Set Prescribed Treatment Duration (e.g., `4.0 hours`).
 - Set Ultrafiltration Goal (UF Goal in mL, e.g., `2500 mL`).
 - Specify Anticoagulation: Heparin Loading Dose (IU) and Maintenance Infusion (IU/hr) or Saline Flushes (for heparin-free dialysis).
- In the **Session Consumables Checklist**:
 - Check all single-use consumables utilized from the **Dialysis Store** (`d38d45c7-20f2-4e1b-9279-8cb5bf567cd1`):
 - `Dialyzer Filter` (Qty: 1)
 - `Arterial & Venous Bloodlines` (Qty: 1)
 - `AV Fistula Needles 16G` (Qty: 2)
 - `Acid Concentrate & Bicarbonate Cartridge` (Qty: 1)
 - `Heparin 5000 IU/mL` (Qty: 1)

- `Normal Saline 1000mL` (Qty: 2)
- `Dialysis Dressing Pack & Gauze` (Qty: 1)

6. Monitor and document hourly intra-dialytic blood pressures, pulse, arterial pressure, venous pressure, TMP, and cumulative UF volume removed.

7. Upon session completion, document:

- Post-Dialysis Weight (kg), Net Ultrafiltration Removed (mL), Post-Dialysis BP.
- Any complications during session: Check applicable flags (*Hypotension, Cramps, Nausea, Access Clotting, Disequilibrium Syndrome*).

8. Click **Complete Session & Generate Billing**.

9. The system invokes `process_dialysis_session_billing`:

- Saves complete session clinical log in `dialysis_sessions`.
- Decrements all selected consumables from the Dialysis Store in `stock_usage`.
- Posts a comprehensive, itemized hemodialysis session bill to the invoice (or packages it for SHA Renal Replacement Claim submission).

Notes:

- For SHA-covered dialysis sessions, verify that the patient has not exceeded their 2 sessions/week statutory SHIF benefit allotment.

10. Accountant / Cashier

Task 10.1: Collect Payment, Issue Receipt & Allocate Funds

Who: `accountant`, `admin`

Steps:

1. Open **Billing & Accounting** (`/accounting` or `/invoices/$id`).
2. Search for the patient by **Invoice Number**, **Patient File Number**, **National ID**, or **Name**.
3. Click on the active invoice to inspect all itemized line items (Consultations, Lab, Imaging, Pharmacy, Procedures, Bed Days).
4. Review the computed totals: `Total Amount`, `Waivers Applied`, `Amount Already Paid`, and `Net Outstanding Balance`.
5. Click **Record Payment**.
6. In the payment modal:
 - Enter **Amount to Pay** (KSh).
 - Select **Payment Method**: `Cash`, `M-Pesa`, `Credit / Debit Card`, `Bank Transfer`, `Insurance Guarantee / Letter of Undertaking`.
 - If M-Pesa / Card / Bank: Enter the unique **Transaction Reference Number** (e.g., M-Pesa Transaction ID `SDF8923JKL`).
7. Click **Confirm Payment**.
8. The system executes database trigger `recalc_invoice_payments()`:
 - Inserts immutable transaction record into `invoice_payments`.
 - Updates invoice `amount_paid` and recalibrates balance.
 - If balance is zero: Updates invoice status to `paid` and clears encounter locks.
 - Generates printable official hospital receipt with QR verification code.
 - Dispatches automated SMS payment confirmation to patient's mobile phone.

Notes:

- Never accept cash without generating and issuing an official printed system receipt immediately.

Task 10.2: Apply an Indigent Patient Waiver or Credit Note

Who: `accountant`, `admin` (Requires supervisory authorization)

Steps:

1. Open the target patient invoice in `/invoices/$id`.

2. Click the **Apply Waiver / Adjustment** button.
3. Enter the **Waiver Amount** (KSh).
4. Select the **Waiver Category**: `Indigent Patient`, `Hospital Board Waiver`, `Clinical Service Waiver`, `Billing Correction`.
5. Enter the mandatory **Supervisory Approval Reference** and justification text.
6. Click **Authorize Waiver**.
7. The system updates `invoices.waiver_amount`, logs `waiver_approved_by` with your user ID and timestamp, recalculates the net balance, and writes an audit event to `audit_log`.

Notes:

-  **Audit Scrutiny:** All waivers are audited during institutional financial reviews and MoH accounting inspections. Never apply a waiver without written executive management approval.

Task 10.3: Execute End-of-Day (EOD) Cash Reconciliation

Who: `accountant`, `director`

Steps:

1. At the close of each cashier shift (e.g., 17:00 or shift change), navigate to **Accounting** → **End-of-Day Reconciliation**.
 2. Select your cashier user name and today's date range.
 3. The system compiles total collections categorized by channel:
 - Total Cash Collected (KSh)
 - Total M-Pesa Receipts (KSh)
 - Total Credit Card / Electronic Funds (KSh)
 - Total Waivers Applied (KSh)
 - Total Outstanding Invoices (KSh)
 4. Count physical currency in the cash drawer and enter the **Physical Cash Count** amount.
 5. The system calculates any variance between physical cash and system receipts.
 6. Click **Generate & Close EOD Shift Report**.
 7. Print the reconciled EOD summary, attach banking deposit slips, sign the report, and submit to the Chief Accountant / Hospital Director.
-

11. Mortician

Task 11.1: Admit an Internal Hospital Body to Mortuary

Who: `mortician`

Steps:

1. Open the **Mortuary Department** room (`214aca6c-cbde-40ce-af5a-92cc7815bda9`).
 2. Click **Admit Deceased Body**.
 3. Select **Body Source**: `Internal Hospital Death`.
 4. Search for the deceased patient by File Number or Inpatient Admission ID.
 5. Demographic details, date/time of death, ward of death, and certifying medical officer are auto-populated.
 6. Assign **Mortuary Cold Storage Unit / Vault Number** (e.g., `Cold Room B - Slot 14`).
 7. Enter receiving mortician notes (presence of valuables, personal effects, physical body condition).
 8. Click **Admit to Mortuary**.
 9. The system executes `assign_mortuary_reference()` (generating a unique mortuary reference, e.g., `MORT-2026-0042`) and links the mortuary stay to the patient's existing hospital invoice.
-

Task 11.2: Admit an External Body (Police Case / Home Death)

Who: `mortician`

Steps:

1. In the Mortuary screen, click **Admit Deceased Body**.
2. Select **Body Source:** `External (Brought in Dead / Police Case / Home Death)`.
3. Input deceased demographic details: Full Name (or *Unknown Deceased Male/Female*), Estimated Age, Gender.
4. Input Intake Legal Metadata:
 - Brought In By: Name, ID Number, Phone, Relationship / Title.
 - If Police Case: Accompanying Police Officer Name, Rank, Service Number, Police Station, and Police OB (Occurrence Book) Number.
 - Place of Death, Date and Time of Death.
5. Assign **Cold Storage Unit / Slot**.
6. Click **Register External Body**.
7. The system:
 - Creates a record in `mortuary_records`.
 - Generates mortuary reference number.
 - Invokes `create_external_mortuary_invoice()` to provision a dedicated commercial mortuary invoice linked to the next-of-kin / police file.

Task 11.3: Document Mortuary Services & Release Body

Who: `mortician`, `accountant`

Steps:

1. As mortuary services are rendered during storage:
 - Click **Add Mortuary Service** on the deceased record.
 - Add billable services: *Embalming / Preservation, Post-Mortem Examination Attendance, Refrigeration Storage Days, Body Dressing & Beautification, Hearse Transfer*.
 - The system automatically appends line items to the mortuary invoice.
 - Nightly daemon `accrue_daily_mortuary_charges()` automatically accrues standard daily refrigeration storage fees.
2. **Body Release Procedure:**
 - Next-of-kin presents official **Burial Permit / Death Certificate** and national ID.
 - Next-of-kin proceeds to Cashier to settle the final mortuary invoice balance.
 - Mortician verifies invoice status is `paid` (or authorized waiver attached).
 - Click **Release Body**.
 - Input collecting next-of-kin name, national ID, vehicle registration, and burial destination.
 - Click **Confirm Body Handover & Discharge**. The storage slot is marked `available` and the mortuary record is officially closed.

Notes:

-  *Release Lock*: The system will strictly block body release if an outstanding financial balance exists without an authorized waiver.

12. Store Keeper

Task 12.1: Receive Commercial Supplier Delivery into Main Store

Who: `store_keeper`, `admin`

Steps:

1. Navigate to **Deliveries & Stock Intake** (`/deliveries`).
2. Click **New Delivery / Stock Receipt**.
3. Enter Delivery Metadata:

- Supplier / Vendor Name (e.g., *KEMSA, MEDS, Harleys Ltd*).
- Supplier Invoice Number / Delivery Note Number.
- Local Purchase Order (LPO) Reference Number.
- Receiving Location: Select **Main Store** (`a0000000-0000-0000-0000-000000000001`).

4. In the **Delivered Items** table, for each delivered commodity:

- Search `stock_items` by name or SKU.
- Enter **Quantity Received** (Pack count / Units).
- Enter **Batch / Lot Number** and **Expiry Date**.
- Enter Unit Buying Price (KSh) and confirm selling tariff.

5. Click **Post Delivery to Stock**.

6. The system executes `delivery_to_stock()` :

- Records delivery in `stock_deliveries` .
- Creates positive ledger entries in `stock_movements` .
- Updates stock item current available quantities and logs weighted average buying costs.

Notes:

- Carefully check physical seals, batch numbers, and expiry dates against the delivery note before clicking submit. Never accept stock expiring within 6 months without supervisor sign-off.

Task 12.2: Transfer Stock to Departmental Sub-Stores

Who: `store_keeper` , `pharmacist`

Steps:

1. Navigate to **Stock Management** (`/stock`).
2. Click **Inter-Store Stock Transfer**.
3. Select **Source Location:** `Main Store` .
4. Select **Destination Location:**
 - `Central Pharmacy Store` (`a0000000-0000-0000-0000-000000000003`)
 - `ICU Store` (`a99583cd-9354-470a-9299-73457734284d`)
 - `Dialysis Store` (`d38d45c7-20f2-4e1b-9279-8cb5bf567cd1`)
 - `Laboratory Store`
5. Select the commodity and input the **Transfer Quantity**.
6. Add transfer note (e.g., *"Weekly stock requisition for ICU"*).
7. Click **Execute Stock Transfer**.
8. The system invokes `transfer_stock_between_locations()` RPC:
 - Deducts quantity from Source Store in `stock_movements` .
 - Credits quantity to Destination Store in `stock_movements` .
 - Generates printable inter-store transfer voucher.

13. System Administrator (Admin)

Task 13.1: Manage Staff User Accounts, Roles & Permissions

Who: `admin` , `system_admin`

Steps:

1. Navigate to **Admin → Users & Access** (`/admin/users`).
2. To approve a newly registered staff member:

- Locate user under **Pending Account Approvals**.
 - Click **Approve Account** (`is_approved = true`).
3. To assign roles:
- Click **Edit Roles** on the user card.
 - Assign applicable roles: `doctor`, `clinical_officer`, `nurse`, `triage_nurse`, `lab_tech`, `radiologist`, `pharmacist`, `accountant`, `insurance_agent`, `mortician`, `store_keeper`, `records_officer`, `admin`.
 - Click **Save Role Assignments**.
4. To grant room access:
- Navigate to **Admin** → **Rooms** (`/admin/rooms`).
 - Select consultation room or diagnostic unit and assign user to `user_room_access`.
5. To configure practitioner council details:
- In user profile, enter **Professional Council Type** (`KMPDC`, `COC`, `NCK`, `PPB`, `KMLTTB`) and **Council Registration Number** for statutory encounter signing.

Notes:

- Follow the principle of least privilege. Never assign `admin` or `sign_encounter` permissions to unauthorized non-clinical personnel.

Task 13.2: Configure Facility Metadata, Wards, Beds & Tariffs**Who:** `admin`**Steps:**

1. **Facility Settings** (`/admin/settings`):
 - Verify Facility Name, KMHFL Code, SHA Facility ID, MoH Level (1–6), County, Contact Phone, and Official Email. Click **Save Settings**.
2. **Wards & Beds** (`/admin/wards`):
 - Create inpatient wards, specify ward category, and assign standard **Daily Bed Rate (KSh)**.
 - Add individual physical bed numbers to each ward.
3. **Master Pricing Catalog** (`/admin/services` & `/admin/pricing`):
 - Set standard cash prices for general consultations, specialty clinics, procedures, and nursing services.
 - Configure **Insurer Contracted Tariffs** (`insurance_contracted_prices`) to establish contracted rate overrides for SHA, Jubilee, Britam, and other underwriters.

14. Medical Director / Executive Management**Task 14.1: Review Executive Dashboard, Epidemiological Trends & Health****Who:** `director`, `admin`**Steps:**

1. Log into the system and navigate to **Dashboard** (`/dashboard`).
2. **Clinical KPI Cards:**
 - Review Today's Outpatient Attendance, Total Active Inpatient Admissions, Bed Occupancy Rate (%), Emergency Casualty Volume, and Lab/Radiology Turnaround Times.
3. **MOH Epidemiological Surveillance:**
 - Inspect the **Top 10 Morbidity Disease Chart** generated by `dashboard_top_diseases()` RPC to detect community disease outbreaks (e.g., spikes in Malaria, Gastroenteritis, or Upper Respiratory Infections).
 - Review the 7-day OPD vs. Inpatient Admission Trend graph (`dashboard_admitted_opd_trend()`).
4. **Financial Summary:**
 - Inspect daily gross revenue, cash collections, insurance claims submitted, outstanding aging balances, and total waivers approved.

5. System Health & Integrations:

- Check status badges for Supabase Database connectivity, Africa's Talking SMS credit balance, and DHA Outbound Queue transmission latency.

Task 14.2: Generate & Export Statutory MoH Reports (705A, 705B, 717)

Who: `director`, `records_officer`, `admin`

Steps:

1. Navigate to **MOH Reporting Suite** (`/moh` or `/moh/705`, `/moh/717`).
 2. Select the target **Reporting Period** (Month and Year, e.g., *July 2026*).
 3. Select Report Type:
 - **MOH 705A:** Under 5 Years Outpatient Morbidity Summary.
 - **MOH 705B:** Over 5 Years Outpatient Morbidity Summary.
 - **MOH 706:** Laboratory Diagnostic Return.
 - **MOH 707:** Inpatient Bed Utilization & Mortality Return.
 - **MOH 717:** Institutional Workload Return.
 4. Click **Compile MoH Report**.
 5. The system executes `get_moh_705_report(start_date, end_date, form_type)` against database aggregate tags.
 6. Review the tabular return displaying disease classification, ICD-11 codes, male cases, female cases, and total incidence.
 7. Click **Export to CSV / Excel** or **Print Official MoH PDF** for submission to the Sub-County Health Records Information Officer (SCHRIO) and upload to the national DHIS2 / KHIS portal.
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AegisCare HMS — Internal API & RPC Reference

Document Title: PostgreSQL Database RPC Functions & Supabase Edge Functions Reference

Document Version: 1.0.0

Runtime Environment: Supabase PostgreSQL 15+ & Deno Serverless Edge Runtime

Target Environment: Kenya MoH Level 1–6 Facilities

Part 1: Supabase PostgreSQL RPC Functions

accrue_daily_bed_charges

Purpose: Iterates over all active inpatient admissions across general, maternity, pediatric, and surgical wards, calculating and posting a daily bed charge line item to each patient's open invoice.

Parameters: None (`void`).

Returns: `void`

Called from: Nightly `pg_cron` daemon (`0 21 * * *` UTC = 00:00 EAT) and manual administrative batch trigger (`src/routes/admin.wards.tsx`).

Business logic:

- Queries `admissions` where `status = 'admitted'`.
- Joins `wards` to extract the active `daily_rate` and `ward_name`.
- Finds the primary open invoice linked to `admissions.encounter_id`.
- Appends a line item to `invoice_line_items` with `item_type = 'bed_day'`, `source_id = admission_id`, `quantity = 1`, and `amount = daily_rate`.
- Description formatted as: `"[Ward Name] - bed charge (YYYY-MM-DD)"`.

accrue_daily_icu_charges

Purpose: Accrues specialized daily intensive care monitoring and equipment staging surcharges for all patients currently occupying beds within the Intensive Care Unit (ICU) or High Dependency Unit (HDU).

Parameters: None (`void`).

Returns: `void`

Called from: Nightly `pg_cron` daemon.

Business logic:

- Queries active admissions specifically located in wards where `ward_type = 'icu'`.
- Applies the intensive care unit daily tariff to the active encounter invoice.
- Distinguishes specialized ICU monitoring charges from standard baseline bed charges.

accrue_daily_mortuary_charges

Purpose: Automatically calculates and posts daily cold-storage refrigeration fees for all deceased bodies currently held in the hospital mortuary.

Parameters: None (`void`).

Returns: `void`

Called from: Nightly `pg_cron` daemon (`5 21 * * *` UTC = 00:05 EAT) and Mortuary Module (`supabase/migrations/20260811000006_sprint_b3_mortuary_external_billing.sql`).

Business logic:

- Iterates over `mortuary_records` where `released_at IS NULL`.
- If the body originated internally (`body_source = 'internal'`), locates the linked hospital invoice.

- If the body was admitted externally (`body_source = 'external'`), locates or provisions the dedicated commercial mortuary invoice via `create_external_mortuary_invoice`.
- Inserts a line item into `invoice_line_items` with `item_type = 'mortuary_storage'` and standard daily storage fee (default KSh 1,000/day).

`archive_old_audit_logs`

Purpose: Archives audit log records older than two years by moving them to the long-term `audit_log_archive` table, satisfying Kenya's 20-year medical record retention mandate while optimizing primary transactional query performance.

Parameters: None (`void`).

Returns: `void`

Called from: Nightly `pg_cron` daemon (`0 23 * * *` UTC = 02:00 EAT) (`supabase/migrations/20260806101050_e8638a1f-75b6-42c3-9a64-be6489877920.sql`).

Business logic:

- Calculates cutoff threshold: `v_cutoff := now() - INTERVAL '2 years'`.
- Deletes records where `changed_at < v_cutoff` from `public.audit_log` using a `RETURNING` clause.
- Inserts deleted rows atomically into `public.audit_log_archive`.
- Logs execution run metrics (rows moved, oldest timestamp, newest timestamp, execution status) into `audit_archive_runs`.

`charge_icu_admission_fee`

Purpose: Posts the mandatory, one-time specialized institutional ICU admission fee when an inpatient is admitted or transferred into an Intensive Care Unit ward.

Parameters:

- `p_encounter_id` (`uuid`) — The unique identifier of the active clinical encounter.
- `p_ward_id` (`uuid`) — The target ward identifier.

Returns: `void`

Called from: Inpatient Admission Dialog (`src/routes/rooms.$id.tsx` line 3997).

Business logic:

- Validates whether the target ward has `ward_type = 'icu'`.
- Locates the active invoice linked to `p_encounter_id`.
- Checks whether an `icu_admission_fee` line item has already been posted to avoid duplicate surcharges.
- Inserts a line item into `invoice_line_items` with `item_type = 'procedure'`, description `'ICU Admission & Specialized Staging Surcharge'`, and standard institutional rate (e.g., KSh 5,000).

`process_dialysis_session_billing`

Purpose: Processes completed hemodialysis session clinical logs, records consumable inventory deductions, and posts itemized billing charges in a single atomic database transaction.

Parameters:

- `p_patient_id` (`uuid`) — The patient UUID.
- `p_room_id` (`uuid`) — The Dialysis Unit room UUID.
- `p_session_id` (`uuid`) — The completed `dialysis_sessions` record UUID.
- `p_items` (`jsonb`) — An array of consumable objects, each containing `item_id`, `name`, `quantity`, and `unit_price`.

Returns: `void`

Called from: Dialysis Nursing Workbench (`src/routes/rooms.$id.tsx` line 6597).

Business logic:

- Resolves the active encounter for the patient in the dialysis room.
- Iterates over the `p_items` JSON array:
 - Invokes `record_stock_usage` to deduct each consumable from the Dialysis Store (`d38d45c7-20f2-4e1b-9279-8cb5bf567cd1`).

- Appends corresponding consumable line items to `invoice_line_items`.
- Appends the base Hemodialysis Session Procedure Fee line item.
- Recalculates invoice totals via database triggers.

`auto_generate_sha_claim`

Purpose: Automatically compiles a structured Social Health Authority (SHA) claim when an encounter is signed or closed for a patient registered under `payment_mode = 'sha_shif'`.

Parameters: Trigger function (executes on `UPDATE OF status ON encounters`).

Returns: `trigger` (or executed via RPC).

Called from: Database trigger `trg_auto_generate_sha_claim` on `encounters` (`supabase/migrations/20260813000003_sprint4_sha_claim_guards.sql`).

Business logic:

- Evaluates whether `NEW.payment_mode = 'sha_shif'` and `NEW.status = 'signed'`.
- Validates that at least one ICD-11 primary diagnosis exists in `encounter_diagnoses`.
- Classifies fund subtype (`phf`, `shif`, `eccif`) based on facility MoH level and clinical service room.
- Creates a record in `sha_claims` with status `'draft'`.
- Iterates over billable diagnostic and procedural items, mapping them to SHA tariff codes in `sha_claim_items`.
- Invokes `build_fhir_claim()` to compile the attached FHIR Claim resource JSON.

`log_break_glass_access`

Purpose: Executes an emergency security override, allowing authorized medical practitioners to access restricted or un-consented patient health records during acute emergencies while logging an immutable forensic audit event.

Parameters:

- `p_patient_id` (`uuid`) — The patient record identifier accessed.
- `p_justification` (`text`) — Mandatory written clinical justification for the override.
- `p_accessed_by` (`uuid`) — The authenticated practitioner UUID.
- `p_accessor_email` (`text`) — The practitioner's authenticated email address.

Returns: `void`

Called from: Clinical Consultation UI (`src/routes/rooms.$id.tsx`, `supabase/migrations/20260806000001_sprint8a_break_glass.sql`).

Business logic:

- Validates that `p_justification` is non-null and not empty; raises an exception if missing.
- Inserts an immutable security event into `public.audit_log` with `action = 'BREAK_GLASS'`, `table_name = 'patients'`, `record_id = p_patient_id`, and `new_data` capturing justification text, accessor email, and timestamp.
- Function defined as `SECURITY DEFINER` with execution granted to `authenticated`.

`audit_trigger_fn`

Purpose: Universal PostgreSQL database trigger function that captures table-level data mutations (INSERT, UPDATE, DELETE) and writes immutable before/after snapshots into `audit_log`.

Parameters: Trigger context (`TG_TABLE_NAME`, `TG_OP`, `NEW`, `OLD`).

Returns: `trigger`

Called from: Attached triggers across 15+ database tables (`encounters`, `patients`, `admissions`, `prescriptions`, `lab_results`, `invoices`, etc.).

Business logic:

- Captures operating user ID from `auth.uid()`.
- Captures `OLD` row as `old_data` JSONB on `UPDATE` and `DELETE` operations.
- Captures `NEW` row as `new_data` JSONB on `INSERT` and `UPDATE` operations.
- Inserts row into `audit_log` with transaction timestamp.

audit_ward_transfers

Purpose: Specialized audit trigger tracking inpatient movements between hospital wards and beds, maintaining chain-of-custody and rate change history.

Parameters: Trigger context on `admissions`.

Returns: `trigger`

Called from: Database trigger `trg_audit_ward_transfers` on `admissions` (`UPDATE OF ward_id, bed_id`).

Business logic:

- Detects changes where `OLD.ward_id IS DISTINCT FROM NEW.ward_id` or `OLD.bed_id IS DISTINCT FROM NEW.bed_id`.
- Frees the previous bed (`status = 'available'`) and locks the new bed (`status = 'occupied'`).
- Logs a structured `WARD_TRANSFER` entry in `audit_log` capturing originating ward, destination ward, previous bed, and new bed.

build_fhir_claim

Purpose: Compiles a complete, standardized HL7 FHIR Release 4 `Claim` resource JSON payload for a given SHA claim identifier.

Parameters:

- `p_claim_id` (`uuid`) — The unique SHA claim identifier in `sha_claims`.

Returns: `jsonb` (The compiled FHIR R4 Claim resource).

Called from: Edge Function `claims-dispatcher` and database trigger `auto_build_fhir_claim` (`supabase/migrations/20260813000006_sprint8_fhir_claim_resource.sql`).

Business logic:

- Retrieves claim details, patient demographics, facility KMHFL and SHA provider numbers, encounter metadata, and itemized claim lines.
- Builds FHIR JSON structure containing:
 - `resourceType: "Claim"`, `status: "active"`, `use: "claim"`.
 - `patient`: Reference to FHIR `Patient`.
 - `provider`: Organization reference with KMHFL identifier.
 - `diagnosis`: Array of ICD-11 conditions mapped from `encounter_diagnoses`.
 - `item`: Array of billable service lines with tariff codes, quantities, and unit prices.
 - `total`: Total claimed monetary amount in KSh.

can_access_room

Purpose: Evaluates whether a specific user is authorized to enter, view queues, and conduct clinical work within a designated hospital room.

Parameters:

- `_user` (`uuid`) — The user UUID (`auth.uid()`).
- `_room` (`uuid`) — The room UUID.

Returns: `boolean` (`true` if authorized, `false` otherwise).

Called from: RLS policies on room queues and Room View components (`src/routes/rooms.$id.tsx`).

Business logic:

- Returns `true` if `_user` possesses the `admin` role in `user_roles`.
- Returns `true` if an explicit record exists in `user_room_access` matching `user_id = _user` and `room_id = _room`.
- Returns `false` otherwise.

create_encounter_from_appointment

Purpose: Converts a pre-booked outpatient appointment into an active clinical encounter upon patient arrival.

Parameters:

- `p_appointment_id` (`uuid`) — The appointment record UUID.

Returns: `uuid` (The newly provisioned `encounters.id`).

Called from: Appointment Calendar Workbench (`src/routes/appointments.tsx`).

Business logic:

- Fetches appointment details (patient ID, clinic provider ID, reason for visit).
- Inserts a new row into `encounters` with `status = 'waiting'`, `encounter_type = 'outpatient'`, and notes pre-filled from appointment reasons.
- Updates appointment status to `checked_in` and links `appointment.encounter_id`.
- Returns the new encounter UUID for immediate room queue routing.

`create_external_mortuary_invoice`

Purpose: Generates a dedicated commercial master invoice for a deceased body brought to the mortuary from outside the hospital (police case / home death).

Parameters:

- `p_record_id` (`uuid`) — The `mortuary_records` identifier.
- `p_created_by` (`uuid` , optional) — The receiving mortician user ID.

Returns: `uuid` (The newly provisioned `invoices.id`).

Called from: Mortuary Intake Flow (`src/routes/rooms.$id.tsx` Mortuary Room).

Business logic:

- Generates sequential invoice number (`INV-YYYY-XXXXX`).
- Inserts a row into `invoices` with `patient_id = NULL` (external deceased) and links mortuary reference metadata.
- Posts initial standard intake registration and cold-storage line items.
- Returns invoice UUID.

`dashboard_top_diseases`

Purpose: Aggregates and returns the top 10 clinical diagnoses recorded across the facility within a specified date window, stratified by age band.

Parameters:

- `p_start` (`date`) — Start date of analysis window.
- `p_end` (`date`) — End date of analysis window.

Returns: Table of `{ age_band text, disease_count integer, icd11_title text }`.

Called from: Executive Dashboard (`src/routes/dashboard.tsx`).

Business logic:

- Joins `encounter_diagnoses`, `encounters`, and `patients`.
- Filters records between `p_start` and `p_end`.
- Groups by ICD-11 condition title and computed age band (`Under 5`, `5 to 59`, `60+`).
- Orders by incidence count descending, returning top morbidities for epidemiological surveillance.

`dashboard_admitted_opd_trend`

Purpose: Computes daily comparative volumes of general outpatient visits versus inpatient admissions over a selected date range.

Parameters:

- `p_start` (`date`) — Start date.
- `p_end` (`date`) — End date.

Returns: Table of `{ day text, opd_count integer, admitted_count integer }`.

Called from: Executive Dashboard (`src/routes/dashboard.tsx`).

Business logic:

- Generates continuous date series between `p_start` and `p_end`.
- Tallies outpatient encounters (`encounter_type = 'outpatient'`) and inpatient admissions (`admissions.admitted_at`).
- Returns daily comparative array for trend chart visualization.

dashboard_emergency_referrals

Purpose: Aggregates emergency attendance volumes and tracks inward versus outward hospital referrals for statutory MoH 717 reporting.

Parameters:

- `p_start` (`date`) — Analysis start date.
 - `p_end` (`date`) — Analysis end date.
- Returns:** Table of (`emergency_count integer`, `referrals_in integer`, `referrals_out integer`).
- Called from:** Executive Dashboard & Reports (`src/routes/reports.tsx`).
- Business logic:**
- Counts encounters where `is_emergency = true`.
 - Tallies `referral_direction = 'in'` and `referral_direction = 'out'`.

get_contracted_price

Purpose: Resolves the negotiated tariff for a specific medical procedure, consultation, drug, or diagnostic test under an insurance underwriter's contract.

Parameters:

- `p_insurance_provider_id` (`uuid`) — Underwriter UUID.
 - `p_item_id` (`uuid`) — Catalog service or stock item UUID.
 - `p_item_type` (`text`) — Item classification (`service`, `lab`, `radiology`, `stock`).
- Returns:** `numeric` (The contracted tariff amount in KSh).
- Called from:** Billing engine and insurance clearance workflows.
- Business logic:**
- Queries `contracted_prices` for matching provider, item ID, and type.
 - If a contracted price is defined and active, returns the negotiated rate.
 - If no contract override exists, falls back to the item's standard baseline price in `lab_test_catalog` or `stock_items`.

get_moh_705_report

Purpose: Compiles certified statutory Ministry of Health Outpatient Morbidity Reports (MOH 705A for Under 5 Years or MOH 705B for Over 5 Years) across standard disease classifications.

Parameters:

- `p_start_date` (`date`) — Reporting period start date.
- `p_end_date` (`date`) — Reporting period end date.
- `p_form_type` (`text`, optional) — `'705A'` (Under 5) or `'705B'` (Over 5). Defaults to `'705A'`.

Returns: Table of:

- `row_number` (`integer`) — Official MoH row index (Rows 1 to 45).
- `disease_name` (`text`) — Official MoH disease classification title.
- `icd11_code` (`text`) — Mapped ICD-11 reference code.
- `male_cases` (`integer`) — Tally of male presentations.
- `female_cases` (`integer`) — Tally of female presentations.
- `total_cases` (`integer`) — Total aggregate morbidity count.

Called from: MOH 705 Reporting Suite (`src/routes/moh.705.tsx`, `supabase/migrations/20260722120000_moh_production_sync.sql`).

Business logic:

- Queries `moh_705_disease_mappings` for the full statutory disease catalog.
- Evaluates `encounter_diagnoses` linked to encounters between `p_start_date` and `p_end_date`.
- Filters patient cohort based on age criteria (`< 5 years` for 705A; `>= 5 years` for 705B).
- Groups and sums case counts by patient biological sex (`patients.sex`).

send_lab_results_to_requesting_room

Purpose: Automatically routes a patient back into the requesting doctor's consultation room queue upon technologist verification of laboratory test results.

Parameters:

- `p_encounter_id` (`uuid`) — The active encounter UUID.
- Returns:** `text` (Target room identifier or status confirmation).
- Called from:** Laboratory Workbench (`src/routes/laboratory.$id.tsx` , `supabase/migrations/20260728120000_radiology_room_routing.sql`).

Business logic:

- Queries `encounter_room_visits` for `p_encounter_id` to identify the originating consultation room prior to the laboratory referral.
- Updates `encounters.current_room_id` to the originating doctor's room.
- Resets `encounters.status = 'waiting'` , creating an automated notification badge on the doctor's screen.

send_radiology_results_to_requesting_room

Purpose: Automatically routes a patient back into the requesting clinician's room queue upon finalization of a diagnostic radiology imaging report.

Parameters:

- `p_encounter_id` (`uuid`) — The active encounter UUID.
- Returns:** `text` (Target room identifier).
- Called from:** Radiology Workbench (`src/routes/radiology.$id.tsx`).

Business logic:

- Resolves the originating consultation room from `encounter_room_visits` .
- Updates `encounters.current_room_id` to the clinician's room and sets status to `waiting` .

transfer_stock_between_locations

Purpose: Executes an inter-store warehouse inventory transfer from a source warehouse to a destination departmental sub-store.

Parameters:

- `source_location_id` (`uuid`) — Originating store UUID (e.g., Main Store).
- `destination_location_id` (`uuid`) — Destination store UUID (e.g., Pharmacy or ICU Store).
- `target_item_id` (`uuid`) — Stock item UUID.
- `transfer_quantity` (`numeric`) — Physical quantity to transfer.
- `note` (`text` , optional) — Requisition note or batch reference.

Returns: `void`

Called from: Stock Management (`src/routes/stock.tsx`).

Business logic:

- Verifies that `transfer_quantity > 0` .
- Inserts a negative movement into `stock_movements` for `source_location_id` (`reason = 'transfer_out'`).
- Inserts a positive movement into `stock_movements` for `destination_location_id` (`reason = 'transfer_in'`).
- Creates a record in `stock_transfers` documenting the transaction and initiating user.

record_stock_usage

Purpose: Deducts consumable supplies and single-use medical devices consumed during patient clinical care or surgical procedures.

Parameters:

- `source_location_id` (`uuid`) — Store location UUID.
- `target_item_id` (`uuid`) — Stock item UUID.
- `used_quantity` (`numeric`) — Quantity consumed.
- `target_encounter_id` (`uuid` , optional) — Associated clinical encounter.

- `usage_reason` (`text` , optional) — Clinical procedure or department reason.
 - `note` (`text` , optional) — Descriptive remarks.
- Returns:** `void`
- Called from:** ICU, Dialysis, and Surgical Room Workbenches.
- Business logic:**
- Inserts record into `stock_usage`.
 - Deducts quantity from warehouse via `stock_movements` with `reason = 'usage'`.

`grant_stock_location_access`

Purpose: Grants specific warehouse sub-store access permissions to a designated staff user.

Parameters:

- `target_user_id` (`uuid`) — User UUID.
 - `target_location_id` (`uuid`) — Stock location UUID.
 - `allow_view` (`boolean` , default `true`) — Permission to view inventory balances.
 - `allow_request` (`boolean` , default `false`) — Permission to submit requisitions.
 - `allow_approve` (`boolean` , default `false`) — Permission to approve requisitions.
 - `allow_issue` (`boolean` , default `false`) — Permission to issue stock.
 - `allow_receive` (`boolean` , default `false`) — Permission to receive supplier stock.
- Returns:** `void`
- Called from:** Admin Inventory Access Suite (`src/routes/admin.permissions.tsx`).
- Business logic:**
- Inserts or updates user privileges in `user_stock_location_access`.

`has_role`

Purpose: Security helper function verifying whether an authenticated user is assigned a specific hospital role.

Parameters:

- `_user_id` (`uuid`) — User UUID.
 - `_role` (`public.app_role`) — Role name (e.g., `'admin'`, `'doctor'`, `'nurse'`, `'pharmacist'`).
- Returns:** `boolean` (`true` if assigned, `false` otherwise).
- Called from:** RLS policies across all database tables.
- Business logic:**
- Checks for matching row in `public.user_roles` where `user_id = _user_id` and `role = _role`.

`user_has_permission`

Purpose: Granular security helper checking whether a user possesses an explicit permission through their assigned roles.

Parameters:

- `_user` (`uuid`) — User UUID.
 - `_perm` (`text`) — Permission string (e.g., `'sign_encounter'`, `'register_patient'`, `'accounting'`).
- Returns:** `boolean`
- Called from:** RLS policies and frontend navigation guards.
- Business logic:**
- Returns `true` if user has `admin` role.
 - Joins `user_roles` and `role_permissions` to match `user_id = _user` and `permission = _perm`.

`is_approved`

Purpose: Foundational security gate verifying whether a staff account has been approved by an administrator.

Parameters:

- `_user_id` (`uuid`) — User UUID (`auth.uid()`).
Returns: `boolean` (`true` if account approved, `false` otherwise).
Called from: Global RLS policies across all 35+ tables.
Business logic:
- Queries `profiles` for `is_approved = true` or `user_roles` for active roles. Rejects unapproved user requests.

`validate_and_get_icd11`

Purpose: Verifies whether an ICD-11 code is locally cached and returns its official WHO MMS title and URI.

Parameters:

- `search_code` (`text`) — ICD-11 alphanumeric code (e.g., `'1B10'`).
Returns: Table of { `p_code text`, `p_title text`, `p_uri text`, `p_is_cached boolean` } .
Called from: Diagnosis validation triggers (`clean_and_validate_diagnosis_insert`).
Business logic:
- Checks local cache table `icd11_codes` . Returns cached record if found; otherwise indicates uncached state for Edge Function lookup.

`verify_patient_identity`

Purpose: Stub RPC function for verifying patient citizen identity against the National Integrated Population Registration System (IPRS) via DHA API proxy.

Parameters:

- `p_patient_id` (`uuid`) — Patient UUID.
- `p_national_id` (`text`) — Citizen National ID or Passport Number.
- `p_id_type` (`text`) — ID classification (`national_id` , `passport` , `birth_certificate`).
- `p_verified_by` (`uuid`) — Staff user initiating verification.
Returns: `jsonb` (Verification status and citizen demographic payload).
Called from: Patient Registration Form (`src/routes/register-patient.tsx`).
Business logic:
- Currently operational in STUB mode (logs verification in `patients.identity_verified`).
- Phase 3 will execute live HTTPS query to DHA IPRS gateway.

`verify_practitioner`

Purpose: Stub RPC validating healthcare practitioner credentials against Kenya Health Worker Regulatory Councils (KMPDC, COC, NCK, PPB, KMLTTB).

Parameters:

- `p_profile_id` (`uuid`) — Profile UUID.
- `p_council_type` (`text`) — Regulatory council name.
- `p_registration_number` (`text`) — Council license/registration number.
Returns: `jsonb` (Practitioner accreditation status).
Called from: User Profile Management (`src/routes/admin.users.tsx`).
Business logic:
- Updates `profiles.council_registration_number` and validates formatting against national council patterns.

Part 2: Supabase Edge Functions Reference

1. send-sms

- **Location:** `supabase/functions/send-sms/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Dispatches real-time SMS messages via Africa's Talking API for OTP consent codes, lab test completion notifications, and billing receipts.
- **Request Headers:**
 - `Authorization: Bearer <SUPABASE_ANON_OR_SERVICE_KEY>`
 - `Content-Type: application/json`
- **Request Payload:**

```
{
  "to": "0712345678",
  "message": "Dear Patient, your AegisCare OTP verification code is 492810. Valid for 10 minutes."
}
```

- **Business Logic:**
 - Formats Kenyan telephone numbers starting with `0` or `01` to international format (`+254...`).
 - Reads secrets `AT_USERNAME` and `AT_API_KEY`.
 - Routes to Africa's Talking sandbox (`https://api.sandbox.africastalking.com/...`) if username is `sandbox`; otherwise routes to live production gateway (`https://api.africastalking.com/...`).
- **Response Format:**

```
{
  "SMSMessageData": {
    "Message": "Sent to 1/1 Total Cost: KES 0.8000",
    "Recipients": [
      { "number": "+254712345678", "status": "Success", "cost": "KES 0.8000", "messageId": "ATXid_..." }
    ]
  }
}
```

2. icd11-search

- **Location:** `supabase/functions/icd11-search/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Proxies live diagnostic search requests to the World Health Organization (WHO) ICD-11 API, securing API secrets server-side and caching OAuth2 bearer tokens.
- **Request Payload:**

```
{
  "query": "Malaria"
}
```

- **Business Logic:**
 - Enforces minimum query length of 2 characters.
 - Maintains an in-memory OAuth2 token cache with automated renewal before expiry against `https://icdaccessmanagement.who.int/connect/token`.
 - Queries WHO ICD-11 MMS 2024 linearization endpoint (`https://id.who.int/icd/release/11/2024-01/mms/search`) with `useFlexisearch=true`.
 - Sanitizes and strips HTML markup from WHO entity titles.
- **Response Format:**

```
{
  "results": [
    {
      "code": "1F40",
      "title": "Malaria due to Plasmodium falciparum",
      "uri": "http://id.who.int/icd/release/11/2024-01/mms/1F40"
    }
  ]
}
```

3. `claims-dispatcher`

- **Location:** `supabase/functions/claims-dispatcher/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Central claim and health information exchange router. Evaluates concluded encounters, verifies patient HIE consent, compiles FHIR bundles, and routes payloads to `dha_outbound_queue`.
- **Request Payload:**

```
{
  "encounter_id": "8fa85ble-7b12-4c28-98e1-0d32b5f7e4a1",
  "patient_id": "4b6c7d8e-9f01-2a3b-4c5d-6e7f8a9b0c1d",
  "insurer_type": "sha_shif",
  "trigger": "encounter_closed"
}
```

- **Business Logic:**
 - Authenticates user JWT.
 - Verifies patient `hie_data_sharing_consented` in `patient_consents`. If unconsented, skips HIE synchronization.
 - If `insurer_type = 'sha_shif'`: Resolves draft claim, builds FHIR claim bundle via `build_fhir_claim`, and queues in `dha_outbound_queue`.
 - If `insurer_type = 'private'`: Queues under `private_claim` handler.
 - If `insurer_type = 'cash'`: Flags internal cash ledger completion.

- **Response Format:**

```
{
  "success": true,
  "encounter_id": "8fa85ble-7b12-4c28-98e1-0d32b5f7e4a1",
  "results": [
    { "queue_type": "fhir_sync", "handler": "FhirSyncHandler", "status": "queued", "queue_id": "..."},
    { "queue_type": "sha_claim", "handler": "ShaClaimsHandler", "status": "queued", "queue_id": "..."}
  ],
  "note": "STUB MODE - Payloads queued locally awaiting Phase 3 DHA credentials."
}
```

4. `fhir-patient`

- **Location:** `supabase/functions/fhir-patient/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Generates a standardized HL7 FHIR Release 4 `Patient` resource JSON payload from database demographic tables.
- **Request Payload:**

```
{
  "patient_id": "4b6c7d8e-9f01-2a3b-4c5d-6e7f8a9b0c1d"
}
```

```
}

```

• Response Format:

```
{
  "resourceType": "Patient",
  "id": "4b6c7d8e-9f01-2a3b-4c5d-6e7f8a9b0c1d",
  "meta": { "profile": ["http://hl7.org/fhir/StructureDefinition/Patient"] },
  "identifier": [
    {
      "use": "official",
      "system": "https://hiskanya.org/facility/12345/patients",
      "value": "P000142"
    }
  ],
  "name": [{ "use": "official", "text": "John Kamau Mwangi", "family": "Mwangi", "given": ["John", "Kamau"] }],
  "gender": "male",
  "birthDate": "1988-04-12",
  "telecom": [{ "system": "phone", "value": "+254712345678", "use": "mobile" }],
  "address": [{ "use": "home", "city": "Nairobi", "district": "Nairobi County", "country": "KE" }],
  "managingOrganization": { "display": "AegisCare Facility" }
}
```

5. `fhir-encounter`

- **Location:** `supabase/functions/fhir-encounter/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Generates a standardized HL7 FHIR Release 4 `Encounter` resource detailing service class, arrival status, linked conditions, and service organization.
- **Request Payload:**

```
{
  "encounter_id": "8fa85b1e-7b12-4c28-98e1-0d32b5f7e4a1"
}
```

• Response Format:

```
{
  "resourceType": "Encounter",
  "id": "8fa85b1e-7b12-4c28-98e1-0d32b5f7e4a1",
  "meta": { "profile": ["http://hl7.org/fhir/StructureDefinition/Encounter"] },
  "status": "finished",
  "class": {
    "system": "http://terminology.hl7.org/CodeSystem/v3-ActCode",
    "code": "AMB",
    "display": "ambulatory"
  },
  "subject": { "reference": "Patient/4b6c7d8e-9f01-2a3b-4c5d-6e7f8a9b0c1d" },
  "diagnosis": [
    {
      "condition": { "reference": "Condition/9c8b7a6e-5d4c-3b2a-1f0e-9d8c7b6a5e4d" },
      "rank": 1
    }
  ],
  "serviceProvider": {
    "identifier": { "system": "https://hiskanya.org/facility", "value": "12345" },
    "display": "AegisCare Facility"
  }
}
```

6. `fhir-condition`

- **Location:** `supabase/functions/fhir-condition/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Returns an array of HL7 FHIR Release 4 `Condition` resources mapped to WHO ICD-11 concept URIs for a specified encounter.
- **Request Payload:**

```
{
  "encounter_id": "8fa85ble-7b12-4c28-98e1-0d32b5f7e4a1"
}
```

- **Response Format:**

```
[
  {
    "resourceType": "Condition",
    "id": "9c8b7a6e-5d4c-3b2a-1f0e-9d8c7b6a5e4d",
    "meta": { "profile": ["http://hl7.org/fhir/StructureDefinition/Condition"] },
    "verificationStatus": {
      "coding": [{ "system": "http://terminology.hl7.org/CodeSystem/condition-ver-status", "code": "confirmed" }]
    },
    "code": {
      "coding": [
        {
          "system": "http://id.who.int/icd/release/11/mms",
          "code": "1F40",
          "display": "Malaria due to Plasmodium falciparum"
        }
      ],
      "text": "Malaria due to Plasmodium falciparum"
    },
    "subject": { "reference": "Patient/4b6c7d8e-9f01-2a3b-4c5d-6e7f8a9b0c1d" },
    "encounter": { "reference": "Encounter/8fa85ble-7b12-4c28-98e1-0d32b5f7e4a1" }
  }
]
```

7. `fhir-bundle`

- **Location:** `supabase/functions/fhir-bundle/index.ts`
- **HTTP Method:** `POST`
- **Purpose:** Assembles a comprehensive HL7 FHIR Release 4 `Bundle` (type = `collection`) consolidating Patient, Encounter, EpisodeOfCare, Condition[], and MedicationDispense[] resources for submission to the DHA Shared Health Record (SHR).
- **Request Payload:**

```
{
  "encounter_id": "8fa85ble-7b12-4c28-98e1-0d32b5f7e4a1"
}
```

- **Response Format:**

```
{
  "resourceType": "Bundle",
  "id": "b1c2d3e4-f5a6-7b8c-9d0e-1f2a3b4c5d6e",
  "type": "collection",
  "timestamp": "2026-08-22T12:00:00Z",
  "entry": [
    { "fullUrl": "urn:uuid:patient-id", "resource": { "resourceType": "Patient", "id": "..." } },
    { "fullUrl": "urn:uuid:encounter-id", "resource": { "resourceType": "Encounter", "id": "..." } },
    { "fullUrl": "urn:uuid:condition-id", "resource": { "resourceType": "Condition", "id": "..." } },
  ]
}
```

```
{ "fullUrl": "urn:uuid:dispense-id", "resource": { "resourceType": "MedicationDispense", "id": "..."} }
]
```

AegisCare HMS — Database Schema & System Architecture

Document Title: Database Schema Specification & System Topology

Document Version: 1.0.0

Database Engine: PostgreSQL 15+ (Supabase Managed with RLS)

Target Environment: Kenya MoH Level 1–6 Facilities

Section 1 — Entity Relationship Diagram (Mermaid ERD)

```
erDiagram
    %% =====
    %% PATIENT CORE & ADMISSIONS
    %% =====
    PATIENTS ||--o{ ENCOUNTERS : "has"
    PATIENTS ||--o{ PATIENT_CONSENTS : "grants"
    PATIENTS ||--o{ CONSENT_OTPS : "receives"
    PATIENTS ||--o{ APPOINTMENTS : "books"
    PATIENTS ||--o{ ADMISSIONS : "undergoes"
    PATIENTS ||--o{ INVOICES : "billed"
    PATIENTS ||--o{ MORTUARY_RECORDS : "admitted_postmortem"

    WARDS ||--o{ BEDS : "contains"
    WARDS ||--o{ ADMISSIONS : "hosts"
    BEDS ||--o{ ADMISSIONS : "occupies"
    ENCOUNTERS ||--o{ ADMISSIONS : "triggers"

    %% =====
    %% CLINICAL & DIAGNOSTICS
    %% =====
    ENCOUNTERS ||--o{ CLINICAL_NOTES : "documents"
    ENCOUNTERS ||--o{ ENCOUNTER_DIAGNOSES : "coded_with"
    ENCOUNTERS ||--o{ LAB_ORDERS : "orders"
    ENCOUNTERS ||--o{ RADIOLOGY_ORDERS : "orders"
    ENCOUNTERS ||--o{ PRESCRIPTIONS : "prescribes"
    ENCOUNTERS ||--o{ ENCOUNTER_ROOM_VISITS : "routes"
    ENCOUNTERS ||--o{ ENCOUNTER_INDICATOR_TAGS : "tagged"

    LAB_ORDERS ||--o{ LAB_RESULTS : "yields"
    LAB_TEST_CATALOG ||--o{ LAB_ORDERS : "defines"
    RADIOLOGY_ORDERS ||--o{ RADIOLOGY_RESULTS : "yields"
    ADMISSIONS ||--o{ MEDICATION_ADMINISTRATIONS : "administers"
    PRESCRIPTIONS ||--o{ MEDICATION_ADMINISTRATIONS : "schedules"

    %% =====
    %% SPECIALIZED UNITS (ICU & DIALYSIS)
    %% =====
    ADMISSIONS ||--o{ ICU_HOURLY_CHARTS : "tracks_hourly"
    ENCOUNTERS ||--o{ DIALYSIS_SESSIONS : "conducts"
    MACHINES ||--o{ DIALYSIS_SESSIONS : "operates_on"
    MACHINES ||--o{ MACHINE_LOGS : "logs_service"

    %% =====
    %% STOCK & PHARMACY
    %% =====
    STOCK_LOCATIONS ||--o{ STOCK_MOVEMENTS : "stores"
    STOCK_LOCATIONS ||--o{ STOCK_TRANSFERS : "transfers_from_or_to"
    STOCK_LOCATIONS ||--o{ STOCK_USAGE : "consumes_from"
    STOCK_ITEMS ||--o{ STOCK_MOVEMENTS : "adjusts"
    STOCK_ITEMS ||--o{ STOCK_TRANSFERS : "transferred"
    STOCK_ITEMS ||--o{ STOCK_USAGE : "used"
    STOCK_ITEMS ||--o{ PRESCRIPTIONS : "fulfills"

    %% =====
    %% BILLING & INVOICING
```

```

%% =====
ENCOUNTERS ||--o{ INVOICES : "provisions"
INVOICES ||--o{ INVOICE_LINE_ITEMS : "contains"
INVOICES ||--o{ INVOICE_PAYMENTS : "settled_by"
MORTUARY_RECORDS ||--o{ INVOICES : "billed_via"

%% =====
%% INSURANCE, SHA & HIE INTEROPERABILITY
%% =====
INSURANCE_PROVIDERS ||--o{ INSURANCE_BENEFIT_PLANS : "offers"
INSURANCE_PROVIDERS ||--o{ CONTRACTED_PRICES : "negotiates"
INSURANCE_BENEFIT_PLANS ||--o{ INSURANCE_BENEFIT_CATEGORIES : "defines_benefits"
ENCOUNTERS ||--o{ SHA_CLAIMS : "generates_claim"
SHA_CLAIMS ||--o{ SHA_CLAIM_ITEMS : "itemizes"
ENCOUNTERS ||--o{ DHA_OUTBOUND_QUEUE : "queues_sync"
ENCOUNTERS ||--o{ EPISODE_OF_CARE : "grouped_by"

%% =====
%% AUTH, ROLES & ROOMS
%% =====
PROFILES ||--o{ USER_ROLES : "assigned"
ROOMS ||--o{ ENCOUNTER_ROOM_VISITS : "visited"
ROOMS ||--o{ USER_ROOM_ACCESS : "grants_access"
ROOMS ||--o{ STOCK_LOCATIONS : "houses_inventory"

%% =====
%% ENTITY DEFINITIONS & ATTRIBUTES
%% =====
PATIENTS {
    uuid id PK
    string file_number UK
    string first_name
    string family_name
    string national_id
    date date_of_birth
    boolean dob_known
    string sex
    string phone
    string county
    jsonb next_of_kin
    boolean is_deceased
    timestamp created_at
}

ENCOUNTERS {
    uuid id PK
    uuid patient_id FK
    string encounter_type
    string payment_mode
    uuid insurance_provider_id FK
    string insurance_policy_number
    string status
    boolean is_emergency
    jsonb vitals
    uuid current_room_id FK
    timestamp created_at
    timestamp updated_at
}

ADMISSIONS {
    uuid id PK
    uuid encounter_id FK
    uuid patient_id FK
    uuid ward_id FK
    uuid bed_id FK
    string status
    timestamp admitted_at
    timestamp discharged_at
    text discharge_summary
    string admitting_doctor
}

WARDS {

```



```

        uuid id FK
        string name
        string ward_type
        numeric daily_rate
        integer capacity
        boolean is_active
    }

    BEDS {
        uuid id FK
        uuid ward_id FK
        string bed_number
        string status
    }

    CLINICAL_NOTES {
        uuid id FK
        uuid encounter_id FK
        uuid admission_id FK
        string note_type
        text content
        uuid authored_by FK
        timestamp authored_at
    }

    ENCOUNTER_DIAGNOSES {
        uuid id FK
        uuid encounter_id FK
        string icd11_code
        string icd11_title
        string icd11_uri
        string diagnosis_type
        integer sequence
    }

    LAB_ORDERS {
        uuid id FK
        uuid encounter_id FK
        uuid patient_id FK
        uuid catalog_id FK
        string priority
        string status
        uuid ordered_by FK
        timestamp ordered_at
    }

    LAB_RESULTS {
        uuid id FK
        uuid order_id FK
        jsonb result_data
        string qualitative_result
        string interpretation
        string status
        uuid verified_by FK
        timestamp verified_at
    }

    RADIOLOGY_ORDERS {
        uuid id FK
        uuid encounter_id FK
        uuid patient_id FK
        string modality
        string anatomical_site
        string priority
        string status
    }

    RADIOLOGY_RESULTS {
        uuid id FK
        uuid order_id FK
        text findings
        text impression
        string radiologist
    }

```

```

        jsonb image_paths
        timestamp reported_at
    }

    PRESCRIPTIONS {
        uuid id PK
        uuid encounter_id FK
        uuid stock_item_id FK
        string drug_name
        string dosage
        string frequency
        string duration
        numeric quantity
        string status
        uuid dispensed_by FK
        timestamp dispensed_at
    }

    MEDICATION_ADMINISTRATIONS {
        uuid id PK
        uuid admission_id FK
        uuid prescription_id FK
        string drug_name
        string dose_given
        string route
        uuid administered_by FK
        timestamp administered_at
    }

    ICU_HOURLY_CHARTS {
        uuid id PK
        uuid admission_id FK
        integer hour_number
        numeric arterial_bp_systolic
        numeric arterial_bp_diastolic
        numeric cvp
        integer gcs_total
        integer rass_score
        numeric peep
        numeric fio2_percent
        numeric urine_output_ml
        jsonb inotropes
        uuid logged_by FK
        timestamp recorded_at
    }

    DIALYSIS_SESSIONS {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        uuid machine_id FK
        numeric pre_weight_kg
        numeric post_weight_kg
        numeric uf_goal_ml
        numeric uf_achieved_ml
        numeric blood_flow_rate
        numeric dialysate_flow_rate
        string dialyzer_model
        string access_type
        jsonb complications
        string status
        timestamp session_start
        timestamp session_end
    }

    MORTUARY_RECORDS {
        uuid id PK
        string mortuary_reference UK
        string body_source
        string deceased_name
        string gender
        date date_of_death
        string storage_slot
    }

```

```

    string brought_in_by_name
    string police_ob_number
    uuid released_to_national_id
    timestamp admitted_at
    timestamp released_at
}

STOCK_ITEMS {
    uuid id PK
    string name
    string category
    string unit
    numeric current_quantity
    numeric reorder_level
    numeric unit_price
    numeric cash_price
    numeric insurance_price
}

STOCK_LOCATIONS {
    uuid id PK
    string name
    string code UK
    uuid room_id FK
    boolean is_active
}

STOCK_MOVEMENTS {
    uuid id PK
    uuid item_id FK
    numeric change
    string reason
    string notes
    uuid created_by FK
    timestamp created_at
}

STOCK_TRANSFERS {
    uuid id PK
    uuid item_id FK
    uuid from_location_id FK
    uuid to_location_id FK
    numeric quantity
    string status
    uuid initiated_by FK
}

STOCK_USAGE {
    uuid id PK
    uuid encounter_id FK
    uuid item_id FK
    uuid location_id FK
    numeric used_quantity
    string usage_reason
    timestamp used_at
}

INVOICES {
    uuid id PK
    string invoice_number UK
    uuid encounter_id FK
    uuid patient_id FK
    numeric subtotal
    numeric discount
    numeric waiver_amount
    numeric amount_paid
    numeric balance
    string status
    timestamp created_at
}

INVOICE_LINE_ITEMS {
    uuid id PK

```

```

        uuid invoice_id FK
        uuid encounter_id FK
        string item_type
        uuid source_id
        string description
        numeric quantity
        numeric unit_price
        numeric amount
    }

    INVOICE_PAYMENTS {
        uuid id PK
        uuid invoice_id FK
        numeric amount
        string method
        string reference
        uuid received_by FK
        timestamp paid_at
    }

    INSURANCE_PROVIDERS {
        uuid id PK
        string name
        string code UK
        numeric coverage_percentage
        boolean is_active
    }

    INSURANCE_BENEFIT_PLANS {
        uuid id PK
        uuid provider_id FK
        string plan_name
        string plan_code
        numeric copay_amount
        integer visit_limit
    }

    SHA_CLAIMS {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        string claim_type
        string fund_type
        string preauth_code
        numeric claim_amount
        string status
        jsonb fhir_bundle
        timestamp submitted_at
    }

    SHA_CLAIM_ITEMS {
        uuid id PK
        uuid claim_id FK
        string tariff_code
        string item_description
        numeric unit_price
        numeric claimed_amount
    }

    DHA_OUTBOUND_QUEUE {
        uuid id PK
        uuid encounter_id FK
        uuid patient_id FK
        string queue_type
        string insurer_type
        jsonb payload
        string status
        integer attempts
        timestamp created_at
    }

    PROFILES {
        uuid id PK

```

```

    string full_name
    string council_type
    string council_registration_number
    string phone
    string national_id
    boolean is_approved
}

USER_ROLES {
    uuid id PK
    uuid user_id FK
    string role
}

ROLE_PERMISSIONS {
    string role PK
    string permission PK
}

ROOMS {
    uuid id PK
    string name
    string code
    string kind
    boolean is_active
}

MACHINES {
    uuid id PK
    string name
    string model
    string serial_number
    string status
    string location
}

MACHINE_LOGS {
    uuid id PK
    uuid machine_id FK
    string log_type
    date log_date
    string description
    numeric cost
}

APP_SETTINGS {
    string id PK
    string facility_name
    string facility_kmhfl_code
    string facility_sha_id
    integer facility_level
    string county
}

AUDIT_LOG {
    uuid id PK
    string table_name
    uuid record_id
    string action
    jsonb old_data
    jsonb new_data
    uuid changed_by FK
    timestamp changed_at
}

```

Section 2 — Patient Journey Flows (Mermaid Flowcharts)

Flow 1: Cash Outpatient Journey

```

flowchart TD
    A[Patient Arrives at Facility] --> B[Reception: Patient Lookup / Registration]
    B --> C[Consent Acquisition: OTP SMS via Africa's Talking]
    C --> D[Select Payment Mode: Cash]
    D --> E[Route Encounter to Triage Room]
    E --> F[Triage: Vital Signs, BMI, Acuity Scoring]
    F --> G[Route to Doctor Consultation Room]
    G --> H[Doctor Consultation: Clinical Notes & ICD-11 Coding]

    H --> I{Diagnostic Workup Required?}
    I -- Yes: Lab Tests --> J[Lab Order Generated & Billed]
    J --> K[Cashier: Pay Lab Invoice Items]
    K --> L[Lab: Specimen Collection & Result Entry]
    L --> M[Lab Tech Verifies: SMS Sent to Patient]
    M --> H

    I -- Yes: Radiology --> N[Radiology Order Generated & Billed]
    N --> O[Cashier: Pay Imaging Invoice Items]
    O --> P[Radiology: Scan & Radiologist Report]
    P --> Q[Report Signed: SMS Sent to Patient]
    Q --> H

    I -- No --> R[Doctor Issues Prescription]
    R --> S[Doctor Electronically Signs Encounter - Lock Records]
    S --> T[Encounter Tagged for MOH 705A/B Demographics]
    T --> U[Cashier: Settle Final Invoice Balance]
    U --> V[Receipt Issued & SMS Payment Confirmation]
    V --> W[Pharmacy: Dispense Drugs from Pharmacy Store]
    W --> X[Patient Discharged / Departure]
  
```

Flow 2: Insurance Inpatient Journey

```

flowchart TD
    A[Patient Arrives with Private / Corporate Insurance] --> B[Reception: Register & Link Underwriter Policy]
    B --> C[Consent Verification: OTP SMS]
    C --> D[Route to Insurance Desk]
    D --> E[Insurance Agent: Verify Eligibility & Pre-Authorization]
    E --> F[Contracted Tariffs Applied to Encounter]
    F --> G[Triage: Baseline Vitals Capture]
    G --> H[Doctor Consultation: Examination & Diagnosis]
    H --> I[Doctor Orders Inpatient Ward Admission]

    I --> J[Assign Inpatient Ward & Specific Bed]
    J --> K[Bed Status Set to Occupied]
    K --> L[Ward Nursing Care: Medication Administrations MAR]
    L --> M[Nightly pg_cron Daemon: Accrue Daily Bed Charges]

    M --> N{Patient Clinically Ready for Discharge?}
    N -- No --> L
    N -- Yes --> O[Doctor Completes Mandatory Discharge Summary]
    O --> P[Doctor Signs Admission File]
    P --> Q[Pharmacy: Reconcile & Dispense Discharge Medications]
    Q --> R[Billing: Compile Itemized Interim Inpatient Bill]
    R --> S[Insurance Desk: Submit Electronic Claim / Guarantee]
    S --> T[Underwriter Approval Received / Copay Paid]
    T --> U[Discharge Clearance Granted & Bed Set to Available]
    U --> V[Patient Discharged Home]
  
```

Flow 3: SHA SHIF Journey (Including Biometric Step)

```

flowchart TD
    A[Patient Presents for SHA Care] --> B[Reception: Search National ID / SHA Number]
    B --> C[Consent Capture: OTP SMS Verification]
    C --> D[Select Payment Mode: sha_shif / phf / eccif]
    D --> E[Insurance Desk: Query SHA Member Registry API ⌚]
    E --> F[Biometric Smart-Card / Fingerprint Verification ⌚]
    F --> G[SHA Fund Type Classified: PHF / SHIF / ECCIF]
    G --> H[Triage: Vitals Scoring & MoH Demographic Tagging]
    H --> I[Doctor Consultation: WHO ICD-11 Diagnosis Search]

    I --> J[Doctor Prescribes Drugs & Diagnostic Workups]
    J --> K[SHA Standard Benefit Packages & Tariffs Linked]
    K --> L[Doctor Electronically Signs Encounter]

    L --> M[Database Trigger: auto_generate_sha_claim]
    M --> N[RPC: build_fhir_claim Compiles FHIR Claim Resource]
    N --> O[Edge Function: claims-dispatcher Routes Payload]
    O --> P[Queued in dha_outbound_queue with mediator_id]
    P --> Q[DHA AfyaLink HIE / SHA Claims Gateway ⌚]
    Q --> R[Pharmacy Dispenses SHIF-Covered Medications]
    R --> S[Patient Discharged]

```

Flow 4: ICU Admission Journey

```

flowchart TD
    A[Critically Ill Patient in Casualty / Operating Theatre] --> B[Doctor Orders Emergency ICU Admission]
    B --> C[Assign Bed in Intensive Care Unit Ward]
    C --> D[RPC: charge_icu_admission_fee Auto-Posts Admission Tariff]
    D --> E[Bed Occupancy Set to Occupied in ICU Ward]
    E --> F[ICU Nursing Station: Continuous Patient Monitoring]

    F --> G[Hourly Flow-Sheet Entry: Arterial BP, CVP, ABG]
    G --> H[Ventilator Parameters: Mode, PEEP, FiO2, VT]
    H --> I[Neurological: GCS /15 & RASS Sedation Scoring]
    I --> J[Bedside Drug Dispensing: Deduct from ICU Store]

    J --> K[Nightly pg_cron: Accrue Daily ICU Charges]
    K --> L{Patient Hemodynamically Stable?}
    L -- No --> F
    L -- Yes: Transfer to General Ward --> M[Trigger: audit_ward_transfers Logs Transfer]
    M --> N[Bed Status in ICU Reset to Cleaning/Available]
    N --> O[Patient Continues Care in General Ward]

```

Flow 5: Dialysis Session Journey

```

flowchart TD
    A[Renal Patient Arrives for Scheduled Hemodialysis] --> B[Reception / Dialysis Unit Check-In]
    B --> C[Pre-Dialysis Assessment: Pre-Weight, Dry Weight, Access Site]
    C --> D[Dialysis Nurse Assigns Dialysis Machine Number]
    D --> E[Dialyzer Prescription: UF Goal mL, BFR, DFR, Treatment Time]
    E --> F[Select Session Consumables: Bloodlines, Needles, Acid/Bicarb]

    F --> G[Initiate Hemodialysis Run]
    G --> H[Hourly Monitoring: Intra-dialytic BP, Pulse, TMP, Net UF]
    H --> I[Session Concluded: Post-Weight & Complications Logged]

    I --> J[RPC: process_dialysis_session_billing Executed]
    J --> K[Deduct Consumables from Dialysis Store in stock_usage]
    K --> L[Post Itemized Dialysis Session Charge to Invoice]
    L --> M[Payment Channel]
    M -- SHA SHIF Renal Package --> N[Queue Renal Claim in dha_outbound_queue]
    M -- Private Insurance --> O[Insurance Clearance & Voucher Settlement]

```

```

M -- Cash --> P[Cashier: Pay Dialysis Invoice]
N --> Q[Patient Discharged with Follow-Up Schedule]
O --> Q
P --> Q

```

Flow 6: Mortuary (External Body) Journey

```

flowchart TD
    A[Deceased Brought in Dead: Police Case / Home Death] --> B[Mortuary Department Intake]
    B --> C[Mortician Captures Intake Metadata & Police OB Number]
    C --> D[Trigger: assign_mortuary_reference Generates Reference Number]
    D --> E[Assign Cold Storage Refrigeration Slot]
    E --> F[RPC: create_external_mortuary_invoice Provisions Master Bill]

    F --> G[Preservation / Embalming / Autopsy Pathology Logged]
    G --> H[Nightly pg_cron: accrue_daily_mortuary_charges Posts Daily Fees]

    H --> I[Next-of-Kin Arrives with Burial Permit & National ID]
    I --> J[Cashier: Reconcile & Settle Mortuary Invoice Balance]
    J --> K[Invoice Paid in Full or Authorized Waiver?]
    K -- No --> L[System Release Gate: Body Release Locked]
    L --> J
    K -- Yes --> M[Mortician Documents Handover & Vehicle Registration]
    M --> N[Mortuary Record Status Set to Released]
    N --> O[Cold Storage Slot Reset to Available]

```

Section 3 — System Architecture Diagram (Mermaid)

```

flowchart TB
    subgraph CLIENT_LAYER ["Frontend Client Tier (Browser)"]
        UI["React 18 Single-Page Application"]
        ROUTER["TanStack Router (Type-Safe Routing)"]
        QUERY["TanStack Query (Optimistic State & Cache)"]
        SHADCN["Tailwind CSS + Shadcn UI Component Suite"]
        UI --- ROUTER --- QUERY --- SHADCN
    end

    subgraph HOSTING_EDGE ["Hosting & CDN Tier"]
        VERCEL["Vercel Edge Network / Lovable Cloud"]
        BUN["Bun Runtime & Build Tooling"]
        VERCEL --- BUN
    end

    subgraph SUPABASE_BACKEND ["Supabase Cloud Platform"]
        AUTH["Supabase Auth (JWT, Role Mapping)"]
        REALTIME["Supabase Realtime (WebSocket Queues & Beds)"]

        subgraph POSTGRES_DB ["PostgreSQL 15 Database Engine"]
            TABLES["35+ Relational Tables"]
            RLS["Row-Level Security (RLS) Policies"]
            TRIGGERS["PL/pgSQL Business Triggers & Rules"]
            CRON["pg_cron Background Scheduling Daemons"]
            AUDIT["Append-Only Audit Logging & Archive"]
            TABLES --- RLS --- TRIGGERS --- CRON --- AUDIT
        end

        subgraph EDGE_FUNCTIONS ["Supabase Deno Edge Functions"]
            SMS_FN["send-sms (Africa's Talking Gateway)"]
            ICD_FN["icd11-search (WHO Cloud Proxy)"]
            DISPATCH_FN["claims-dispatcher (HIE & Claims Router)"]
            FHIR_PAT["fhir-patient (R4 Resource Builder)"]
            FHIR_ENC["fhir-encounter (R4 Resource Builder)"]
            FHIR_COND["fhir-condition (R4 Resource Builder)"]
        end
    end

```



```

    FHIR_BND["fhir-bundle (R4 Collection Builder)"]
end
end

subgraph EXTERNAL_INTEGRATIONS ["External Gateways & Regulators"]
  AT_SMS["Africa's Talking SMS API (Production ✓)"]
  WHO_API["WHO ICD-11 MMS Cloud API (Production ✓)"]
  DHA_HIE["Kenya DHA AfyaLink HIE (Pending Credentials ⚠)"]
  SHA_GATEWAY["SHA SHIF / PHF / ECCIF Claims (Pending ⚠)"]
  MPESA["Safaricom M-Pesa Daraja Gateway (Planned 📅)"]
end

UI -->|HTTPS / REST & GraphQL| POSTGRES_DB
UI -->|WSS Realtime Feeds| REALTIME
UI -->|JWT Auth Handshake| AUTH
UI -->|Edge Invocations| EDGE_FUNCTIONS

SMS_FN -->|HTTPS REST| AT_SMS
ICD_FN -->|OAuth2 / REST| WHO_API
DISPATCH_FN -->|FHIR R4 REST| DHA_HIE
DISPATCH_FN -->|FHIR Claim REST| SHA_GATEWAY
POSTGRES_DB -->|Nightly Cron Execution| CRON

```

Section 4 — Table Descriptions & Security Schema

1. **patients**

- **Purpose:** Master Patient Index (MPI) repository storing demographic identity, contact details, next-of-kin, and vital status.
- **Key Columns:** `id` (UUID PK), `file_number` (Unique Text, e.g., `P000142`), `first_name`, `middle_name`, `family_name`, `patient_name`, `national_id`, `date_of_birth`, `dob_known` (Boolean), `estimated_age`, `sex` (`male` / `female`), `phone`, `email`, `county`, `next_of_kin` (JSONB), `is_deceased` (Boolean).
- **Relationships:** One-to-many with `encounters`, `admissions`, `invoices`, `appointments`, `patient_consents`, `consent_otps`.
- **RLS Policy Summary:**
 - `SELECT`: Permitted for all authenticated and approved hospital staff (`is_approved(auth.uid())`).
 - `INSERT` / `UPDATE`: Restricted to users possessing `register_patient`, `records_create`, or `admin` permissions.
 - `DELETE`: Explicitly prohibited (`USING (false)`).

2. **encounters** (Aliased as **patient_registrations**)

- **Purpose:** Central operational table tracking patient episodes spanning OPD registration, triage vitals, consultation, and discharge.
- **Key Columns:** `id` (UUID PK), `patient_id` (FK `patients.id`), `encounter_type` (`outpatient`, `inpatient`, `emergency`), `payment_mode` (`cash`, `insurance`, `sha_shif`, `exemption`), `insurance_provider_id` (FK `insurance_providers.id`), `insurance_policy_number`, `status` (`waiting`, `in_progress`, `done`, `signed`, `cancelled`), `is_emergency` (Boolean), `vitals` (JSONB), `current_room_id` (FK `rooms.id`), `subtotal`, `amount_paid`, `created_at`, `updated_at`.
- **Relationships:** Belongs to `patients`; Has many `clinical_notes`, `encounter_diagnoses`, `lab_orders`, `radiology_orders`, `prescriptions`, `invoices`, `admissions`, `sha_claims`.
- **RLS Policy Summary:**
 - `SELECT`: Permitted for approved staff.
 - `INSERT`: Permitted for reception, triage, and clinical staff.
 - `UPDATE`: Permitted for attending clinicians and room staff; mutations strictly blocked once `status = 'signed'` via `enforce_encounter_lock` trigger.

3. **admissions**

- **Purpose:** Tracks inpatient hospitalizations from bed allocation through clinical ward care to discharge summary.
- **Key Columns:** `id` (UUID PK), `encounter_id` (FK `encounters.id`), `patient_id` (FK `patients.id`), `ward_id` (FK `wards.id`), `bed_id` (FK `beds.id`), `admitted_at`, `discharged_at`, `admitting_doctor`, `admission_reason`, `admission_type`, `status` (`admitted`, `discharged`, `transferred`), `discharge_summary` (Text).
- **Relationships:** Belongs to `encounters`, `patients`, `wards`, `beds`; Has many `icu_hourly_charts`, `medication_administrations`.
- **RLS Policy Summary:**
 - `SELECT`: Approved staff.
 - `INSERT` / `UPDATE`: Restricted to clinical staff (`doctor`, `clinical_officer`, `nurse`, `admin`). Discharge requires non-null `discharge_summary`.

4. `wards` & `beds`

- **Purpose:** Physical inpatient infrastructure management, tracking bed capacities, occupancy states, and daily ward rates.
- **Key Columns:**
 - `wards`: `id` (UUID PK), `name`, `ward_type` (`general`, `maternity`, `pediatric`, `surgical`, `icu`, `hdu`), `daily_rate` (Numeric), `capacity` (Integer), `is_active` (Boolean).
 - `beds`: `id` (UUID PK), `ward_id` (FK `wards.id`), `bed_number` (Text), `status` (`available`, `occupied`, `maintenance`, `cleaning`).
- **Relationships:** `wards` has many `beds` and `admissions`. `beds` belongs to `wards`.
- **RLS Policy Summary:**
 - `SELECT`: Approved staff.
 - `INSERT` / `UPDATE` / `DELETE`: Restricted to `admin` role.

5. `clinical_notes`

- **Purpose:** Structured clinical documentation entered by attending doctors and clinical officers.
- **Key Columns:** `id` (UUID PK), `encounter_id` (FK `encounters.id`), `admission_id` (FK `admissions.id`), `note_type` (`chief_complaint`, `hpi`, `ros`, `examination`, `plan`, `progress_note`), `content` (Text), `authored_by` (FK `auth.users`), `authored_at`.
- **Relationships:** Belongs to `encounters`, `admissions`.
- **RLS Policy Summary:**
 - `SELECT`: Approved medical and nursing staff.
 - `INSERT`: Attending clinicians. Updates locked once parent encounter is signed.

6. `encounter_diagnoses`

- **Purpose:** International diagnostic disease codes linked to encounters, mapped to WHO ICD-11 MMS terminology.
- **Key Columns:** `id` (UUID PK), `encounter_id` (FK `encounters.id`), `icd11_code` (Varchar), `icd11_title` (Text), `icd11_uri` (Text), `diagnosis_type` (`primary`, `secondary`, `working`, `differential`), `sequence` (Integer), `notes` (Text).
- **Relationships:** Belongs to `encounters`.
- **RLS Policy Summary:**
 - `SELECT`: Approved clinical staff.
 - `INSERT` / `UPDATE`: Attending clinicians. Protected by `clean_and_validate_diagnosis_insert()` and locked upon encounter signing.

7. `lab_orders` & `lab_results`

- **Purpose:** Laboratory diagnostic requisitions, analyzer result records, reference range validation, and technologist verification.
- **Key Columns:**
 - `lab_orders`: `id` (UUID PK), `encounter_id`, `patient_id`, `catalog_id` (FK `lab_test_catalog.id`), `priority` (`routine`, `urgent`), `status` (`ordered`, `in_progress`, `completed`, `cancelled`), `ordered_by`.

- `lab_results` : `id` (UUID PK), `order_id` (FK `lab_orders.id`), `result_data` (JSONB), `qualitative_result` (Text), `interpretation` (Text), `status` (`draft` , `verified`), `verified_by` (FK `auth.users`), `verified_at` .
 - **Relationships:** `lab_orders` belongs to `encounters` ; has one `lab_results` .
 - **RLS Policy Summary:**
 - `lab_orders` : Clinicians insert; Lab techs update status.
 - `lab_results` : `lab_tech` role has exclusive insert/update rights. Verified results are immutable.
-

8. `radiology_orders` & `radiology_results`

- **Purpose:** Medical imaging workflow management from clinician requisition to radiologist diagnostic report.
 - **Key Columns:**
 - `radiology_orders` : `id` (UUID PK), `encounter_id` , `patient_id` , `modality` (`xray` , `ultrasound` , `ct` , `mri` , `ecg`), `anatomical_site` , `priority` , `status` , `clinical_indication` .
 - `radiology_results` : `id` (UUID PK), `order_id` (FK `radiology_orders.id`), `findings` (Text), `impression` (Text), `radiologist` (Text), `image_paths` (JSONB), `reported_at` .
 - **Relationships:** `radiology_orders` belongs to `encounters` ; has one `radiology_results` .
 - **RLS Policy Summary:**
 - `radiology_orders` : Clinicians insert; Radiologists update.
 - `radiology_results` : Restricted to `radiologist` and `doctor` roles.
-

9. `prescriptions` & `medication_administrations`

- **Purpose:** Clinical pharmaceutical orders, pharmacy dispensing validations, and inpatient bedside nursing administration (MAR).
 - **Key Columns:**
 - `prescriptions` : `id` (UUID PK), `encounter_id` (or `registration_id`), `stock_item_id` (FK `stock_items.id`), `drug_name` , `dosage` , `frequency` , `duration` , `quantity` , `status` (`pending` , `dispensed` , `cancelled`), `dispensed_by` , `dispensed_at` .
 - `medication_administrations` : `id` (UUID PK), `admission_id` , `prescription_id` , `drug_name` , `dose_given` , `route` , `administered_by` , `administered_at` .
 - **Relationships:** `prescriptions` belongs to `encounters` , `stock_items` ; has many `medication_administrations` .
 - **RLS Policy Summary:**
 - `prescriptions` : Clinicians create; `pharmacist` updates to `dispensed` (triggering stock decrement).
 - `medication_administrations` : Restricted to `nurse` and `doctor` roles.
-

10. `icu_hourly_charts`

- **Purpose:** High-frequency critical care monitoring records for ICU inpatients.
 - **Key Columns:** `id` (UUID PK), `admission_id` (FK `admissions.id`), `hour_number` (Integer), `arterial_bp_systolic` , `arterial_bp_diastolic` , `cvp` , `gcs_total` (3–15), `rass_score` (-5 to +4), `peep` , `fio2_percent` , `urine_output_ml` , `inotropes` (JSONB), `logged_by` , `recorded_at` .
 - **Relationships:** Belongs to `admissions` .
 - **RLS Policy Summary:**
 - `SELECT` : Approved ICU clinical staff.
 - `INSERT` / `UPDATE` : Restricted to `nurse` , `doctor` , and `admin` roles in ICU room.
-

11. `dialysis_sessions`

- **Purpose:** Clinical flow-sheet for hemodialysis treatments and automated consumable billing.
- **Key Columns:** `id` (UUID PK), `encounter_id` , `patient_id` , `machine_id` (FK `machines.id`), `pre_weight_kg` , `post_weight_kg` , `uf_goal_ml` , `uf_achieved_ml` , `blood_flow_rate` , `dialysate_flow_rate` , `dialyzer_model` , `access_type` , `complications` (JSONB),

`status`, `session_start`, `session_end`.

- **Relationships:** Belongs to `encounters`, `patients`, `machines`.
- **RLS Policy Summary:**
 - Restricted to `nurse`, `doctor`, and `admin` assigned to the Dialysis Unit.

12. `mortuary_records`

- **Purpose:** Administrative tracking of deceased bodies admitted internally or externally.
- **Key Columns:** `id` (UUID PK), `mortuary_reference` (Unique Text, e.g., `MORT-2026-0012`), `body_source` (`internal`, `external`), `deceased_name`, `gender`, `date_of_death`, `storage_slot`, `brought_in_by_name`, `police_ob_number`, `released_to_national_id`, `admitted_at`, `released_at`.
- **Relationships:** Has one linked `invoices` record.
- **RLS Policy Summary:**
 - Restricted to `mortician`, `accountant`, and `admin` roles. Body release blocked if invoice has an unpaid balance.

13. `stock_items`, `stock_locations`, `stock_movements`, `stock_transfers`, `stock_usage`

- **Purpose:** Real-time multi-location perpetual inventory management across Main Warehouse, Pharmacy, ICU, and Dialysis stores.
- **Key Columns:**
 - `stock_items`: `id` (UUID PK), `name`, `category`, `unit`, `current_quantity`, `reorder_level`, `unit_price`, `cash_price`, `insurance_price`.
 - `stock_locations`: `id` (UUID PK), `name`, `code` (Unique), `room_id` (FK `rooms.id`), `is_active`.
 - `stock_movements`: `id` (UUID PK), `item_id`, `change` (Numeric), `reason` (`delivery`, `dispense`, `transfer_in`, `transfer_out`, `usage`, `adjustment`), `notes`, `created_by`, `created_at`.
 - `stock_transfers`: `id` (UUID PK), `item_id`, `from_location_id`, `to_location_id`, `quantity`, `status`.
 - `stock_usage`: `id` (UUID PK), `encounter_id`, `item_id`, `location_id`, `used_quantity`, `usage_reason`.
- **Relationships:** `stock_items` has many movements, transfers, and usage records.
- **RLS Policy Summary:**
 - Read: Approved staff.
 - Mutations: Governed by `user_stock_location_access` matrix and executed via secure RPCs (`apply_stock_movement` trigger).

14. `invoices`, `invoice_line_items`, `invoice_payments`

- **Purpose:** Master fiscal ledger maintaining real-time itemized billing, payments, waivers, and aging.
- **Key Columns:**
 - `invoices`: `id` (UUID PK), `invoice_number` (Unique Text, e.g., `INV-2026-00891`), `encounter_id`, `patient_id`, `subtotal`, `discount`, `waiver_amount`, `amount_paid`, `balance`, `status` (`draft`, `pending`, `partial`, `paid`, `waived`, `claimed`).
 - `invoice_line_items`: `id` (UUID PK), `invoice_id`, `encounter_id`, `item_type` (`consultation`, `lab`, `radiology`, `prescription`, `bed_day`, `procedure`, `consumable`), `source_id`, `description`, `quantity`, `unit_price`, `amount`.
 - `invoice_payments`: `id` (UUID PK), `invoice_id`, `amount`, `method` (`cash`, `mpesa`, `card`, `insurance`, `bank`), `reference`, `received_by`, `paid_at`.
- **Relationships:** `invoices` has many `invoice_line_items` and `invoice_payments`.
- **RLS Policy Summary:**
 - Select: Approved staff.
 - Payments / Waivers: Restricted to `accountant` and `admin` roles. Line items automatically maintained by database triggers.

15. `sha_claims` & `sha_claim_items`

- **Purpose:** Electronic claim repository for Social Health Authority (SHA) reimbursement packages.

- **Key Columns:**

- `sha_claims`: `id` (UUID PK), `encounter_id`, `patient_id`, `claim_type` (`outpatient`, `inpatient`), `fund_type` (`phf`, `shif`, `eccif`), `preauth_code`, `claim_amount`, `status` (`draft`, `pending`, `submitted`, `approved`, `rejected`), `fhir_bundle` (JSONB), `submitted_at`.
- `sha_claim_items`: `id` (UUID PK), `claim_id` (FK `sha_claims.id`), `tariff_code`, `item_description`, `unit_price`, `claimed_amount`.

- **Relationships:** Belongs to `encounters`, `patients`; has many `sha_claim_items`.

- **RLS Policy Summary:**

- Read: Approved staff.
- Insert / Update: Restricted to `insurance_agent`, `accountant`, and `admin` roles.

16. `dha_outbound_queue`

- **Purpose:** Asynchronous outbound message queue for DHA AfyaLink HIE and SHA FHIR bundles.

- **Key Columns:** `id` (UUID PK), `encounter_id`, `patient_id`, `queue_type` (`fhir_sync`, `sha_claim`, `private_claim`, `cash_receipt`), `insurer_type`, `payload` (JSONB), `status` (`pending`, `processing`, `completed`, `failed`, `skipped`), `attempts` (Integer), `error_message`, `created_at`.

- **Relationships:** Belongs to `encounters`, `patients`.

- **RLS Policy Summary:**

- Select: Approved administrative staff.
- Mutations: Managed exclusively by Edge Functions via `service_role` key.

17. `profiles`, `user_roles`, `role_permissions`, `user_room_access`

- **Purpose:** RBAC security framework enforcing access control and practitioner credential verification.

- **Key Columns:**

- `profiles`: `id` (UUID PK, references `auth.users.id`), `full_name`, `council_type` (`KMPDC`, `COC`, `NCK`, `PPB`, `KMLTTB`), `council_registration_number`, `national_id`, `phone`, `is_approved` (Boolean).
- `user_roles`: `id` (UUID PK), `user_id` (FK `auth.users`), `role` (`app_role` enum).
- `role_permissions`: `role` (`app_role`), `permission` (Text PKs).
- `user_room_access`: `user_id` (FK `auth.users`), `room_id` (FK `rooms.id`).

- **RLS Policy Summary:**

- `profiles`: Users can read/update their own profile; admins can approve accounts.
- `user_roles` & `role_permissions`: Exclusively manageable by `admin` role.

18. `app_settings` & `facility_features`

- **Purpose:** Global institutional settings and dynamic facility-level feature switches.

- **Key Columns:** `id` (Text PK `'global'`), `facility_name`, `facility_kmhfl_code`, `facility_sha_id`, `facility_level` (1–6), `county`, `address`, `phone`, `email`.

- **RLS Policy Summary:**

- Select: Public/authenticated.
- Update: Exclusively restricted to `admin` and `system_admin` roles.

19. `audit_log`, `audit_log_archive`, `audit_archive_runs`

- **Purpose:** Tamper-evident operational audit trail and long-term 20-year compliance archive.

- **Key Columns:**

- `audit_log`: `id` (UUID PK), `table_name`, `record_id`, `action` (`INSERT`, `UPDATE`, `DELETE`, `BREAK_GLASS`), `old_data` (JSONB), `new_data` (JSONB), `changed_by` (FK `auth.users`), `changed_at`.
 - `audit_log_archive`: Mirror schema of `audit_log` with `archived_at` timestamp.
 - `audit_archive_runs`: `id`, `run_at`, `rows_archived`, `status`, `error_message`.
- **RLS Policy Summary:**
 - Append-Only Security: `SELECT` permitted for approved admins; `INSERT`, `UPDATE`, `DELETE` strictly blocked for all user JWTs (`USING (false)`). Mutations occur strictly via `SECURITY DEFINER` database triggers.
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AegisCare HMS — System Dictionary & Glossary

Document Version: 1.0.0

Target Architecture: Kenya MoH Level 1–6 Facilities

Classification: Reference Manual & System Lexicon

Introduction

This glossary provides authoritative definitions, operational context, and architectural mappings for all clinical, business, technical, system, and regulatory terms utilized across the **AegisCare Hospital Management System (HMS)**.

A

Accountant

Category: System / Role

Definition: A system user role assigned to financial officers responsible for handling invoicing, collecting and allocating payments (cash, mobile money, cards), applying manager-approved waivers, issuing credit notes, and executing End-of-Day (EOD) financial reconciliation.

Used in: Billing & Accounting Module, Invoices (`invoices` , `invoice_payments` , `user_roles`)

Accounting

Category: System / Permission

Definition: A granular role permission (`accounting`) that grants authorized staff write access to invoice balance mutations, manual line-item price overrides, and cash collection ledger entries.

Used in: `role_permissions` , Billing & Accounting UI (`/accounting` , `/invoices/$id`)

Admission

Category: Clinical

Definition: The formal clinical and administrative process of admitting an outpatient or emergency patient into an inpatient ward bed (General, Maternity, Pediatric, Surgical, ICU, or HDU) for continuous overnight monitoring, nursing care, and treatment.

Used in: Inpatient Module, Wards Management, Consultation (`admissions` , `beds` , `wards` , `encounters`)

Admission Fee (ICU)

Category: Business / System

Definition: A one-time standard institutional surcharge automatically applied to an inpatient invoice upon admission to an Intensive Care Unit ward, covering specialized sterile isolation and equipment staging.

Used in: Inpatient / ICU Admission Flow, Billing (`charge_icu_admission_fee` RPC, `invoice_line_items`)

Africa's Talking (AT)

Category: Kenya-specific / Technical

Definition: A pan-African telecommunications API platform utilized by AegisCare for automated real-time SMS dispatch, specifically sending patient OTP consent codes, lab test completion alerts, and billing receipts formatted to Kenyan phone numbers (`+254...`).

Used in: Edge Function (`send-sms`) , Patient Registration, Laboratory Result Verification

AfyaLink HIE

Category: Kenya-specific / Technical

Definition: The Kenyan Digital Health Authority (DHA) National Health Information Exchange platform enabling standardized inter-facility clinical data sharing and centralized health registries via FHIR R4 APIs.

Used in: Integration Module, FHIR Synchronization Queue (`dha_outbound_queue` , Edge Function (`fhir-bundle`)

Age Band

Category: Clinical / Kenya-specific

Definition: MoH standardized age stratification groups (Under 5 years, 5–59 years, 60+ years) used for epidemiological disease tracking, clinical triage workflows, and statutory MOH 705A/B reporting.

Used in: Triage, Dashboard Analytics, MOH Reports (`tag_encounter_demographics` , `moh_705_disease_mappings`)

Alert & Notification System

Category: System

Definition: Automated in-app and SMS alerting mechanisms that notify clinical staff of critical laboratory panic values, out-of-range vital signs, or impending drug stock depletion.

Used in: Laboratory (`laboratory.$id.tsx`), Triage, Ward Monitoring, Stock Management

Admin

Category: System / Role

Definition: System administrative user possessing full institutional permissions to manage hospital metadata, staff user profiles, role assignments, department room rosters, ward beds, medical machines, and facility billing tariffs.

Used in: Admin Module (`/admin/*` , `user_roles` , `role_permissions`)

Anuria / Oliguria

Category: Clinical

Definition: Critical indicators of renal failure where urine output is severely suppressed or absent (<100 mL/24h for anuria; <400 mL/24h for oliguria), tracked during ICU hourly monitoring and Hemodialysis fluid balance assessments.

Used in: ICU Hourly Charting, Dialysis Module (`icu_hourly_charts` , `dialysis_sessions`)

Appointment

Category: Clinical / System

Definition: A scheduled forward booking for a patient with a specific clinician, specialized clinic (e.g., MCH, SOPC, GOPC, Dental), or diagnostic service room, which can be seamlessly converted into an active clinical encounter upon patient arrival.

Used in: Appointments Module (`appointments` , `create_encounter_from_appointment` RPC)

App Settings

Category: Technical / System

Definition: The centralized global configuration repository stored in Supabase storing facility identity (KMHFL code, SHA ID, facility level, contact info) and operational switches.

Used in: Database Table `app_settings` , System Configuration (`/admin/settings`)

Audit Log

Category: Technical / System

Definition: An immutable, append-only security log tracking every database insert, update, delete, login, and elevated data access event with cryptographic user ID, table name, record identifier, timestamps, and previous/new state JSON payloads.

Used in: Security Module, Compliance Audits (`audit_log` , `audit_trigger_fn`)

Audit Log Archive

Category: Technical / System

Definition: A secondary long-term storage table where audit log entries older than two years are transferred via automated nightly cron jobs to comply with the 20-year Kenya Data Protection Act statutory retention period without degrading primary transactional query performance.

Used in: `audit_log_archive` , `audit_archive_runs` , `archive_old_audit_logs()` RPC

Auto-Generated SHA Claim

Category: Business / System

Definition: An automated database-driven procedure triggered upon encounter signing or invoice closure for SHA-registered patients that

compiles encounter diagnoses, billable items, and clinical notes into a standardized SHA claim and FHIR claim bundle.

Used in: Insurance Desk, Claims Dispatcher (`sha_claims` , `auto_generate_sha_claim()` trigger)

B

Bed

Category: System / Clinical

Definition: A uniquely identified physical sleeping/care unit within a hospital ward or room, maintaining real-time occupancy state (`available` , `occupied` , `maintenance` , `cleaning`) and linked to admitted patients.

Used in: Inpatient Module, Wards Management (`beds` , `admissions`)

Bed Day

Category: Business / System

Definition: A standard hospital billing unit representing 24 hours of inpatient ward occupancy, charged automatically on a daily schedule based on the assigned ward rate.

Used in: Inpatient Billing, `accrue_daily_bed_charges()` RPC, `invoice_line_items`

Biometric Verification

Category: Kenya-specific / Business

Definition: The planned integration with DHA/SHA biometric smart card readers and fingerprint scanners to verify patient identity and SHA benefit eligibility at the registration desk prior to service delivery.

Used in: Patient Registration, Insurance Desk (`patients.biometric_verified` , ⌚ Phase 3 DHA Integration)

Break Glass Access

Category: System / Security

Definition: An emergency security protocol allowing clinicians to bypass standard consent or privacy barriers to immediately view a patient's electronic health record during life-threatening emergencies. Every break-glass activation mandates a documented clinical justification and immediately creates an immutable `BREAK_GLASS` high-priority audit record.

Used in: Consultation, Patient History (`log_break_glass_access` RPC, `audit_log`)

Bun

Category: Technical

Definition: An ultra-fast, modern JavaScript/TypeScript runtime, bundler, and package manager used for local development, fast build tooling, and test execution within the AegisCare software engineering workflow.

Used in: System Infrastructure, `bun.lock` , Dev Server Runtime

C

Cash Payment Mode

Category: Business

Definition: The direct self-pay workflow where a patient pays out-of-pocket for consultations, laboratory diagnostics, radiology, procedures, and dispensed pharmaceuticals before or immediately after service rendering.

Used in: Patient Registration, Billing & Accounting (`encounters.payment_mode = 'cash'` , `invoice_payments`)

Claims Dispatcher

Category: Technical / Business

Definition: A serverless Supabase Edge Function that routes completed medical insurance claims and FHIR bundles to their respective external clearinghouses (SHA HIE, private insurance gateways, or local archival queues) based on the encounter payer configuration.

Used in: Supabase Edge Function `claims-dispatcher` , Insurance Module, `dha_outbound_queue`

Clearance (Insurance Clearance)

Category: Business / System

Definition: The administrative authorization process executed by the Insurance Desk prior to triage or consultation, validating policy active status, pre-authorization approval, member validity, and copay requirements.

Used in: Insurance Desk (`/admin/insurance` , `encounters.insurance_status`)

Clinical Notes

Category: Clinical

Definition: Structured clinical documentation entered by healthcare providers encompassing Chief Complaints, History of Presenting Illness (HPI), Review of Systems (ROS), Physical Examination, Clinical Impressions, and Treatment Plans.

Used in: Consultation / Clinical Module (`clinical_notes` , `rooms.$id.tsx`)

Clinical Officer

Category: System / Role

Definition: A licensed healthcare practitioner registered under the Clinical Officers Council (COC) of Kenya authorized to examine outpatients and inpatients, formulate ICD-11 diagnoses, prescribe pharmaceuticals, request diagnostic workups, perform medical procedures, and sign clinical encounters.

Used in: Consultation, Inpatient, Triage (`user_roles` , `clinical_notes` , `encounter_diagnoses`)

Consultation

Category: Clinical

Definition: The core clinical encounter stage where a clinician (Doctor, Clinical Officer, or Specialist) reviews patient history, examines the patient, documents findings, assigns diagnoses, and initiates diagnostic orders, medications, or ward admissions.

Used in: Consultation Module (`rooms.$id.tsx` , `clinical_notes`)

Contracted Price / Tariff

Category: Business / System

Definition: A pre-negotiated fee schedule agreed upon between the healthcare facility and specific insurance underwriters (e.g., SHA, Jubilee, Britam, CIC), overriding baseline cash prices for consultations, bed stays, procedures, and tests.

Used in: Billing Engine, Insurance Pricing (`insurance_contracted_prices` , `get_contracted_price()` RPC)

Copay

Category: Business

Definition: A fixed out-of-pocket sum or percentage required by an insurer to be paid directly by the insured patient before accessing covered outpatient consultations or specialized procedures.

Used in: Insurance Desk, Cash Office (`encounters.copay_amount` , `invoices`)

Cron Job (pg_cron)

Category: Technical

Definition: Background scheduled jobs executed automatically within the Supabase Postgres database engine at precise intervals (e.g., nightly at 02:00 EAT) to accrue daily bed, ICU, and mortuary charges, and archive stale audit trails.

Used in: Database Infrastructure (`pg_cron` , `supabase/migrations/*_cron.sql`)

D

Daily Charge (Bed / ICU / Mortuary)

Category: Business / System

Definition: Automated fee accrual calculated daily for active inpatient beds, intensive care ward stays, and mortuary cold-storage slots until formal clinical discharge or physical release.

Used in: Inpatient, ICU, Mortuary Billing (`accrue_daily_bed_charges()` , `accrue_daily_mortuary_charges()`)

Delivery (Stock Delivery)

Category: System / Business

Definition: The formal receiving and warehouse logging of pharmaceutical drugs, clinical supplies, and general consumables from external suppliers, increasing current inventory quantities and recording purchase batch numbers and expiry dates.

Used in: Stock Module, Pharmacy (`stock_deliveries` , `stock_movements` , `delivery_to_stock()` RPC)

Deliveries

Category: System / Permission

Definition: A specialized permission (`deliveries`) allowing storekeepers and pharmacy personnel to accept, inspect, and book commercial supplier deliveries into the main warehouse stock ledger.

Used in: `role_permissions` , Stock Management (`/deliveries`)

Dental Officer

Category: System / Role

Definition: A licensed dental surgeon or dental clinical officer authorized to examine oral cavity pathology, perform dental extractions and restorative procedures, prescribe medications, and sign specialized dental encounters.

Used in: Dental Clinic Room (`user_roles` , `encounters` , `prescriptions`)

DHA (Digital Health Authority)

Category: Kenya-specific

Definition: The state corporation established under the Kenya Digital Health Act, 2023, mandated to establish, maintain, and regulate national digital health infrastructure, health data standards, health information exchanges (HIE), and system certifications.

Used in: Regulatory Compliance, DHA Compliance Suite, FHIR Interfaces

DHA Outbound Queue

Category: Technical / System

Definition: An asynchronous database transactional message queue storing outgoing FHIR bundles and insurance claims destined for DHA AfyaLink HIE and SHA servers with automatic retry, backoff, and logging semantics.

Used in: System Core, `dha_outbound_queue` , Edge Function `claims-dispatcher`

Dialysis Nurse

Category: System / Role

Definition: A specialized registered renal nurse authorized to initiate, monitor, and document hemodialysis sessions, capture ultrafiltration volumes and vitals, log vascular access status, record complications, and dispense dialysis consumables.

Used in: Dialysis Unit (`rooms.$id.tsx` , `dialysis_sessions` , `dialysis_session_consumables`)

Dialysis Session

Category: Clinical / System

Definition: A structured clinical record capturing a complete hemodialysis treatment run, including dialyzer model, blood flow rate (BFR), dialysate flow rate (DFR), pre/post-dialysis blood pressure, weight changes, fluid removal goals (UF), and anticoagulant dosages.

Used in: Dialysis Unit Room (`dialysis_sessions` , `process_dialysis_session_billing()` RPC)

Dialysis Store

Category: System / Stock

Definition: A dedicated sub-inventory location (`d38d45c7-20f2-4e1b-9279-8cb5bf567cd1`) housing specialized renal consumables such as dialyzers, bloodlines, AV fistula needles, bicarbonate cartridges, acid concentrates, and heparin.

Used in: Stock Management, Dialysis Billing (`stock_locations` , `stock_items`)

Differential Diagnosis

Category: Clinical

Definition: A list of alternative potential diseases or conditions considered by a clinician during the diagnostic evaluation process before reaching

a confirmed or working diagnosis.

Used in: Consultation Module (`encounter_diagnoses.diagnosis_type = 'differential'`)

Director

Category: System / Role

Definition: Executive management persona possessing read-only executive visibility across all institutional dashboards, financial aggregations, MoH disease reports, clinical throughput metrics, and system audit logs.

Used in: Executive Dashboard, Reports (`/dashboard` , `/reports` , `user_roles`)

Discharge

Category: Clinical

Definition: The clinical and administrative conclusion of an inpatient hospital stay, requiring a verified discharge summary, reconciled pharmacy discharge medications, completed final invoice clearance, and bed status reset.

Used in: Inpatient Module (`admissions.status = 'discharged'` , `beds.status = 'available'`)

Discharge Summary

Category: Clinical / System

Definition: A mandatory clinical summary document authored by the attending medical officer at the conclusion of an inpatient admission, detailing admission reasons, significant findings, therapeutic procedures performed, discharge condition, and follow-up instructions.

Used in: Inpatient Module (`admissions.discharge_summary` , `require_discharge_summary` trigger)

Doctor

Category: System / Role

Definition: A licensed Medical Practitioner (Medical Officer, Physician, Surgeon, or Specialist) registered with the KMPDC authorized to conduct clinical evaluations, perform invasive surgical procedures, order and interpret complex diagnostics, prescribe all drug schedules, admit/discharge inpatients, and electronically sign encounters.

Used in: Clinical Consultation, Wards, ICU, Theatre (`user_roles` , `clinical_notes` , `encounters`)

Drug Formulation & Route

Category: Clinical / Pharmacy

Definition: The physical form (e.g., tablet, suspension, vial, ampoule, cream) and pharmaceutical route of administration (e.g., Oral, IV, IM, SC, Topical, Inhalation) specified on clinical prescriptions.

Used in: Pharmacy Module, Inpatient Medication Administration (`prescriptions` , `medication_administrations`)

E

ECCIF (Emergency, Chronic and Critical Illness Fund)

Category: Kenya-specific / Business

Definition: A dedicated universal healthcare statutory fund established under the Social Health Insurance Act (SHIA) of Kenya, designed to cover emergency treatments, intensive care stays, and severe chronic condition interventions beyond basic primary care thresholds.

Used in: Insurance Desk, Billing, SHA Benefit Plans (`insurance_benefit_categories` , `sha_claims`)

Edge Function

Category: Technical

Definition: A serverless TypeScript/Deno computational module running on Supabase's globally distributed network, used in AegisCare for low-latency operations such as SMS messaging, external WHO ICD-11 lookups, and DHA FHIR payload compilation.

Used in: `supabase/functions/*` (`send-sms` , `icd11-search` , `claims-dispatcher` , `fhir-bundle`)

Emergency Encounter

Category: Clinical / System

Definition: An urgent clinical registration pathway that allows patients in acute distress to bypass normal intake queues and cash payment checks,

routing them immediately to the resuscitation room or casualty doctor.

Used in: Patient Registration, Triage, Casualty (`encounters.is_emergency = true` , `encounters.encounter_type = 'emergency'`)

Emergency Referral

Category: Clinical / MOH

Definition: The formal transfer of an unstable emergency patient to a higher-level facility (referral out) or receipt of an acute case from a lower-tier dispensary/health centre (referral in), logged for epidemiological MOH 717 reporting.

Used in: Patient Registration, Consultation, MOH Reporting (`encounters.referral_direction` , `dashboard_emergency_referrals()`)

Encounter (Patient Registration)

Category: Clinical / System

Definition: A single, discrete interaction episode between a patient and healthcare service providers spanning registration, triage, consultation, laboratory tests, radiology imaging, pharmaceutical dispensing, and discharge.

Used in: Database Table `encounters` (aliased as `patient_registrations`), Throughout All Clinical Modules

Encounter Diagnosis

Category: Clinical

Definition: A formal diagnostic condition code linked to a specific encounter, mapped to WHO ICD-11 MMS nomenclature, tagged by sequence (Primary, Secondary, Comorbidity), and categorized by clinical certainty.

Used in: Consultation, Inpatient, SHA Claims (`encounter_diagnoses` , `clean_and_validate_diagnosis_insert()`)

Encounter Lock / Signing

Category: System / Clinical

Definition: An irreversible cryptographic state change executed by authorized clinicians upon completing an encounter (`status = 'signed'`), locking all clinical notes, diagnoses, and orders against further mutation to comply with legal evidentiary and DHA certification standards.

Used in: Consultation, Inpatient (`enforce_encounter_lock()` trigger, `sprint13a_encounter_signing.sql`)

Episode of Care

Category: Technical / Clinical

Definition: A longitudinal FHIR resource and clinical container (`episode_of_care`) that groups related healthcare encounters and clinical events managed under a single continuous management plan or disease condition over time.

Used in: FHIR Interoperability, Chronic Disease Tracking (`episode_of_care` , Edge Function `fhir-bundle`)

External Body (Mortuary)

Category: Clinical / System

Definition: A deceased individual brought to the hospital mortuary directly from outside the institution (e.g., police case, home death, roadside accident) rather than passing away while admitted in an internal hospital ward, requiring specific legal documentation, next-of-kin verification, and dedicated cash billing invoicing.

Used in: Mortuary Module (`mortuary_records.body_source = 'external'` , `create_external_mortuary_invoice()` RPC)

F

Facility Level (Kenya MoH Level 1–6)

Category: Kenya-specific / System

Definition: The official Kenya Ministry of Health healthcare facility grading system dictating operational scope and service availability:

- **Level 1:** Community Health Unit (Community Health Promoters)
- **Level 2:** Dispensary (Outpatient primary care)
- **Level 3 (3A/3B):** Health Centre (Outpatient + Basic Maternity; no full wards or mortuary)
- **Level 4:** Primary Hospital / Sub-County (Full inpatient, surgical theatre, mortuary, basic imaging)
- **Level 5:** Secondary / County Referral Hospital (Specialized care, ICU, Dialysis, advanced diagnostics)

- **Level 6:** National Referral & Teaching Hospital (Tertiary & quaternary sub-specialties)
Used in: Facility Settings, Feature Gating (`facility_features` , `app_settings.facility_level`)

Facility Features

Category: System

Definition: System configuration flags that dynamically enable or lock entire application modules (e.g., Mortuary, ICU, Dialysis, Inpatient Wards) based on the accredited MoH Level of the host facility.

Used in: `facility_features` , UI Dynamic Navigation

FHIR (Fast Healthcare Interoperability Resources - R4)

Category: Technical

Definition: The international HL7 standard framework for structuring and exchanging electronic health information via RESTful JSON APIs, mandated by Kenya's DHA for national interoperability.

Used in: DHA/SHA Integration, Edge Functions `fhir-patient` , `fhir-encounter` , `fhir-condition` , `fhir-bundle`

FHIR Bundle

Category: Technical

Definition: A single cohesive JSON container collecting multiple related FHIR resources (Patient, Encounter, Condition, MedicationDispense, Observation, Claim) transmitted as a batch or transaction to the DHA Shared Health Record (SHR).

Used in: Edge Function `fhir-bundle` , `dha_outbound_queue`

FHIR Condition

Category: Technical

Definition: A standardized FHIR R4 resource representing clinical diagnoses, complaints, and comorbidities mapped directly to WHO ICD-11 MMS URIs and concept codes.

Used in: Edge Function `fhir-condition` , Consultation Sync

FHIR Encounter

Category: Technical

Definition: A standardized FHIR R4 resource detailing patient arrival, service class (Ambulatory, Inpatient, Emergency), service provider organization, attending practitioner, and clinical duration.

Used in: Edge Function `fhir-encounter` , `generate_fhir_encounter()` RPC

FHIR Patient

Category: Technical

Definition: A standardized FHIR R4 resource representing demographic and administrative patient identity attributes (official identifiers, full names, gender, date of birth, contact telecom, and residence address).

Used in: Edge Function `fhir-patient` , Patient Registration Sync

Final Diagnosis

Category: Clinical

Definition: The conclusive diagnostic determination established by the attending medical officer after evaluating all physical findings, laboratory workups, and radiological imaging results.

Used in: Consultation, Inpatient Discharge (`encounter_diagnoses.diagnosis_type = 'final'`)

Form 705A (Under 5 Outpatient)

Category: Kenya-specific / MOH

Definition: The official statutory Ministry of Health monthly epidemiological surveillance report capturing outpatient morbidity patterns for children aged under five years across standard disease classifications.

Used in: MOH Reporting Module (`/moh/705` , `get_moh_705_report(..., '705A')`) RPC

Form 705B (Over 5 Outpatient)

Category: Kenya-specific / MOH

Definition: The official statutory Ministry of Health monthly epidemiological summary documenting outpatient morbidity for patients aged five years and above.

Used in: MOH Reporting Module (`/moh/705` , `get_moh_705_report(..., '705B')`) RPC

Form 706 (Laboratory Monthly Summary)

Category: Kenya-specific / MOH

Definition: The standard MoH statutory reporting template summarizing aggregate laboratory diagnostic tests performed (malaria smears, HIV testing, urinalysis, full hemogram, lipid profiles) categorized by clinical outcome.

Used in: MOH Reporting Module (`/moh/706` , `moh_indicator_definitions`)

Form 707 (Inpatient Monthly Summary)

Category: Kenya-specific / MOH

Definition: The statutory MoH monthly report tracking inpatient admissions, bed occupancy rates, average length of stay (ALOS), maternal/neonatal outcomes, surgical interventions, and inpatient mortality.

Used in: MOH Reporting Module (`/moh/707` , `refresh_moh_707_monthly_aggregates()`)

Form 717 (Hospital Workload Summary)

Category: Kenya-specific / MOH

Definition: The comprehensive institutional workload return capturing aggregate monthly throughput across general outpatient clinics, specialized clinics, casualty visits, dental attendances, imaging procedures, and mortuary storage days.

Used in: MOH Reporting Module (`/moh/717`)

FP (Family Planning) Indicators

Category: Kenya-specific / MOH

Definition: Standardized reproductive health metrics tracking client intake and dispensing of contraceptive modalities (Implants, Oral Pills, Injectable Depo-Provera, IUCDs, Emergency Contraceptives, and Condoms).

Used in: MOH FP Report (`/moh/fp` , `moh_indicator_definitions`)

G

Glasgow Coma Scale (GCS)

Category: Clinical

Definition: An objective neurological scoring scale (ranging from 3 to 15) assessing acute levels of consciousness based on Eye Opening (1–4), Verbal Response (1–5), and Motor Response (1–6).

Used in: Triage, Casualty, ICU Hourly Charting (`icu_hourly_charts.gcs_total` , Triage Vitals)

Grant Stock Location Access

Category: System / Permission

Definition: An administrative function granting specific warehouse sub-store access privileges (`allow_view` , `allow_request` , `allow_approve` , `allow_issue` , `allow_receive`) to designated staff users.

Used in: Stock Security (`grant_stock_location_access()`) RPC, `user_stock_location_access`)

H

HIE (Health Information Exchange)

Category: Kenya-specific / Technical

Definition: The national digital health network managed by the DHA that enables electronic medical record systems nationwide to exchange standardized FHIR clinical documents securely.

Used in: System Architecture, DHA Outbound Queue

HTS (HIV Testing Services) Counsellor

Category: System / Role

Definition: A certified healthcare worker dedicated to conducting pre-test counseling, point-of-care rapid HIV screening, post-test counseling, and linkage of reactive clients to Comprehensive Care Centres (CCC).

Used in: VCT / HTS Clinic (`user_roles` , `encounters`)

I

ICD-11 (International Classification of Diseases 11th Revision)

Category: Clinical / Technical

Definition: The World Health Organization's global clinical classification standard for systematic recording, reporting, analysis, and comparison of mortality and morbidity data, natively embedded into AegisCare.

Used in: Consultation, Inpatient, Edge Function (`icd11-search` , `icd11_codes`)

ICD-11 MMS (Mortality and Morbidity Statistics)

Category: Clinical / Technical

Definition: The specific WHO linearization of ICD-11 curated for general clinical coding and health statistics reporting, queried in real time via WHO's cloud API.

Used in: Consultation Coding, Edge Function (`icd11-search`)

ICU (Intensive Care Unit)

Category: Clinical

Definition: A specialized, high-dependency clinical ward providing advanced life support, continuous invasive hemodynamic monitoring, mechanical ventilation, and dedicated one-on-one nursing for critically ill patients.

Used in: Inpatient / ICU Module, Wards (`wards.ward_type = 'icu'`), Room (`a839ad3e-8721-4428-a825-6eeelf75207b`)

ICU Hourly Chart

Category: Clinical / System

Definition: A structured electronic flow-sheet updated hourly by ICU nurses, capturing invasive arterial/CVP pressures, mechanical ventilator settings (PEEP, FiO2, Tidal Volume), RASS sedation depth, GCS, pupil reactivity, urine output, inotrope infusion rates, and arterial blood gas (ABG) values.

Used in: ICU Ward Interface (`icu_hourly_charts` , `rooms.$id.tsx`)

ICU Nurse

Category: System / Role

Definition: A critical care certified nurse assigned to the ICU ward authorized to maintain hourly critical flow-sheets, titrate vasoactive infusions, manage mechanical ventilators, and dispense medications directly from the specialized ICU sub-inventory store.

Used in: ICU Clinical Room (`user_roles` , `icu_hourly_charts` , `stock_usage`)

ICU Store

Category: System / Stock

Definition: A dedicated departmental drug and consumable sub-location (`a99583cd-9354-470a-9299-73457734284d`) storing emergency critical-care drugs (Adrenaline, Noradrenaline, Midazolam, Fentanyl, Propofol, Vecuronium), endotracheal tubes, and central lines.

Used in: Stock Management, ICU Ward Dispensing (`stock_locations` , `stock_items`)

Identity Verification (IPRS)

Category: Kenya-specific / Technical

Definition: The automated validation of citizen identities against the Kenya Integrated Population Registration System database via DHA APIs, verifying National ID or Passport details against official state population registers.

Used in: Patient Registration (`verify_patient_identity()`) RPC,  DHA Integration)

Inpatient (IPD)

Category: Clinical

Definition: Clinical care provided to patients admitted to a hospital ward bed for comprehensive medical, surgical, pediatric, or obstetric treatment requiring one or more overnight stays.

Used in: Inpatient Module (`/inpatient` , `admissions` , `wards` , `beds`)

Insurance Agent

Category: System / Role

Definition: A hospital desk officer responsible for verifying patient insurance policies (SHA, Private, Corporate), capturing authorization numbers, confirming coverage benefit limits, and submitting insurance claims.

Used in: Insurance Desk (`/admin/insurance` , `user_roles` , `sha_claims`)

Insurance Benefit Category

Category: Business / System

Definition: A specific clinical benefit envelope defined under an insurance plan (e.g., Outpatient Consultation, Basic Lab, Specialized Imaging, Inpatient Normal Delivery, ICU Bed Care, Hemodialysis).

Used in: Insurance Setup (`insurance_benefit_categories` , `insurance_benefit_plans`)

Insurance Benefit Plan

Category: Business / System

Definition: A configured insurance policy product associated with an insurance underwriter, detailing co-payment amounts, visit limits, annual maximum financial caps, and pre-authorization triggers.

Used in: Insurance Desk, Billing (`insurance_benefit_plans` , `insurance_providers`)

Insurance Limit / Visit Cap

Category: Business / System

Definition: A hard limit established by insurance contracts capping the number of outpatient visits or maximum financial exposure allowed per patient per specified period.

Used in: Insurance Desk (`enforce_insurance_visit_limit()` trigger)

Insurance Provider

Category: Business / System

Definition: An underwriter, fund, or scheme (e.g., Social Health Authority - SHA, Jubilee Insurance, CIC, Britam, First Assurance) covering healthcare costs for enrolled beneficiaries.

Used in: Insurance Desk, Invoicing (`insurance_providers`)

Invoice

Category: Business / System

Definition: The consolidated fiscal bill generated for every patient encounter or mortuary booking, aggregating all billable line items (consultations, lab tests, radiology, pharmaceuticals, bed charges, procedures), tracking payments received, waivers applied, and outstanding balances.

Used in: Billing & Accounting Module (`invoices` , `invoice_line_items` , `invoice_payments`)

Invoice Line Item

Category: Business / System

Definition: An individual charged service, drug, diagnostic test, bed day, or consumable entry appearing on a patient invoice with quantity, unit price, source reference, and calculated subtotal.

Used in: Invoicing (`invoice_line_items` , `sync_invoice_line_item_from_prescription()`)

Invoice Payment

Category: Business / System

Definition: A financial transaction ledger entry recording funds received against an open invoice, tagged with payment method (Cash, M-Pesa,

Card, Bank Transfer, SHA Claim, Private Insurance Guarantee), transaction reference, and cashier user ID.

Used in: Cash Office, Accounting (`invoice_payments` , `recalc_invoice_payments()` trigger)

IPRS (Integrated Population Registration System)

Category: Kenya-specific

Definition: The national government identity registry database maintained by the Ministry of Interior, serving as the single source of truth for citizen identity verification in Kenya.

Used in: Patient Intake, Identity Verification

J

JWT (JSON Web Token)

Category: Technical

Definition: A cryptographically signed token issued by Supabase Auth upon successful user authentication, passed in HTTP Authorization headers to enforce database Row-Level Security policies and secure Edge Function execution.

Used in: Security Architecture, Supabase Auth

K

KMHFL (Kenya Master Health Facility List) Code

Category: Kenya-specific

Definition: The unique national numeric identifier (e.g., `12345`) assigned to every licensed healthcare facility by the Ministry of Health, mandatory for national health reporting, SHA claims, and DHA interoperability.

Used in: Facility Settings (`app_settings.facility_kmhfl_code`), DHA/SHA Payloads

KMPDC (Kenya Medical Practitioners and Dentists Council)

Category: Kenya-specific

Definition: The statutory regulatory council established by Cap 253 of the Laws of Kenya that licenses medical practitioners, dental surgeons, and all public/private healthcare institutions nationwide.

Used in: Staff Registration Profiles (`profiles.council_registration_number` , `verify_practitioner()`)

KSh (Kenyan Shilling)

Category: Business / Kenya-specific

Definition: The official national currency of Kenya (`KES` / `KSh`), used universally across AegisCare for price catalogs, invoice line items, cash collections, and insurance claims.

Used in: Billing Engine, Financial Reports, Price Catalogs

L

Laboratory

Category: Clinical / System

Definition: The clinical diagnostic department responsible for analyzing biological specimens (blood, urine, stool, CSF, sputum, tissue) to assist clinicians in disease diagnosis, staging, and monitoring.

Used in: Laboratory Module (`/laboratory` , `lab_orders` , `lab_results`)

Laboratory Order

Category: Clinical / System

Definition: A formal diagnostic request initiated by a clinician during an outpatient or inpatient encounter, specifying tests to be performed,

clinical indications, and priority status.

Used in: Laboratory Module (`lab_orders` , `laboratory.$id.tsx`)

Laboratory Result

Category: Clinical / System

Definition: Quantitative values or qualitative interpretations recorded by a lab technologist for a requested test, compared against reference ranges, flagged for critical abnormal values, and electronically verified.

Used in: Laboratory Module (`lab_results` , `send_lab_result_to_room()`) RPC

Laboratory Test Catalog

Category: Clinical / System

Definition: The institutional master catalog of all accredited diagnostic laboratory investigations offered by the facility, specifying test names, department categories, specimen types, normal reference ranges, turn-around times, and base cash prices.

Used in: Laboratory Configuration, Services (`lab_test_catalog`)

Lab Tech (Laboratory Technologist)

Category: System / Role

Definition: A certified laboratory professional registered under the Kenya Medical Laboratory Technicians and Technologists Board (KMLTB), authorized to accept specimens, run clinical analyzers, record test outcomes, verify results, and manage lab reagent stocks.

Used in: Laboratory Module (`user_roles` , `lab_orders` , `lab_results`)

Lab View / Lab Update / Lab Results Create

Category: System / Permission

Definition: Granular security permissions governing laboratory operations:

- `lab_view` : Read laboratory workbenches and completed results.
- `lab_update` : Edit specimen reception and processing statuses.
- `lab_results_create` : Input clinical assay results and reference ranges.

Used in: `role_permissions` , Laboratory UI Access

Line Item

Category: Business

Definition: A distinct row item within an invoice or purchase order representing a single billable entity with associated unit price, quantity, and total cost.

Used in: Billing, Procurement (`invoice_line_items`)

Location Access (Stock)

Category: System / Stock

Definition: Granular security access controls assigning specific hospital users permissions to view, request, approve, issue, or receive inventory within designated warehouse locations.

Used in: Stock Management (`user_stock_location_access` , `user_can_access_stock_location()`) RPC

LPO (Local Purchase Order)

Category: Business

Definition: An official commercial procurement document issued by the hospital procurement department to an approved supplier, authorizing the purchase and delivery of specified pharmaceuticals or clinical consumables at agreed prices.

Used in: Stock Procurement (`stock_deliveries.invoice_number` , Procurement Workflows)

M

Machine

Category: System / Clinical

Definition: A specialized diagnostic, therapeutic, or life-support medical device (e.g., Automated Hematology Analyzer, Chemistry Analyzer, Digital X-Ray, Ultrasound, Hemodialysis Machine, ICU Ventilator) registered in the system and monitored for operational uptime and maintenance logs.

Used in: Machines Management (`/machines` , `machines` , `machine_logs`)

Machine Log

Category: System / Technical

Definition: A maintenance, service, calibration, or error history entry logged against a specific hospital medical device to maintain regulatory quality control standards.

Used in: Machines Module (`machine_logs`)

Machines

Category: System / Permission

Definition: A system security permission (`machines`) authorizing technical staff and biomedical engineers to create machine records, schedule maintenance windows, and log calibration records.

Used in: `role_permissions` , Equipment Management

Main Store

Category: System / Stock

Definition: The central hospital warehouse inventory repository (`a0000000-0000-0000-0000-000000000001`) where all bulk supplier deliveries are received, verified, and held prior to inter-departmental distribution.

Used in: Stock Module (`stock_locations` , `stock_transfers`)

MCH (Maternal and Child Health)

Category: Clinical / MOH

Definition: Specialized primary care services encompassing Antenatal Care (ANC), Postnatal Care (PNC), Child Welfare & Immunization (EPI), and Prevention of Mother-to-Child Transmission of HIV (PMTCT).

Used in: MCH Clinic Room, MOH MCH Reports (`/moh/mch` , `moh_indicator_definitions`)

Medication Administration Record (MAR)

Category: Clinical

Definition: An electronic nursing record documenting the actual bedside administration of scheduled and PRN (as-needed) medications to admitted inpatients, including drug, dosage, route, exact timestamp, and administering nurse signature.

Used in: Inpatient Ward Module (`medication_administrations` , `inpatient_.$admissionId.tsx`)

Medication Dispense (FHIR)

Category: Technical / Clinical

Definition: A standardized FHIR R4 resource representing the physical supply and dispensing of a pharmaceutical product to a patient, including quantity dispensed, days supply, and dosage instructions.

Used in: Pharmacy Module, Edge Function (`fhir-bundle` , `trg_medication_dispense_fhir()`)

Mediator ID

Category: Technical / DHA

Definition: The unique transaction tracing identifier assigned to an outbound FHIR bundle submitted to the DHA AfyaLink Health Information Mediator gateway.

Used in: `dha_outbound_queue.mediator_id` , FHIR Interoperability

Mermaid Diagram

Category: Technical

Definition: A plain-text diagramming language embedded within Markdown files to render dynamic entity-relationship diagrams (ERDs), process flowcharts, and system architectural topologies natively.

Used in: System Documentation, `docs/schema.md`

MOH (Ministry of Health Kenya)

Category: Kenya-specific

Definition: The national executive ministry responsible for healthcare policy, public health standards, health indicator registries, and statutory epidemiology reporting across all 47 counties.

Used in: MOH Reporting Suite, System Compliance

MOH Indicator Definition

Category: Kenya-specific / System

Definition: Configuration metadata defining specific disease or service tally criteria mapped to statutory MoH monthly report templates (MOH 705A/B, 706, 707, 717, FP, MCH).

Used in: MOH Module (`moh_indicator_definitions` , `process_encounter_indicators()`)

MOH Monthly Aggregates

Category: Kenya-specific / System

Definition: Pre-computed database summary tables storing aggregated patient visit and diagnosis tallies per month, accelerating statutory reporting performance.

Used in: Database Table (`moh_monthly_aggregates` , `refresh_moh_monthly_aggregates()`) RPC

Mortician

Category: System / Role

Definition: A licensed funeral service practitioner authorized to manage mortuary cold-room storage units, log internal and external body admissions, document embalming procedures and post-mortem examinations, manage viewing schedules, and execute body release clearance.

Used in: Mortuary Module (`/rooms/$mortuaryRoomId` , `mortuary_records` , `user_roles`)

Mortuary Module

Category: Clinical / System

Definition: A dedicated facility module for tracking the reception, refrigeration bay allocation, preservation, autopsy documentation, storage charge calculation, and formal handover of deceased bodies to next-of-kin or state authorities.

Used in: Mortuary Department (`mortuary_records` , `accrue_daily_mortuary_charges()`)

Mortuary Record

Category: Clinical / System

Definition: A comprehensive administrative record for a deceased body, detailing full name, age, gender, date/time of death, cause of death, admitting body source (internal ward vs. external police/home intake), storage slot, next-of-kin details, and release clearance status.

Used in: Mortuary Module (`mortuary_records` , `assign_mortuary_reference()` trigger)

Mortuary Room

Category: System

Definition: The physical mortuary facility department entity (`214aca6c-cbde-40ce-af5a-92cc7815bda9`) within the hospital room hierarchy.

Used in: Facility Architecture, Room Routing

M-Pesa Integration (Daraja API)

Category: Kenya-specific / Technical

Definition: The planned Phase 3 integration with Safaricom's Daraja M-Pesa API, supporting automated STK Push payment requests at cashier desks and automated B2C refund disbursements.

Used in: Cash Office, Billing Module ( Phase 3 Roadmap)

MUAC (Mid-Upper Arm Circumference)

Category: Clinical

Definition: A quick nutritional screening measurement taken in pediatric triage to identify acute malnutrition (Red <11.5 cm: Severe Acute Malnutrition; Yellow 11.5–12.5 cm: Moderate Acute Malnutrition; Green >12.5 cm: Normal).

Used in: Triage Module, Pediatric Consultation (`encounters.vitals`)

N

NLMIS Code (National Lab Management Information System)

Category: Kenya-specific / System

Definition: Standardized commodity codes assigned by the Kenya National Lab Management Information System to lab reagents and test kits for harmonized national supply chain reporting.

Used in: Laboratory Test Catalog (`lab_test_catalog.nlmis_code` , `sprint11c_nlmis_code.sql`)

Nurse

Category: System / Role

Definition: A licensed general nursing practitioner registered under the Nursing Council of Kenya (NCK), authorized to capture vital signs, administer prescribed oral/parenteral medications, perform wound dressings, manage ward inpatients, and record nursing progress notes.

Used in: Triage, Wards, Inpatient Module (`user_roles` , `medication_administrations`)

Nutritionist

Category: System / Role

Definition: A licensed healthcare professional specializing in clinical dietetics and nutritional assessment, managing therapeutic feeds, calculating caloric requirements, and running outpatient nutrition clinics.

Used in: Specialized Clinics, Inpatient Care (`user_roles` , `clinical_notes`)

O

ODPC (Office of the Data Protection Commissioner)

Category: Kenya-specific / Compliance

Definition: The statutory supervisory authority established under Kenya's Data Protection Act, 2019, regulating the lawful processing of personal and health data, mandating patient consent, purpose specification, and access auditability.

Used in: System Security, Consent Framework, Data Privacy Policies

OPD (Outpatient Department)

Category: Clinical

Definition: The hospital clinical division delivering non-emergency diagnostic, therapeutic, and preventive care to patients who do not require overnight inpatient admission.

Used in: Reception, Queue, Consultation (`encounters.encounter_type = 'outpatient'`)

OTP (One-Time Password) Consent

Category: System / Security

Definition: An automated Kenya Data Protection Act compliance workflow where an encrypted 6-digit numeric verification token is sent via SMS to the patient's registered mobile phone to obtain explicit consent for digital health record processing and national HIE sharing.

Used in: Patient Registration, Consent Management (`consent_otps` , `patient_consents` , Edge Function `send-sms`)

P

Patient Consent

Category: System / Compliance

Definition: An explicit, timestamped electronic record confirming a patient's approval for clinical treatment, billing disclosures, research use, or data sharing with the national DHA Health Information Exchange.

Used in: Patient Registration (`patient_consents`)

Patient File Number

Category: System / Clinical

Definition: A unique, sequential institutional patient identifier (e.g., `P000123`) automatically assigned to every newly registered patient and stamped on all physical cards and digital records.

Used in: Patient Registration (`patients.file_number` , `set_patient_file_number()` trigger)

Patient Registration

Category: Clinical / System

Definition: The initial intake administrative workflow capturing patient legal identification, demographic profile, next-of-kin contacts, insurance membership, and consent preferences.

Used in: Patient Registration Module (`/register-patient` , `patients` , `encounters`)

Payment Mode (Cash / Insurance / SHA / Corporate / Exemption)

Category: Business / System

Definition: The financial categorization applied to an encounter determining billing rules and reimbursement mechanisms:

- `cash` : Out-of-pocket self-pay.
- `insurance` : Private or corporate medical underwriter.
- `sha_shif` : Social Health Authority fund.
- `exemption` : Vulnerable patient waived under government waiver programs.

Used in: Patient Registration, Billing Engine (`encounters.payment_mode`)

PHF (Primary Healthcare Fund)

Category: Kenya-specific / Business

Definition: A universal health coverage fund managed under Kenya's Social Health Insurance Act, financing essential primary care outpatient services at MoH Level 2 (Dispensaries) and Level 3 (Health Centres) without copayment.

Used in: Insurance Desk, Benefit Packages (`insurance_benefit_categories`)

Pharmacist

Category: System / Role

Definition: A licensed pharmaceutical professional registered with the Pharmacy and Poisons Board (PPB) of Kenya, authorized to verify clinical prescriptions, screen for drug-drug interactions, dispense medications from outpatient and inpatient pharmacies, and manage pharmacy warehouse stock.

Used in: Pharmacy Module (`user_roles` , `prescriptions` , `stock_movements`)

Pharmacy Module

Category: Clinical / Stock

Definition: The operational department handling prescription queues, pharmacist verification, batch-specific drug dispensing, patient dosage counseling, and real-time inventory deductions.

Used in: Pharmacy Room (`prescriptions` , `dispense_prescription_stock()` trigger)

Pharmacy Store

Category: System / Stock

Definition: The dedicated dispensing pharmacy sub-location (`a0000000-0000-0000-0000-000000000003`) from which general outpatient and ward prescriptions are dispensed to patients.

Used in: Stock Management, Pharmacy (`stock_locations`)

Physiotherapist

Category: System / Role

Definition: A licensed physical rehabilitation specialist providing therapeutic exercise, manual therapy, and post-operative mobility rehabilitation for outpatients and ward inpatients.

Used in: Specialized Therapy Rooms (`user_roles` , `clinical_notes`)

Pre-Authorization (Preauth)

Category: Business

Definition: Formal advance approval granted by an insurance provider (e.g., SHA or private insurer) authorizing scheduled elective admissions, advanced radiological imaging (CT/MRI), or specialized surgical procedures before delivery.

Used in: Insurance Desk, Inpatient Admission (`sha_claims.preauth_code` , `encounters.preauth_number`)

Prescription

Category: Clinical / Pharmacy

Definition: A formal medical instruction issued by a licensed clinician detailing a drug name, formulation, dosage strength, frequency, route, and treatment duration for a specific patient encounter.

Used in: Consultation, Inpatient, Pharmacy (`prescriptions` , `rooms.$id.tsx`)

Primary Diagnosis

Category: Clinical

Definition: The chief condition or disease entity established at the conclusion of clinical evaluation to be predominantly responsible for the patient seeking medical care.

Used in: Consultation, MOH Reports, SHA Claims (`encounter_diagnoses.sequence = 1`)

Profiles

Category: Technical / System

Definition: The database user profile extension table storing practitioner professional metadata (full legal name, professional council type, registration number, national ID, phone, department).

Used in: User Management (`profiles` , `verify_practitioner()`)

Q

Queue Management

Category: System

Definition: The operational routing workflow that assigns arriving patients to specific service rooms (Triage, Insurance, Doctor, Lab, Radiology, Pharmacy, Billing) and manages real-time waiting lists.

Used in: Queue Module (`/queue` , `/admin/queue` , `encounters.room_id` , `encounters.queue_status`)

R

Radiology Module

Category: Clinical / System

Definition: The medical diagnostic imaging department responsible for performing X-rays, ultrasounds, CT scans, and mammograms, generating structured imaging reports, and notifying ordering doctors.

Used in: Radiology Module (`/radiology` , `radiology_orders` , `radiology_results`)

Radiology Order

Category: Clinical / System

Definition: An electronic imaging requisition created by an attending clinician specifying anatomical region, projection modality, clinical indications, and priority.

Used in: Consultation, Inpatient, Radiology (`radiology_orders`)

Radiology Result

Category: Clinical / System

Definition: A comprehensive diagnostic interpretation and formal radiological report entered by a radiologist or sonographer.

Used in: Radiology Module (`radiology_results` , `send_radiology_results_to_requesting_room()`) RPC

Radiology View / Radiology Update / Radiology Results Create

Category: System / Permission

Definition: Security permissions governing imaging operations:

- `radiology_view` : View incoming imaging orders and finalized reports.
- `radiology_update` : Update modality examination statuses.
- `radiology_results_create` : Author and sign formal diagnostic imaging reports.

Used in: `role_permissions` , Radiology UI Access

Radiologist

Category: System / Role

Definition: A medical specialist or imaging technologist certified to operate diagnostic imaging modalities, interpret radiographs, scans, and sonograms, and author diagnostic radiology reports.

Used in: Radiology Department (`user_roles` , `radiology_orders` , `radiology_results`)

RASS (Richmond Agitation-Sedation Scale)

Category: Clinical

Definition: A standardized 10-point clinical scale ranging from -5 (un arousable sedation) through 0 (alert and calm) to +4 (combative agitation), used in ICU hourly charting to assess sedation depth.

Used in: ICU Hourly Charting (`icu_hourly_charts.rass_score`)

Receptionist

Category: System / Role

Definition: Front-office administrative staff responsible for searching patient archives, creating new patient master files, capturing OTP consent, assigning payment modes, and dispatching patients to triage or the insurance desk.

Used in: Reception Desk, Queue (`/register-patient` , `user_roles`)

Records Create / Records View

Category: System / Permission

Definition: Permissions governing patient master file management:

- `records_view` : Search and view patient demographic and registration histories.
- `records_create` : Register new patient profiles and open new clinical encounters.

Used in: `role_permissions` , Patient Registration

Records Officer

Category: System / Role

Definition: A medical records technician (Health Information Management Officer) responsible for patient registration data quality, physical and electronic file tracking, ICD-11 coding reviews, and monthly MoH data reconciliation.

Used in: Registration, Health Records, MOH Reporting (`user_roles`)

Referral Direction (In / Out)

Category: Clinical / MOH

Definition: The movement classification of transferred patients:

- `in` : Patient referred into the facility from a peripheral dispensary or clinic.
- `out` : Patient referred out to a higher-tier secondary or tertiary hospital.

Used in: Patient Registration, Consultation, MOH 717 (`encounters.referral_direction` , `encounters.referral_out_facility`)

Register Patient

Category: System / Permission

Definition: The system security permission (`register_patient`) permitting staff to access the patient intake wizard, issue patient file numbers, and initiate billable encounters.

Used in: `role_permissions`, Front Desk UI

Report Permissions (`reports.finance`, `reports.registrations`, `reports.stock`, `reports.tests`)

Category: System / Permission

Definition: Granular security permissions governing report generation:

- `reports.finance` : Access revenue, payment, waiver, and aging ledgers.
- `reports.registrations` : View patient demographic and attendance volumes.
- `reports.stock` : View inventory movements, consumption, and stock valuations.
- `reports.tests` : View laboratory diagnostic throughput summaries.

Used in: `role_permissions`, Reports Module (`/reports`)

Realtime (Supabase Realtime)

Category: Technical

Definition: Supabase's WebSocket-based real-time listener architecture allowing the AegisCare user interface to update room queues, lab results, bed allocations, and vital alerts instantly without manual page refreshing.

Used in: Queue Management, Bed Tracking, Lab Workbenches

RLS (Row-Level Security)

Category: Technical / Security

Definition: PostgreSQL database engine security policies that enforce granular data access boundaries at the individual database row level based on the authenticated user's JWT role, approved status, and departmental assignment.

Used in: All Supabase Database Tables

Room

Category: System

Definition: A defined physical operational clinical or service room within the hospital (e.g., Room 1 OPD, Consultation Room, Triage, Minor Theatre, Lab, Radiology, Cash Office, Mortuary).

Used in: Queue Routing, Room Management (`rooms`, `encounter_room_visits`)

Room Routing

Category: System / Clinical

Definition: The automated or manual transfer of a patient encounter from one service room to another across their care journey.

Used in: `route_registration_to_service_room()` RPC, `encounter_room_visits`

RPC (Remote Procedure Call)

Category: Technical

Definition: A server-side PostgreSQL function executable directly from the frontend client via Supabase RPC, executing complex transactional business operations atomically.

Used in: Database Business Logic (`charge_icu_admission_fee`, `process_dialysis_session_billing`)

S

Secondary Diagnosis

Category: Clinical

Definition: An additional condition or co-existing disease present during a clinical encounter that co-exists with or develops during treatment, influencing medical management.

Used in: Consultation, Inpatient, SHA Claims (`encounter_diagnoses.sequence > 1`)

Service Room

Category: System / Clinical

Definition: A room configured to provide diagnostic, therapeutic, or administrative services rather than general clinical consultation (e.g., Laboratory, Ultrasound, Pharmacy, Cashier).

Used in: `rooms.room_kind = 'service'`, Room Routing

Services Manage

Category: System / Permission

Definition: An administrative permission (`services.manage`) enabling authorized supervisors to edit the master service price catalog, lab test menu, and procedural tariffs.

Used in: `role_permissions`, Admin Settings (`/admin/services`)

SHA (Social Health Authority)

Category: Kenya-specific

Definition: The statutory state corporation established under the Social Health Insurance Act, 2023, succeeding the former National Hospital Insurance Fund (NHIF) to administer universal social health insurance across Kenya.

Used in: Insurance Desk, Claims Engine, DHA Interoperability

SHA Claim

Category: Business / Technical

Definition: An electronic reimbursement claim record generated for SHA-insured patients, containing encounter diagnosis codes, validated procedure tariffs, itemized pharmacy claims, pre-authorization codes, and attached FHIR claim bundles.

Used in: Insurance Desk, Claims Dispatcher (`sha_claims`, `sha_claim_items`)

SHA Claim Item

Category: Business / Technical

Definition: An itemized billing line item included within an SHA insurance claim, mapped to SHA standardized tariff codes and package categories.

Used in: Claims Management (`sha_claim_items`)

SHA SHIF (Social Health Insurance Fund)

Category: Kenya-specific / Business

Definition: The primary mandatory social insurance fund under the SHA, financed via statutory payroll and voluntary contributions to provide universal outpatient, inpatient, and specialized surgical care across Kenya.

Used in: Patient Registration, Insurance Desk (`encounters.payment_mode = 'sha_shif'`)

SHR (Shared Health Record)

Category: Kenya-specific / Technical

Definition: The central national electronic clinical repository operated by the DHA where longitudinal citizen health summaries, FHIR encounter bundles, diagnostic observations, and dispensed medications are consolidated nationwide.

Used in: System Architecture, DHA Compliance, Edge Function (`fhir-bundle`)

Sign Encounter

Category: System / Permission

Definition: A privileged clinical permission (`sign_encounter`) restricted to licensed Doctors, Clinical Officers, and Dental Officers, allowing them to finalize, cryptographically lock, and legally sign clinical encounters.

Used in: `role_permissions`, Consultation Completion

SSR (Server-Side Rendering)

Category: Technical

Definition: Frontend architectural rendering mode provided by TanStack Start where initial page HTML is pre-rendered on the server to optimize loading performance, SEO, and initial render speeds.

Used in: Frontend Application Architecture

Staff

Category: System / Role

Definition: A baseline general staff role providing foundational read and operational access across general queue tracking, patient file lookup, and internal communications.

Used in: `user_roles`, Baseline System Access

Stock Item

Category: Stock / System

Definition: A registered pharmaceutical product, medical supply, laboratory reagent, or surgical consumable maintained in the inventory database with stock keeping unit (SKU) codes, unit of measure, re-order levels, buying prices, and selling prices.

Used in: Stock Module, Pharmacy (`stock_items`)

Stock Keeper

Category: System / Role

Definition: An inventory management officer responsible for receiving warehouse deliveries, conducting periodic physical stock counts, and approving inter-departmental store transfers.

Used in: Stock Management (`/stock`, `/deliveries`, `user_roles`)

Stock Location

Category: Stock / System

Definition: A distinct physical warehouse or departmental sub-inventory store within the hospital (e.g., Main Store, Central Pharmacy, ICU Store, Dialysis Store, Lab Store).

Used in: Stock Module (`stock_locations`)

Stock Movement

Category: Stock / System

Definition: An immutable inventory ledger transaction recording physical additions or subtractions of stock quantities, tagged by movement type

(`delivery`, `dispense`, `transfer_out`, `transfer_in`, `usage`, `adjustment`, `loss`).

Used in: Stock Module (`stock_movements`, `apply_stock_movement()` trigger)

Stock Transfer

Category: Stock / System

Definition: An authorized inter-store movement of inventory items from a source store (e.g., Main Store) to a destination departmental store (e.g., Pharmacy Store, ICU Store).

Used in: Stock Module (`stock_transfers`, `transfer_stock_between_locations()` RPC)

Stock Usage

Category: Stock / System

Definition: The direct logging of consumed supplies and single-use medical devices used during patient care (e.g., ICU catheters, Dialysis lines, Theatre sutures) linked to an encounter or department.

Used in: Clinical Units, Stock Management (`stock_usage`, `record_stock_usage()` RPC)

Supabase

Category: Technical

Definition: The open-source backend-as-a-service platform powering AegisCare, integrating a managed PostgreSQL relational database, Row-Level Security, Supabase Auth, realtime WebSocket feeds, and Edge Functions.

Used in: Entire Backend Infrastructure

System Admin

Category: System / Role

Definition: The highest technical infrastructure administrator persona possessing root privileges across database policies, edge function settings,

integration keys, security configurations, and user identity lifecycles.

Used in: Database Management, System Infrastructure

T

TanStack Router / TanStack Start

Category: Technical

Definition: Modern, 100% type-safe routing and full-stack React application frameworks utilized to structure AegisCare's frontend user interface, server-side data loaders, and component rendering trees.

Used in: Frontend Web Architecture (`src/routes/*`)

Tariffs

Category: Business

Definition: The standardized official pricing schedule applied to consultations, bed days, surgical operations, diagnostic investigations, and medical consumables.

Used in: Billing, Insurance Clearance (`insurance_contracted_prices`)

Triage

Category: Clinical / System

Definition: The essential nursing assessment workflow performed immediately following registration to measure vital signs, evaluate clinical acuity, calculate pediatric nutrition scores, and prioritize patients for doctor consultation.

Used in: Triage Module (`rooms.$id.tsx` , `encounters.vitals`)

Triage Nurse

Category: System / Role

Definition: A nurse stationed at the outpatient triage desk responsible for capturing blood pressure, pulse, temperature, respiratory rate, oxygen saturation, random blood glucose, weight, height, and pain scores.

Used in: Outpatient Triage (`user_roles` , `encounters`)

U

Ultrafiltration (UF) Rate / Goal

Category: Clinical / Dialysis

Definition: The prescribed target volume of excess fluid (in milliliters) to be removed from a hemodialysis patient's vascular and interstitial spaces during a dialysis run, calculated from pre-dialysis weight and dry weight.

Used in: Dialysis Unit (`dialysis_sessions.uf_goal_ml` , `dialysis_sessions.uf_rate`)

User Has Permission

Category: Technical / Security

Definition: A PostgreSQL security function (`user_has_permission(user uuid, _perm text)`) that evaluates whether an authenticated user possesses a designated permission via their assigned system roles.

Used in: RLS Security Policies, UI Access Guards

User Role

Category: Technical / Security

Definition: The database association linking a user profile in `auth.users` to one or more functional hospital roles (`app_role`), determining functional interface permissions and clinical access rights.

Used in: User Management (`user_roles` , `has_role()` RPC)

User Room Access

Category: Technical / Security

Definition: Security access mapping records explicitly assigning authorized healthcare workers to specific hospital consultation rooms, diagnostic suites, or wards.

Used in: Room Security (`user_room_access` , `can_access_room()`) RPC

V

Vitals (Vital Signs)

Category: Clinical

Definition: The objective physiological measurements of baseline body functions recorded at triage and bedside:

- Systolic & Diastolic Blood Pressure (mmHg)
- Heart / Pulse Rate (bpm)
- Respiratory Rate (breaths/min)
- Body Temperature (°C)
- Pulse Oximetry / SpO2 (%)
- Random Blood Sugar (mmol/L)
- Weight (kg) & Height (cm) → Body Mass Index (BMI)

Used in: Triage, Consultation, Inpatient MAR, ICU Flow-sheets (`encounters.vitals`)

Vercel

Category: Technical

Definition: The cloud hosting platform providing automated continuous deployment, edge delivery network, and production hosting for the AegisCare TanStack Start / React web application.

Used in: Deployment Infrastructure

W

Waive / Waiver

Category: Business

Definition: A formal administrative reduction or full cancellation of an outstanding invoice balance granted to indigent or vulnerable patients, requiring verified supervisory approval and mandatory audit justification.

Used in: Billing & Accounting (`invoices.waiver_amount` , `invoices.waiver_approved_by`)

Ward

Category: Clinical / System

Definition: A distinct inpatient residential nursing unit (e.g., Male Medical, Female Surgical, Maternity, Pediatric, ICU, HDU) containing assigned beds, dedicated nursing rosters, and daily bed rates.

Used in: Inpatient Management (`wards` , `beds` , `admissions`)

Ward Transfer

Category: Clinical / System

Definition: The clinical and operational transfer of an admitted inpatient from one hospital ward or bed to another (e.g., transferring a stabilizing patient from the ICU to the General Medical Ward).

Used in: Inpatient Module (`admissions` , `beds` , `audit_ward_transfers` trigger)

Working Diagnosis

Category: Clinical

Definition: The provisional clinical diagnostic hypothesis formulated by an attending doctor upon initial examination, serving as the clinical basis

for ordering diagnostic workups and initiating initial therapies.

Used in: Consultation Module (`encounter_diagnoses.diagnosis_type = 'working'`)

Facility Onboarding Guide

System: Aegiscare

Model: One deployment per facility (Model A)

Overview

Aegiscare is designed so each facility gets its own isolated deployment:

- Own GitHub repository (forked from master)
- Own Supabase project (own database, own auth, own storage)
- Own Vercel deployment (own URL)

No facility can see another facility's patient data.

Steps to onboard a new facility

Step 1 — Fork the repository

1. Go to `https://github.com/fmurage6331-dev/confit-core`
2. Click **Fork** → create under the facility's GitHub account
3. Clone the fork locally

Step 2 — Create a new Supabase project

1. Go to `supabase.com` → New project
2. Name it after the facility (e.g. `aegiscare-facility-name`)
3. Note the **Project URL** and **anon key** and **service role key**

Step 3 — Run migrations

In the new Supabase project's SQL Editor, run all files in `supabase/migrations/` in filename order (oldest first).

Step 4 — Deploy to Vercel

1. Go to `https://vercel.com` → New Project → Import from GitHub
2. Select the forked repository
3. Set environment variables in Vercel dashboard:
 - `VITE_SUPABASE_URL` — your Supabase project URL
 - `VITE_SUPABASE_ANON_KEY` — your Supabase anon key
4. Click **Deploy** → Vercel builds and publishes automatically
5. Note your deployment URL (e.g. `https://aegiscare-facilityname.vercel.app`)

Step 5 — Configure facility details

1. Log in to the deployed app as admin
2. Go to **Admin** → **Settings** → **Facility tab**
3. Fill in:
 - Facility name
 - KMHFL code (find at `hiskanya.org`)
 - County, address, phone, email

- SHA facility ID and provider number (when available)

4. Click **Save facility details**

Step 6 — Configure Edge Function secrets

In Supabase → Edge Functions → Secrets, set:

- `ICD_CLIENT_ID` — WHO ICD-11 API client ID (Task F)
- `ICD_CLIENT_SECRET` — WHO ICD-11 API client secret (Task F)

Step 7 — Create admin user

In Supabase → Authentication → Users → Invite user Set their role to `admin` in the `user_roles` table.

Kenya Data Residency & Cloud Migration

Current Hosting (Development Phase)

- Supabase Cloud: servers in `us-east-1` (USA) — acceptable for development
- Vercel: global CDN — acceptable for frontend

Required for ODPC + DHA Compliance (Production)

Kenya law (ODPC Data Protection Act 2019) requires health data to remain in Kenya.

Recommended Kenya-compliant hosting options:

Provider	Type	Notes
Safaricom Cloud	Kenya-based VPS	Preferred — local support
Azure East Africa	Johannesburg region	Closest Azure region
AWS af-south-1	Cape Town	Closest AWS region
Angani	Kenya-based	Local Kenyan cloud provider

Migration Steps (When Ready)

Step 1 — Self-host Supabase on Kenya server:

```
# On your Kenya cloud server
git clone https://github.com/supabase/supabase
cd supabase/docker
cp .env.example .env
# Edit .env with your settings
docker compose up -d
Step 2 - Export existing Supabase data:

Bash

# Export from current Supabase project
supabase db dump -p your-password > aegiscare-backup.sql

# Import to new Kenya-hosted Supabase
psql -h your-kenya-server -U postgres -d postgres < aegiscare-backup.sql
Step 3 - Update environment variables:

text

VITE_SUPABASE_URL=https://your-kenya-server.com
VITE_SUPABASE_ANON_KEY=your-new-anon-key
## Pushing updates to a deployed facility
```

```

When the master repo gets updates:

```bash
In the facility's forked repo
git remote add upstream https://github.com/fmurage6331-dev/confit-core
git fetch upstream
git merge upstream/main
git push origin main
</div>

<div style="page-break-before:always;">

<!-- DOCUMENT 7: Integration & Activation Manual -->

cat > docs/integration-activation-manual.md << 'EOF'
AegisCare / LabTrack - Integration Activation Manual

Version: 1.0
Date: 2026-08-05
Author: AegisCare Development Team
Status: Stubs built - awaiting credentials

Overview

This manual documents every external integration that has been
architected and stubbed in the system. When API credentials and
onboarding are complete for each service, follow the exact steps
in this manual to activate it.

Golden rule: Never activate an integration in production until
you have tested it successfully in the sandbox/test environment first.

Integration Status Dashboard

| # | Integration | Built | Credentials | Sandbox | Live |
|---|---|---|---|---|---|
| 1 | IPRS Patient Identity |  |  |  |  |
| 2 | SHA Member Registry |  |  |  |  |
| 3 | SHA Notification Number |  |  |  |  |
| 4 | SHA Claims API |  |  |  |  |
| 5 | AfyaLink HIE (FHIR) |  |  |  |  |
| 6 | Kenya HWR Practitioners |  |  |  |  |
| 7 | Africa's Talking SMS |  |  |  |  |
| 8 | M-Pesa Daraja | Planned |  |  |  |
| 9 | DHA Certification |  ready |  |  |  |

INTEGRATION 1 - IPRS Patient Identity Verification

What It Does
Verifies patient identity against the National Population Register
using their National ID or Passport number. Returns full name,
date of birth, gender, and address. Replaces manual data entry
and prevents duplicate or fraudulent registrations.

What We Have Built
- verify_patient_identity() PL/pgSQL stub RPC in Supabase
- "Verify identity & SHA status" button on registration form
- Consent OTP dialog before calling the API
- patients.identity_verified column
- patients.identity_verified_at column
- patients.identity_verified_by FK to auth.users

```

```

Credentials Needed
- DHA developer account: https://developers.dha.go.ke
- IPRS API client ID and secret
- Facility KMHFL code (already in app_settings)
- DHA Provisional Interoperability Accreditation certificate

Sandbox Environment
- URL: https://sandbox.dha.go.ke/iprs/v1/verify
- Test NID numbers provided by DHA upon sandbox access

Activation Steps

Step 1 – Get credentials
1. Visit https://developers.dha.go.ke
2. Register facility using KMHFL code
3. Request IPRS API access
4. Receive: client_id, client_secret, sandbox URL

Step 2 – Add secrets in Supabase Dashboard
Go to Edge Functions → Secrets and add:
- IPRS_CLIENT_ID
- IPRS_CLIENT_SECRET
- IPRS_API_URL = https://sandbox.dha.go.ke/iprs/v1/verify
- FACILITY_KMHFL_CODE = your KMHFL code

Step 3 – Create Edge Function
Create: supabase/functions/verify-identity/index.ts

The function must:
1. Validate caller JWT
2. Get OAuth token from DHA
3. POST to IPRS API with payload:
 { identifier, consent: "1", consent_collected_by: KMHFL_CODE }
4. If patient_id provided, update patients table with verified data
5. Return: { connected: true, found: true, iprs_data: {...} }

Step 4 – Update frontend
In src/routes/register-patient.tsx, update doVerification():
Replace supabase.rpc("verify_patient_identity") with:
supabase.functions.invoke("verify-identity", { body: {...} })

Step 5 – Deploy and test
supabase functions deploy verify-identity

INTEGRATION 2 – SHA Member Registry

What It Does
Verifies a patient's SHA membership status in real time.
Returns: member number, relationship to principal,
active/inactive status, fund type (PHF/SHIF), dependants.

What We Have Built
- patients.sha_membership_status column
- patients.sha_membership_verified_at column
- patients.sha_member_number column
- SHA status badge on patient profile, accounting, insurance desk
- Warning in InsuranceDialog if status is inactive

Credentials Needed
- SHA provider portal: https://provider.sha.go.ke
- SHA facility provider number (in app_settings)
- SHA API OAuth2 credentials

Sandbox Environment
- URL: https://api.sandbox.sha.go.ke/v1/members/verify

Activation Steps

Step 1 – Register on SHA provider portal
1. Visit https://provider.sha.go.ke
2. Request API credentials

```

3. Receive: SHA\_CLIENT\_ID, SHA\_CLIENT\_SECRET

**\*\*Step 2 – Add secrets\*\***

- SHA\_CLIENT\_ID
- SHA\_CLIENT\_SECRET
- SHA\_API\_URL = https://api.sandbox.sha.go.ke/v1
- SHA\_PROVIDER\_NO = your provider number

**\*\*Step 3 – Add SHA call inside verify-identity Edge Function\*\***

After the IPRS call, add a parallel SHA API call:

POST to SHA\_API\_URL/members/verify

Body: { id\_number, id\_type }

Returns: member\_number, status, relationship, fund\_type

Update patients table with SHA fields after successful response.

---

## INTEGRATION 3 – SHA Notification Number

### What It Does

Every SHA encounter requires a unique notification number generated by the SHA backend at visit initiation. This number is the primary reference **for** all downstream claims.

### What We Have Built

- encounters.sha\_notification\_number column
- Temporary placeholder: TEMP-SHA-{timestamp}-{encounter\_id}
- InsuranceDialog captures and saves the number
- FHIR encounter includes sha\_notification\_number as identifier

### Activation Steps

**\*\*Step 1 – Create sha-notify Edge Function\*\***

At encounter creation, call SHA eligibility API to get the real notification number.

**\*\*Step 2 – Update encounter creation flow\*\***

After creating encounter (registration or check-in), call sha-notify and save the returned notification\_number to encounters.sha\_notification\_number immediately.

**\*\*Step 3 – Background migration\*\***

After going live, run a script to swap all TEMP- prefixed notification numbers with real SHA transaction IDs **for** any encounters created during the stub period.

```sql

-- Identify encounters needing real notification numbers

SELECT id, sha_notification_number

FROM encounters

WHERE sha_notification_number LIKE 'TEMP-SHA-%'

AND payment_mode = 'insurance'

ORDER BY created_at DESC;

INTEGRATION 4 – SHA Claims API

What It Does

Submits encounter claims to SHA **for** reimbursement. The claims-dispatcher Edge Function already routes SHA claims to the

What We Have Built

claims-dispatcher Edge Function (deployed, stub mode)

ShaClaimsHandler writes to dha_outbound_queue

InsuranceDialog "Submit claim" button

FHIR R4 payload included **in** every claim submission

Claim status tracking: pending/submitted/approved/rejected/paid

Admin Claims Queue Monitor shows all pending items

Activation Steps

Step 1 – Update ShaClaimsHandler **in** claims-dispatcher File: supabase/functions/claims-dispatcher/index.ts

Replace the stub ShaClaimsHandler with live implementation:

Get OAuth token from SHA API

POST to SHA_API_URL/claims/submit Body: { notification_number, member_number, fhir_encounter, provider_number, claim_data }

Returns: claim_number, status

```

Update encounters: claim_number, claim_status='submitted', claim_submitted_at
Update dha_outbound_queue status to 'sent'
Step 2 – Deploy supabase functions deploy claims-dispatcher

Step 3 – Test flow

Register SHA patient
Complete consultation with ICD-11 diagnosis
Open SHA/Insurance Desk → Submit claim
Check Admin → Claims Queue → status = sent
Log into SHA provider portal → verify claim received
INTEGRATION 5 – AfyaLink HIE (FHIR Sync)
What It Does
Syncs clinical records to the national Health Information Exchange. Only fires when patient has consented (hie_data_shar

What We Have Built
generate_fhir_encounter(UUID) PL/pgSQL RPC (FHIR R4 compliant)
fhir-patient Edge Function (deployed)
fhir-encounter Edge Function (deployed)
fhir-condition Edge Function (deployed)
FhirSyncHandler in claims-dispatcher (stub with consent check)
patient_consent.hie_data_sharing_consented boolean
FHIR preview in InsuranceDialog and Admin Queue Monitor
What DHA Will Validate
FHIR R4 Patient resource (fhir-patient)
FHIR R4 Encounter resource (fhir-encounter) including:
ICD-11 diagnoses from encounter_diagnoses
SHA notification number as identifier
SHA fund type as serviceType
Practitioner reference (created_by user)
Facility KMHFL code as serviceProvider
FHIR R4 Condition resource (fhir-condition)
Credentials Needed
DHA AfyaLink: https://afyalink.dha.go.ke
OAuth2 client credentials
FHIR server base URL
DHA certification (Form HMIS 4)
Activation Steps
Step 1 – Apply for DHA certification

Submit Form HMIS 4 at https://certification.dha.go.ke/process
Generate sample FHIR output to attach: SELECT generate_fhir_encounter('any-encounter-uuid-here');
Validate at https://validator.fhir.org
DHA review takes 4-8 weeks
Step 2 – Receive Provisional Interoperability Accreditation Valid 6-12 months. Grants AfyaLink sandbox access.

Step 3 – Add secrets

AFYALINK_CLIENT_ID
AFYALINK_CLIENT_SECRET
AFYALINK_FHIR_BASE_URL = https://sandbox.afyalink.dha.go.ke/fhir
AFYALINK_TOKEN_URL = https://sandbox.afyalink.dha.go.ke/oauth/token
Step 4 – Update FhirSyncHandler in claims-dispatcher

Verify patient consent (already coded)
Get OAuth token from AfyaLink
Build FHIR Bundle (Patient + Encounter + Condition)
POST Bundle to AfyaLink FHIR endpoint
Record HIE response ID in dha_outbound_queue.response
Update queue status to 'sent' or 'failed'
Step 5 – Deploy and test supabase functions deploy claims-dispatcher

Test: Register patient with HIE consent → complete encounter → trigger sync → verify in AfyaLink sandbox portal.

INTEGRATION 6 – Kenya Health Worker Registry (HWR)
What It Does
Verifies healthcare worker credentials against their professional council register. Ensures only licensed practitioners

KMPDC – Kenya Medical Practitioners and Dentists Council
NCK – Nursing Council of Kenya
KMLTTB – Kenya Medical Laboratory Technologists Board
What We Have Built
verify_practitioner() PL/pgSQL stub RPC

```

"Verify with Registry" button on Account page
 profiles.council_registration_number column
 profiles.council_type column (KMPDC/NCK/KMLTTB)
 profiles.council_verified boolean
 profiles.council_full_name, council_qualification, council_status
 Credentials Needed
 Kenya HWR API: <https://hwr.health.go.ke>
 API key or OAuth credentials
 Activation Steps
 Step 1 – Get credentials Contact <https://hwr.health.go.ke> for API access. Receive: HWR_API_KEY
 Step 2 – Add secrets
 HWR_API_KEY
 HWR_API_URL = <https://hwr.health.go.ke/api/v1>
 Step 3 – Create Edge Function Create: supabase/functions/verify-practitioner/index.ts
 Call: GET HWR_API_URL/practitioners/{council_type}/{reg_number} Headers: X-API-Key, X-Facility: KMHFL_CODE Returns: full
 Update profiles table with verified data.
 Step 4 – Update account.tsx Replace supabase.rpc("verify_practitioner") with: supabase.functions.invoke("verify-practitioner")
 INTEGRATION 7 – Africa's Talking SMS (OTP Consent)
 What It Does
 Sends consent OTP codes to patients via SMS. Replaces the current on-screen OTP display (Track A) with real SMS delivery
 What We Have Built
 consent-dialog.tsx with full OTP flow (Track A working)
 consent_otp table (SHA-256 hashed, 10-min expiry)
 On-screen OTP working end-to-end in production
 Legal disclaimer text hardcoded per SHA Act 2023 S.48
 Legal SMS Text (MANDATORY – SHA Act 2023 S.48)
 AEGISCARE: Your verification code for care visit is {OTP}. Valid for 10 mins. By sharing this code, you consent to treat
 Credentials Needed
 Africa's Talking account: <https://africastalking.com>
 API key
 Sender ID "AEGISCARE" registered and CA Kenya whitelisted (Communications Authority whitelist takes 2-4 weeks)
 Activation Steps
 Step 1 – Register and get credentials
 Visit <https://africastalking.com>
 Create application → get API key
 Go to SMS → Sender IDs → Register "AEGISCARE"
 Submit CA approval with facility registration certificate
 Wait 2-4 weeks for CA whitelist approval
 Step 2 – Add secrets
 AT_API_KEY
 AT_USERNAME
 AT_SENDER_ID = AEGISCARE
 AT_ENVIRONMENT = sandbox (change to production when ready)
 Step 3 – Create Edge Function Create: supabase/functions/send-consent-otp/index.ts
 The function must:
 Generate 6-digit OTP
 SHA-256 hash it
 Update consent_otp table with hash
 POST to Africa's Talking SMS API URL: <https://api.africastalking.com/version1/messaging> Body: { username, to: phone, message }
 Update consent_otp.delivery_status
 Step 4 – Update consent-dialog.tsx Replace on-screen OTP display with Edge Function call. Do NOT show OTP on screen after
 Step 5 – Deploy and test supabase functions deploy send-consent-otp Test with a real Kenyan number in sandbox mode.
 INTEGRATION 8 – M-Pesa (Daraja API)
 What It Will Do
 STK Push for cash payments at point of care. Payment confirmation stored on encounter.
 What Needs To Be Built
 supabase/functions/mpesa-stk-push/
 supabase/functions/mpesa-callback/

```

encounters.mpesa_transaction_id column
"Pay via M-Pesa" button in accounting.tsx
Credentials Needed
Safaricom Developer: https://developer.safaricom.co.ke
Consumer Key, Consumer Secret
M-Pesa till or paybill number
Activation Steps
Step 1 – Register

Visit https://developer.safaricom.co.ke
Create app – get Consumer Key + Consumer Secret
Register M-Pesa shortcode/till number
Step 2 – SQL ALTER TABLE encounters ADD COLUMN IF NOT EXISTS mpesa_transaction_id TEXT;

Step 3 – Build Edge Functions mpesa-stk-push: Initiates payment prompt on patient phone mpesa-callback: Receives payment

Step 4 – Update accounting.tsx Add "Pay via M-Pesa" button that calls mpesa-stk-push.

INTEGRATION 9 – DHA Certification (Form HMIS 4)
What DHA Will Inspect
READY:

ICD-11 coding on all diagnoses (encounter_diagnoses table)
FHIR R4 Patient, Encounter, Condition (functions deployed)
60 audit triggers across 20 tables
Append-only audit log (RLS blocks DELETE/UPDATE)
19 role-based access control roles
Session timeout 30 minutes
Password policy (uppercase + number + special char)
Patient consent management with OTP
SHA fund type classification (PHF/SHIF/ECCIF)
SHA tariff schedule seeded (Legal Notice 146/147)
Facility identifiers in app_settings
STILL NEEDED:

IPRS biometric (Provisional cert covers this gap)
Audit log archiving plan (20-year retention)
AfyaLink sandbox connection
Certification Steps
Complete facility registration on DHA portal https://certification.dha.go.ke/process

Generate sample FHIR output to attach to application: Run in Supabase SQL Editor: SELECT generate_fhir_encounter('your-

Validate output at https://validator.fhir.org

Submit Form HMIS 4 with:

Facility KMHFL code
System documentation (see docs/ folder)
FHIR validation screenshot
Security assessment
DHA technical review (4-8 weeks)

Receive Provisional Interoperability Accreditation Valid 6-12 months Allows: AfyaLink sandbox, OTP as biometric fallback

Complete IPRS integration during provisional period

Full certification granted Unlocks: Production AfyaLink, real SHA notification numbers

Pre-Activation Checklist
Before activating ANY integration in production:

Tested fully in sandbox environment
Patient data encrypted in transit (HTTPS only)
OTP flow works end-to-end
generate_fhir_encounter() output validated at https://validator.fhir.org
audit_log recording all actions
No plain-text secrets in codebase
All secrets in Supabase Edge Function secrets only
Database backed up before go-live
Rollback procedure tested
Staff notified of new workflow changes
Go-Live Log

```

| Integration | Go-Live | Date | Activated By | Notes |
|---|----------------------------|-----------------------------------|--------------|-------|
| IPRS Identity | — | — | — | |
| SHA Registry | — | — | — | |
| SHA Notification | — | — | — | |
| SHA Claims | — | — | — | |
| AfyaLink HIE | — | — | — | |
| HWR Practitioners | — | — | — | |
| Africa's Talking | — | — | — | |
| M-Pesa | — | — | — | |
| DHA Certification | — | — | — | |
| Support Contacts | | | | |
| Service Contact URL | | | | |
| DHA | info@dha.go.ke | https://dha.go.ke | | |
| SHA Provider Portal | providers@sha.go.ke | https://provider.sha.go.ke | | |
| AfyaLink | afyalink@dha.go.ke | https://afyalink.dha.go.ke | | |
| Africa's Talking | support@africastalking.com | https://africastalking.com | | |
| Safaricom Daraja | apisupport@safaricom.co.ke | https://developer.safaricom.co.ke | | |
| KMHFL | kmhfl@health.go.ke | https://kmhfl.health.go.ke | | |
| HWR | hwr@health.go.ke | https://hwr.health.go.ke | | |
| Last updated: 2026-08-05. Update this document each time an integration is activated or credentials are obtained. EOF | | | | |

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<!-- DOCUMENT 8: File Manifest & HIE Activation -->

AegisCare HMS — Complete File Manifest, Inter-File Dependency Map & National HIE Activation Guide

Document Title: Source Code Manifest, Architectural Inter-File Dependency Matrix & DHA AfyaLink HIE / SHA Production

Document Version: 1.0.0

Target Environment: Kenya MoH Level 1-6 Facilities

Classification: Engineering Reference & Deployment Playbook

1. Executive Summary & Purpose

This document provides a comprehensive audit and technical catalog of **every source file** in the AegisCare HMS repository.

1. **The Exact Purpose of Every File** across frontend routes, shared components, business libraries, Supabase integrations, etc.
2. **Inter-File Interaction & Dependency Mapping:** Which files import, call, render, or pass state to each other.
3. **Step-by-Step National HIE & SHA Activation Manual:** The precise code modifications, environment secrets, and data flows required for activation.

2. Directory Structure & File Hierarchy

config-core/ | — package.json # Bun / Node dependencies & script runners | — vite.config.ts # Vite build bundler & TanStack Start SSR config
 | — tsconfig.json # TypeScript compiler paths & strict typing | — components.json # Shadcn UI component registry config | — wrangler.jsonc #
 Cloudflare / Edge deployment parameters | — eslint.config.js # ESLint static analysis configuration | — public/ # Static public assets (logos, icons,
 manifests) | — supabase/ | — config.toml # Supabase local environment & ports config | — migrations/ # 60+ PostgreSQL schema
 migrations & triggers | — functions/ # Deno Serverless Edge Functions | — send-sms/index.ts # Africa's Talking SMS dispatcher | —
 icd11-search/index.ts # WHO ICD-11 Cloud API OAuth2 proxy | — claims-dispatcher/index.ts # HIE & SHA Claims outbound router | — fhir-
 patient/index.ts # FHIR R4 Patient builder | — fhir-encounter/index.ts # FHIR R4 Encounter builder | — fhir-condition/index.ts # FHIR R4
 Condition builder | — fhir-bundle/index.ts # FHIR R4 Collection Bundle builder | — src/ | — start.ts # TanStack Start SSR hydration entry
 | — server.ts # Server entry handler for SSR requests | — router.tsx # TanStack Router instance & context provider | — routeTree.gen.ts # Auto-
 generated type-safe route tree | — styles.css # Global Tailwind CSS & design variables | — integrations/supabase/ | — client.ts # Browser-
 side Supabase client instance | — client.server.ts # Server-side SSR Supabase client | — auth-attacher.ts # Server request authorization
 header injector | — auth-middleware.ts # Server-side auth verification middleware | — types.ts # Auto-generated database TypeScript
 interfaces | — lib/ # Core business logic, contexts & helpers | — auth-context.tsx # Global Auth provider & user role state | — branding-
 context.tsx # Hospital facility branding & logo context | — facility-level.ts # Kenya MoH Level 1-6 feature gating logic | — insurance-calc.ts #
 Insurance coverage & co-pay calculators | — otp-service.ts # Client helper for consent OTP dispatch | — moh-reports.ts # MOH aggregate
 data compilation utilities | — require-access.tsx # Route authorization & role wrapper component | — roles.ts # System roles & permissions

constant definitions | |— server-fn-auth.ts # Server functions authorization guards | |— test-parameters.ts # Lab test parameter definitions & normal ranges | |— test-templates.ts # Lab test template master loaders | |— test-types.ts # Diagnostic laboratory TypeScript types | |— error-capture.ts # Global frontend exception telemetry | |— error-page.ts # SSR error boundary fallback renderer | |— supabase-untyped.ts # Flexible Supabase escape-hatch helper | |— utils.ts # Tailwind class merge (cn) & formatters | |— admin-users.functions.ts # Server functions for staff account approval | |— mcp/ # Model Context Protocol (MCP) server & tools | |— index.ts # MCP registry & tool handler dispatcher | |— tools/ # MCP tool implementations (get-patient, etc.) | |— hooks/ | |— use-mobile.tsx # Responsive viewport breakpoint detector | |— components/ # Reusable UI & presentation components | |— app-shell.tsx # Master hospital layout navigation & header | |— consent-dialog.tsx # ODPC OTP consent verification modal | |— parameter-table.tsx # Dynamic lab result input flow-sheet | |— print-header.tsx # Official MoH hospital letterhead for printing | |— service-picker.tsx # Procedure & service catalog selector | |— moh/ # MOH reporting presentation components | |— moh-indicator-table.tsx # MOH monthly aggregate data grid | |— moh-month-picker.tsx # Year/Month selector for statutory returns | |— moh-week-picker.tsx # Weekly epidemiological surveillance picker | |— moh-report-shell.tsx # Standard wrapper with print & export actions | |— moh-reports-grid.tsx # Visual dashboard grid for all MoH tools | |— ui/ # 40+ Shadcn UI primitives (button, dialog, etc.) | |— routes/ # 50+ Application Views & Workbenches | |— _root.tsx # Root application layout & shell wrapper | |— index.tsx # Landing redirector & role-based home router | |— login.tsx # Staff login & authentication screen | |— register-patient.tsx # Patient registration & OTP intake wizard | |— queue.tsx # Hospital-wide real-time queue manager | |— rooms.\$id.tsx # Universal Clinical Workbench (OPD/ICU/Dialysis) | |— laboratory.index.tsx # Laboratory pending orders list | |— laboratory.\$id.tsx # Lab test accessioning & result entry | |— radiology.index.tsx # Radiology requisition queue | |— radiology.\$id.tsx # Radiology scanning & diagnostic reporting | |— inpatient.tsx # Inpatient ward list & bed occupancy visualizer | |— inpatient.\$admissionId.tsx # Inpatient bedside chart & MAR flow-sheet | |— stock.tsx # Multi-location perpetual inventory ledger | |— deliveries.tsx # Supplier stock delivery receiving desk | |— accounting.tsx # Billing ledger, payments & EOD reconciliation | |— invoices.index.tsx # Master invoice directory & aging | |— invoices.\$id.tsx # Detailed invoice breakdown & receipt printer | |— appointments.tsx # Clinic appointment scheduling calendar | |— machines.tsx # Biomedical equipment registers & logs | |— dashboard.tsx # Real-time executive & clinical KPI dashboard | |— reports.tsx # Clinical & financial report generator | |— patients.index.tsx # Master Patient Index (MPI) search | |— patients.\$id.tsx # Longitudinal electronic health record (EHR) | |— encounter-records..tsx # Historic clinical encounter archives | |— records..tsx # Medical records office file tracker | |— moh.tsx # MOH Reporting Suite root shell | |— moh.index.tsx # MOH statutory returns launcher | |— moh.705.tsx # MOH 705A & 705B Outpatient Morbidity reports | |— moh.706.tsx # MOH 706 Laboratory summary report | |— moh.707.tsx # MOH 707 Inpatient bed utilization report | |— moh.717.tsx # MOH 717 Institutional workload return | |— moh.fp.tsx # Family Planning contraceptive report | |— moh.mch.tsx # Maternal & Child Health clinic report | |— moh.204.tsx # MOH 204 Outpatient register export | |— moh.505.tsx # MOH 505 Disease outbreak surveillance | |— moh.642.tsx # MOH 642 Blood transfusion safety return | |— admin.settings.tsx # Facility KMHFL, SHA ID & MoH Level settings | |— admin.users.tsx # Staff account approvals & profile editor | |— admin.permissions.tsx # Role permissions & inventory location security | |— admin.rooms.tsx # Hospital room manager & staff assignment | |— admin.wards.tsx # Inpatient wards & bed manager with tariffs | |— admin.services.tsx # Master clinical procedure price catalog | |— admin.pricing.tsx # Insurer contracted tariff rate overrides | |— admin.insurance.tsx # Insurance desk, schemes & SHA claim monitor | |— admin.audit-log.tsx # Tamper-evident forensic security log viewer | |— admin.moh-indicators.tsx # MOH disease mapping & indicator rule editor | |— admin.queue.tsx # Central queue control & bottleneck monitor | |— admin.requests.tsx # Staff access request approval workbench | |— admin.test-templates.tsx # Lab test parameter & reference range editor | |— account.tsx # Current staff profile settings | |— change-password.tsx # Password update screen | |— forgot-password.tsx # Password recovery request | |— reset-password.tsx # Password token reset confirmation | |— mcp.ts # MCP HTTP handler endpoint | |— | |— [joauth.consent.tsx # OAuth consent handler (legacy — not used in production) | |— [mcp]/invoke-tool/\$tool.ts # MCP tool invocation gateway | |— [mcp]/list-tools.ts # MCP tool discovery manifest | |— [well-known]/* # OAuth protected resource metadata

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## 3. Detailed File Catalog & Inter-File Interactions

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### 3.1 Application Bootstrap, Server & Routing Layer

#### `src/start.ts`
- **Purpose:** Client-side hydration entry point for TanStack Start SSR. Initializes the browser environment and mounts the
- **Interacts with:** Imports `createRouter` from `src/router.tsx` and passes it to `StartClient`.

#### `src/server.ts`
- **Purpose:** Server-side request handler for TanStack Start. Intercepts incoming HTTP requests, initializes SSR context,
- **Interacts with:** `src/router.tsx`, `src/integrations/supabase/auth-middleware.ts`, `src/lib/error-page.ts`.

#### `src/router.tsx`
- **Purpose:** Central routing coordinator. Constructs the type-safe TanStack Router instance, attaches QueryClient, inject
- **Interacts with:** Imports auto-generated route tree from `src/routeTree.gen.ts`, `src/integrations/supabase/client.ts`,

#### `src/routeTree.gen.ts`
- **Purpose:** Automatically generated by TanStack Router compiler. Maps filesystem route files in `src/routes/` to type-sa
- **Interacts with:** Referenced universally by `src/router.tsx` and all navigation `` calls.

```

```

#### `src/styles.css`
- **Purpose:** Core styling manifest containing Tailwind CSS directives, custom CSS variables for light/dark themes, print styles.
- **Interacts with:** Imported in `src/routes/__root.tsx`.

---

### 3.2 Global Layout & Core Navigation

#### `src/routes/__root.tsx`
- **Purpose:** Root layout component wrapping all pages in the application. Provides TanStack Query client, Toast notifications, and global context providers.
- **Interacts with:** Imports `src/components/app-shell.tsx`, `src/lib/auth-context.tsx`, `src/lib/branding-context.tsx`, `src/lib/roles.ts`.

#### `src/components/app-shell.tsx`
- **Purpose:** Master responsive application shell featuring the top facility branding banner, user profile pill, role indicator, and main content area.
- **Interacts with:** Reads `useAuth()` from `src/lib/auth-context.tsx`, reads `useBranding()` from `src/lib/branding-context.tsx`.

#### `src/routes/index.tsx`
- **Purpose:** Root landing route (`/`). Evaluates the authenticated user's assigned role and redirects them immediately to the appropriate dashboard.
- **Interacts with:** `src/lib/auth-context.tsx`, `src/lib/roles.ts`.

#### `src/routes/login.tsx`
- **Purpose:** Front-facing staff authentication portal. Handles email/password authentication against Supabase Auth, evaluates session, and redirects to the dashboard.
- **Interacts with:** `src/integrations/supabase/client.ts`, `src/lib/auth-context.tsx`.

---

### 3.3 Patient Intake, Consent & Health Records

#### `src/routes/register-patient.tsx`
- **Purpose:** Outpatient registration intake wizard. Searches for existing records, captures citizen identification, manages consent, and creates a new patient record.
- **Interacts with:** Calls `supabase.from('patients').insert()`, `supabase.from('encounters').insert()`, triggers `src/components/consent-dialog.tsx`.

#### `src/components/consent-dialog.tsx`
- **Purpose:** Reusable ODPC-compliant digital consent dialog. Triggers SMS OTP generation via Edge Function `send-sms` and manages consent state.
- **Interacts with:** `src/integrations/supabase/client.ts`, `src/lib/otp-service.ts`, Supabase Edge Function `send-sms`.

#### `src/routes/patients.index.tsx` & `src/routes/patients.$id.tsx`
- **Purpose:** Master Patient Index search directory (`.index.tsx`) and longitudinal electronic medical record viewer (`.id.tsx`).
- **Interacts with:** Queries `patients`, `encounters`, `clinical_notes`, `admissions`, `invoices`, `lab_orders`, `prescriptions`.

#### `src/routes/records.index.tsx`, `records.$id.tsx`, `records.new.tsx`
- **Purpose:** Health Information Management (HIM) department file management suite for tracking physical paper file movement and digital record linking.
- **Interacts with:** `patients`, `encounters`, `user_roles`.

#### `src/routes/encounter-records.index.tsx` & `encounter-records.$id.tsx`
- **Purpose:** Dedicated read-only archive for searching and viewing completed and signed clinical encounters.
- **Interacts with:** `encounters`, `encounter_diagnoses`, `clinical_notes`.

---

### 3.4 Queue Management & Clinical Workbenches

#### `src/routes/queue.tsx`
- **Purpose:** Hospital-wide real-time queue overview displaying live patient counts across Triage, Consultation Rooms, Lab, and Imaging.
- **Interacts with:** Subscribes to Supabase Realtime WebSocket events on `encounters` table; links directly to individual patient records.

#### `src/routes/rooms.$id.tsx`
- **Purpose:** The Universal Clinical Workbench (Core Engine). Dynamically adapts its interface based on room kind:
  - **Consultation Room:** Chief complaints, physical exam, WHO ICD-11 diagnosis search (`icd11-search`), lab/imaging order entry.
  - **Triage Room:** Vital signs capture (BP, Pulse, Temp, RR, SpO2, RBS), BMI calculation, pediatric MUAC nutrition screen.
  - **ICU Room:** Hourly flow-sheets (`icu_hourly_charts`), ventilator monitoring, GCS/RASS scoring, bedside ICU store drug administration.
  - **Dialysis Room:** Hemodialysis session logging (`dialysis_sessions`), dialyzer parameters, ultrafiltration fluid metrics.
  - **Pharmacy Room:** Outpatient prescription dispensing and batch inventory deductions.
  - **Mortuary Room:** Deceased intake, refrigeration slot allocation, and body release clearances.
- **Interacts with:**
  - Subscribes to Realtime queue updates for `room_id`.
  - Calls Edge Function `icd11-search` for live WHO ICD-11 coding.
  - Calls Edge Function `send-sms` for patient alerts.
  - Executes RPCs: `charge_icu_admission_fee`, `process_dialysis_session_billing`, `log_break_glass_access`, `send_lab_results`.
  - Reads `src/lib/auth-context.tsx`, `src/lib/facility-level.ts`.

#### `src/routes/inpatient.tsx` & `src/routes/inpatient.$admissionId.tsx`
- **Purpose:** Inpatient management interface for tracking admissions, care plans, and room assignments.

```

```

- `inpatient.tsx`: Visual bed occupancy map across all wards showing real-time bed statuses (`available`, `occupied`, `cl
- `inpatient_.$admissionId.tsx`: Active inpatient chart containing Medication Administration Records (MAR), nursing vital
- **Interacts with:** Queries `admissions`, `wards`, `beds`, `medication_administrations`, `clinical_notes`, `invoices`. En
---

### 3.5 Diagnostic Suites (Laboratory & Radiology)

#### `src/routes/laboratory.index.tsx` & `src/routes/laboratory.$id.tsx`
- **Purpose:** Laboratory accessioning queue (`.index.tsx`) and result entry flow-sheet (`. $id.tsx`). Validates qualitative
- **Interacts with:** Queries `lab_orders`, `lab_results`, `lab_test_catalog`. Calls `send_lab_results_to_requesting_room`

#### `src/routes/radiology.index.tsx` & `src/routes/radiology.$id.tsx`
- **Purpose:** Diagnostic imaging queue (`.index.tsx`) and radiologist report authoring suite (`. $id.tsx`). Captures organ
- **Interacts with:** Queries `radiology_orders`, `radiology_results`. Calls `send_radiology_results_to_requesting_room` RP
---

### 3.6 Inventory, Pharmacy & Logistics

#### `src/routes/stock.tsx`
- **Purpose:** Multi-store perpetual inventory workbench. Displays warehouse stock balances, reorder level alerts, stock va
- **Interacts with:** Queries `stock_items`, `stock_locations`, `stock_movements`, `stock_transfers`. Executes `transfer_st

#### `src/routes/deliveries.tsx`
- **Purpose:** Commercial supplier receiving dock. Logs supplier deliveries, purchase invoices, LPO references, batch numbe
- **Interacts with:** Queries `stock_deliveries`, `stock_items`. Executes `delivery_to_stock()` database trigger.
---

### 3.7 Billing, Invoicing & Financial Accounting

#### `src/routes/accounting.tsx`
- **Purpose:** Cashier payment processing desk, multi-channel payment recording (Cash, M-Pesa, Card, Bank, Insurance), pati
- **Interacts with:** Queries `invoices`, `invoice_payments`, `invoice_line_items`. Recalculates balances via `recalc_invoi

#### `src/routes/invoices.index.tsx` & `src/routes/invoices.$id.tsx`
- **Purpose:** Master fiscal invoice directory (`.index.tsx`) and detailed itemized bill viewer / receipt printer (`. $id.ts
- **Interacts with:** Queries `invoices`, `invoice_line_items`, `invoice_payments`, `patients`. Uses `src/components/print-
---

### 3.8 Appointments, Machines & Operations

#### `src/routes/appointments.tsx`
- **Purpose:** Outpatient appointment calendar and scheduling interface. Manages clinic capacities, doctor schedules, and c
- **Interacts with:** Queries `appointments`, `rooms`, `patients`. Executes `create_encounter_from_appointment` RPC.

#### `src/routes/machines.tsx`
- **Purpose:** Biomedical device inventory and preventative maintenance log (analyzers, X-ray machines, ventilators, dialys
- **Interacts with:** Queries `machines`, `machine_logs`.
---

### 3.9 Executive Dashboard & MOH Reporting Suite

#### `src/routes/dashboard.tsx`
- **Purpose:** Executive real-time situational dashboard displaying clinical attendance, bed occupancy rates, revenue metri
- **Interacts with:** Executes RPCs `dashboard_top_diseases`, `dashboard_admitted_opd_trend`, `dashboard_emergency_referral

#### `src/routes/reports.tsx`
- **Purpose:** Central clinical and financial report generator supporting CSV/PDF data exports.
- **Interacts with:** Queries `invoices`, `encounters`, `stock_movements`.

#### `src/routes/moh.tsx`, `moh.index.tsx` & Statutory MoH Routes
- **`src/routes/moh.705.tsx`:** Form MOH 705A (Under 5) and MOH 705B (Over 5) Outpatient Morbidity reports. Executes `get_m
- **`src/routes/moh.706.tsx`:** Form MOH 706 Laboratory summary report.
- **`src/routes/moh.707.tsx`:** Form MOH 707 Inpatient bed utilization and mortality report.
- **`src/routes/moh.717.tsx`:** Form MOH 717 Institutional hospital workload return.
- **`src/routes/moh.fp.tsx`:** Family Planning contraceptive commodities report.
- **`src/routes/moh.mch.tsx`:** Maternal and Child Health immunization and antenatal care return.
- **`src/routes/moh.204.tsx`:** MOH 204 Outpatient register line-list export.
- **`src/routes/moh.505.tsx`:** Weekly disease outbreak epidemiological surveillance.

```

```

- **src/routes/moh.642.tsx:** Blood transfusion safety summary.
- **Interacts with:** Uses `src/components/moh/*`, queries `moh_monthly_aggregates`, `encounter_indicator_tags`.

---

### 3.10 System Administration & Security Governance

#### `src/routes/admin.settings.tsx`
- **Purpose:** Hospital metadata configurator: Facility Name, KMHFL Code, SHA Facility ID, MoH Level (1-6), County, and con
- **Interacts with:** Updates `app_settings` table.

#### `src/routes/admin.users.tsx` & `admin.permissions.tsx`
- **Purpose:** Staff user lifecycle manager: Approves new accounts (`is_approved`), assigns system roles (`user_roles`), co
- **Interacts with:** `profiles`, `user_roles`, `role_permissions`, `user_stock_location_access`. Executes `grant_stock_loc

#### `src/routes/admin.rooms.tsx` & `admin.wards.tsx`
- **Purpose:** Roster manager for physical consultation rooms, diagnostic suites, wards, and beds with associated base bill
- **Interacts with:** `rooms`, `wards`, `beds`, `user_room_access`.

#### `src/routes/admin.services.tsx` & `admin.pricing.tsx`
- **Purpose:** Master price catalog manager (`services.tsx`) and insurer contracted rate schedule overrides (`pricing.tsx`)
- **Interacts with:** `lab_test_catalog`, `stock_items`, `contracted_prices`.

#### `src/routes/admin.insurance.tsx`
- **Purpose:** Insurance desk controller: Manages insurance underwriters, benefit packages, and monitors SHA outbound claim
- **Interacts with:** `insurance_providers`, `insurance_benefit_plans`, `sha_claims`, `dha_outbound_queue`.

#### `src/routes/admin.audit-log.tsx`
- **Purpose:** Forensic audit log viewer displaying immutable event streams (`INSERT`, `UPDATE`, `DELETE`, `BREAK_GLASS`).
- **Interacts with:** Queries `audit_log`, `audit_log_archive`, `audit_archive_runs`.

---

### 3.11 Backend Supabase Edge Functions

#### `supabase/functions/send-sms/index.ts`
- **Purpose:** Dispatches SMS messages via Africa's Talking API for OTP consent tokens and lab notifications.
- **Called by:** `src/components/consent-dialog.tsx`, `src/routes/rooms.$id.tsx`, `src/routes/laboratory.$id.tsx`.

#### `supabase/functions/icd11-search/index.ts`
- **Purpose:** Authenticates against WHO Cloud OAuth2 and executes live flexisearch against ICD-11 MMS 2024 linearization.
- **Called by:** `src/routes/rooms.$id.tsx` (Diagnosis search bar).

#### `supabase/functions/claims-dispatcher/index.ts`
- **Purpose:** Evaluates concluded encounters, verifies HIE consent, compiles FHIR bundles, and routes payloads to `dha_out
- **Called by:** `src/routes/rooms.$id.tsx` (Encounter sign action) and `src/routes/admin.insurance.tsx`.

#### `supabase/functions/fhir-patient/index.ts`
- **Purpose:** Returns a standardized HL7 FHIR R4 `Patient` resource JSON for a given `patient_id`.

#### `supabase/functions/fhir-encounter/index.ts`
- **Purpose:** Returns a standardized HL7 FHIR R4 `Encounter` resource JSON for a given `encounter_id`.

#### `supabase/functions/fhir-condition/index.ts`
- **Purpose:** Returns an array of HL7 FHIR R4 `Condition` resources mapped to ICD-11 concept URIs.

#### `supabase/functions/fhir-bundle/index.ts`
- **Purpose:** Compiles a full HL7 FHIR R4 `Bundle` (type = `collection`) consolidating Patient, Encounter, EpisodeOfCare,

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## 4. Inter-File Dependency Matrix

Source File	Imports From / Calls	Invoked / Rendered By	Key DB Tables / RPCs / APIs
`src/routes/__root.tsx`	`app-shell.tsx`, `auth-context.tsx`, `branding-context.tsx`	TanStack Router Root	None (Layo
`src/components/app-shell.tsx`	`auth-context.tsx`, `facility-level.ts`	`__root.tsx`	`app_settings`, `profiles`
`src/routes/register-patient.tsx`	`consent-dialog.tsx`, `facility-level.ts`, `client.ts`	Router (`/register-patient`)	
`src/components/consent-dialog.tsx`	`otp-service.ts`, `client.ts`, `send-sms` Edge Fn	`register-patient.tsx`	`conse
`src/routes/rooms.$id.tsx`	`client.ts`, `auth-context.tsx`, `parameter-table.tsx`	Router (`/rooms/$id`)	`encounters
`src/routes/laboratory.$id.tsx`	`parameter-table.tsx`, `client.ts`	Router (`/laboratory/$id`)	`lab_orders`, `lab_re
`src/routes/radiology.$id.tsx`	`client.ts`, `auth-context.tsx`	Router (`/radiology/$id`)	`radiology_orders`, `radio
`src/routes/inpatient.$admissionId.tsx`	`client.ts`, `print-header.tsx`	Router (`/inpatient/$admissionId`)	`admiss

```

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| `src/routes/stock.tsx` | `client.ts`, `auth-context.tsx` | Router (`/stock`) | `stock_items`, `stock_locations`, `stock_m`
| `src/routes/accounting.tsx` | `client.ts`, `print-header.tsx` | Router (`/accounting`) | `invoices`, `invoice_line_items`
| `src/routes/moh.705.tsx` | `moh-indicator-table.tsx`, `moh-report-shell.tsx` | Router (`/moh/705`) | `moh_705_disease_map`
| `supabase/functions/claims-dispatcher/index.ts` | Supabase JS Client | Client HTTP POST | `dha_outbound_queue`, `sha_clai`
| `supabase/functions/fhir-bundle/index.ts` | Supabase JS Client | Client HTTP POST | `patients`, `encounters`, `encounter_`
```

```
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```

```
## 5. National HIE, DHA & SHA Production Activation Manual
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When official API credentials, certificates, and endpoints are granted by the Digital Health Authority (DHA) and Soci
```

HIE / SHA PRODUCTION ACTIVATION ROADMAP |

Step 1: Credentials | Step 2: Supabase Secrets | Step 3: Code Updates | | Obtain DHA & SHA OAuth2 | Configure Edge Secrets | Activate Live Handlers in | | Client ID & Secrets | in Supabase Dashboard | claims-dispatcher & Edge Functions |

Step 4: Database Verify | Step 5: Sandbox Testing | Step 6: Production Cutover | | Validate Facility KMHFL | Execute End-to-End Test | Switch Endpoint URLs to Production | | & SHA Provider Numbers | Encounter in Sandbox | Gateway & Monitor Queue Logs |

```
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```

Step 1: Obtain Official Regulatory Credentials

Register the facility portal on the national regulatory developer gateways:

- DHA AfyaLink HIE Gateway:** [https://developers.dha.go.ke] (https://developers.dha.go.ke)
 - Submit Form HMIS 4 (Facility Interoperability Accreditation).
 - Obtain: `AFYALINK\_CLIENT\_ID`, `AFYALINK\_CLIENT\_SECRET`, `AFYALINK\_FHIR\_BASE\_URL`.
- SHA Claims Gateway:** [https://providers.sha.go.ke] (https://providers.sha.go.ke)
 - Register facility KMHFL code.
 - Obtain: `SHA\_API\_CLIENT\_ID`, `SHA\_API\_CLIENT\_SECRET`, `SHA\_API\_BASE\_URL`.
- IPRS Identity Proxy:** [https://dha.go.ke/iprs] (https://dha.go.ke/iprs)
 - Obtain: `IPRS\_CLIENT\_ID`, `IPRS\_CLIENT\_SECRET`, `IPRS\_API\_URL`.

```
---
```

Step 2: Configure Supabase Edge Function Secrets

In the Supabase Dashboard, navigate to **Edge Functions → Secrets** (or execute via Supabase CLI):

```
```bash
```

```
1. DHA AfyaLink HIE Secrets
```

```
supabase secrets set AFYALINK_CLIENT_ID="your_afyalink_client_id_here"
supabase secrets set AFYALINK_CLIENT_SECRET="your_afyalink_client_secret_here"
supabase secrets set AFYALINK_TOKEN_URL="https://afyalink.dha.go.ke/oauth/token"
supabase secrets set AFYALINK_FHIR_BASE_URL="https://afyalink.dha.go.ke/fhir/r4"
```

```
2. SHA Claims API Secrets
```

```
supabase secrets set SHA_API_CLIENT_ID="your_sha_client_id_here"
supabase secrets set SHA_API_CLIENT_SECRET="your_sha_client_secret_here"
supabase secrets set SHA_API_BASE_URL="https://api.sha.go.ke/v1"
```

```
3. IPRS Identity Verification Secrets
```

```
supabase secrets set IPRS_CLIENT_ID="your_iprs_client_id_here"
supabase secrets set IPRS_CLIENT_SECRET="your_iprs_client_secret_here"
supabase secrets set IPRS_API_URL="https://api.dha.go.ke/iprs/v1/verify"
```

```
4. Africa's Talking SMS Secrets (Ensure production credentials)
```

```
supabase secrets set AT_USERNAME="your_at_production_username"
supabase secrets set AT_API_KEY="your_at_production_api_key"
```

## Step 3: Activate Live Handlers in `claims-dispatcher` Edge Function

Open file `supabase/functions/claims-dispatcher/index.ts` :

**A. Activate Live DHA AfyaLink `FhirSyncHandler` :**

Replace the local stub block in `FhirSyncHandler` with the live HTTP submission call:

```
// Replace lines in FhirSyncHandler:
async function FhirSyncHandler(supabase, req, payload) {
 // 1. Verify consent (Already enforced)
 const { data: consent } = await supabase
 .from("patient_consents")
 .select("hie_data_sharing_consented")
 .eq("patient_id", req.patient_id)
 .eq("consented", true)
 .maybeSingle();

 if (!consent?.hie_data_sharing_consented) {
 return { status: "skipped", message: "Patient has not consented to HIE sharing." };
 }

 // 2. Fetch compiled FHIR R4 Bundle from fhir-bundle function
 const bundleResponse = await fetch(`${Deno.env.get("SUPABASE_URL")}/functions/v1/fhir-bundle`, {
 method: "POST",
 headers: {
 "Authorization": `Bearer ${Deno.env.get("SUPABASE_SERVICE_ROLE_KEY")}`,
 "Content-Type": "application/json"
 },
 body: JSON.stringify({ encounter_id: req.encounter_id })
 });
 const fhirBundle = await bundleResponse.json();

 // 3. Acquire OAuth2 Token from DHA AfyaLink
 const tokenResp = await fetch(Deno.env.get("AFYALINK_TOKEN_URL")!, {
 method: "POST",
 headers: { "Content-Type": "application/x-www-form-urlencoded" },
 body: new URLSearchParams({
 grant_type: "client_credentials",
 client_id: Deno.env.get("AFYALINK_CLIENT_ID")!,
 client_secret: Deno.env.get("AFYALINK_CLIENT_SECRET")!,
 scope: "system/Bundle.write"
 })
 });
 const { access_token } = await tokenResp.json();

 // 4. POST FHIR Bundle to DHA AfyaLink Server
 const hieResp = await fetch(`${Deno.env.get("AFYALINK_FHIR_BASE_URL")}/Bundle`, {
 method: "POST",
 headers: {
 "Authorization": `Bearer ${access_token}`,
 "Content-Type": "application/fhir+json"
 },
 body: JSON.stringify(fhirBundle)
 });
 const hieResult = await hieResp.json();

 // 5. Update dha_outbound_queue with mediator transaction reference
 await supabase.from("dha_outbound_queue").insert({
 encounter_id: req.encounter_id,
 patient_id: req.patient_id,
 queue_type: "fhir_sync",
 payload: fhirBundle,
 status: hieResp.ok ? "completed" : "failed",
 mediator_id: hieResult.id || fhirBundle.id,
 error_message: hieResp.ok ? null : JSON.stringify(hieResult)
 });

 return { status: "queued", handler: "FhirSyncHandler", message: "Synchronized with DHA AfyaLink." };
}
```

**B. Activate Live SHA `ShaClaimsHandler` :**

Replace the stub in `ShaClaimsHandler` with the live SHA API claim submission endpoint:

```
// Replace lines in ShaClaimsHandler:
async function ShaClaimsHandler(supabase, req, payload) {
 // 1. Resolve compiled FHIR Claim bundle
 const { data: claim } = await supabase
 .from("sha_claims")
 .select("id, fhir_bundle")
 .eq("encounter_id", req.encounter_id)
 .single();

 // 2. Transmit to SHA Claims Gateway
 const shaResp = await fetch(`${Deno.env.get("SHA_API_BASE_URL")}/claims/submit`, {
 method: "POST",
 headers: {
 "Authorization": `Bearer ${Deno.env.get("SHA_API_CLIENT_SECRET")}`,
 "Content-Type": "application/json"
 },
 body: JSON.stringify(claim.fhir_bundle)
 });
 const shaResult = await shaResp.json();

 // 3. Update sha_claims record status
 await supabase.from("sha_claims").update({
 status: shaResp.ok ? "submitted" : "rejected",
 submitted_at: new Date().toISOString()
 }).eq("id", claim.id);

 return { status: "queued", handler: "ShaClaimsHandler", message: "SHA Claim submitted successfully." };
}
```

## Step 4: Re-Deploy Supabase Edge Functions

Execute deployment via Bun / Supabase CLI:

```
supabase functions deploy send-sms
supabase functions deploy icd11-search
supabase functions deploy claims-dispatcher
supabase functions deploy fhir-patient
supabase functions deploy fhir-encounter
supabase functions deploy fhir-condition
supabase functions deploy fhir-bundle
```

## Step 5: Verify Facility Metadata in `app_settings`

In the AegisCare Admin interface (**Admin** → **Settings** → **Facility Tab**), verify that:

1. `facility_name` matches the official gazetted facility name.
2. `facility_kmhfl_code` contains the verified 5-digit KMHFL code (e.g., `12345`).
3. `facility_sha_id` contains the official SHA provider registration number.
4. `facility_level` matches the accredited MoH Level (1–6).

## Step 6: End-to-End Verification & Monitoring

1. **Register a Test Patient:** Complete registration with OTP consent verification.
2. **Conduct Clinical Encounter:** Document consultation with an ICD-11 primary diagnosis and order a lab test.
3. **Sign Encounter:** Click **Sign Encounter** as a Doctor/Clinical Officer.
4. **Inspect Outbound Queue:** Navigate to **Admin** → **Insurance** → **Claims Queue Monitor** ( `/admin/insurance` ) and confirm:
  - `queue_type = 'fhir_sync'` displays `status = 'completed'` with a valid DHA `mediator_id`.

- `queue_type = 'sha_claim'` displays `status = 'submitted'`.

5. **Log into DHA / SHA Portals:** Verify that the test encounter bundle appears in the DHA AfyaLink sandbox browser and the SHA provider claims reconciliation ledger.

---



# MOH Reporting: Tagging Architecture

## Overview

The MOH reporting system uses a two-layer tagging architecture to automatically generate indicator codes when encounters are created or updated.

## The Two Layers

### 1. Demographics Tagging — `tag_encounter_demographics()`

**Purpose:** Universal demographic indicators used across ALL MOH tools.

**Current indicators:**

Code	Criteria
<code>OPD_UNDER5_M</code>	Under-5 male
<code>OPD_UNDER5_F</code>	Under-5 female
<code>OPD_OVER5_M</code>	Age 5-59 male
<code>OPD_OVER5_F</code>	Age 5-59 female
<code>OPD_OVER60</code>	Age 60+ (any gender)
<code>OPD_NEW_M</code>	New patient (first-ever encounter) male
<code>OPD_NEW_F</code>	New patient (first-ever encounter) female
<code>OPD_REVISIT_M</code>	Revisit patient male
<code>OPD_REVISIT_F</code>	Revisit patient female

#### Why hardcoded?

- These demographics are standard across MOH 705, 717, and future tools
- Adding new demographic codes only requires editing one function
- Avoids maintaining definition rows for universal concepts
- Ensures consistency — every encounter gets tagged with the correct demographic

#### Implementation:

- Bound to `trg_tag_demographics` trigger on `encounters` table (INSERT + UPDATE)
- Reads `patients.dob_known`, `patients.date_of_birth`, `patients.estimated_age`, `patients.sex`
- Uses `ON CONFLICT (encounter_id, indicator_code) DO NOTHING` for idempotency

### 2. Tool-Specific Tagging — `process_encounter_indicators()`

**Purpose:** Indicators that are specific to individual MOH tools.

**Current indicators:**

Code	Tool	Criteria Type
<code>LAB_HIV</code>	MOH 706	Lab test (HIV)
<code>LAB_MALARIA</code>	MOH 706	Lab test (Malaria)
<code>FP_IMPLANT</code>	MOH FP	Family planning (implant)

<code>FP_PILLS</code>	MOH FP	Family planning (pills)
<code>FP_POP</code>	MOH FP	Progestin Pills
<code>FP_ECP</code>	MOH FP	Emergency Contraception
<code>FP_INJECTABLE</code>	MOH FP	Injectable Contraceptives
<code>FP_IUCD</code>	MOH FP	IUCD
<code>FP_CONDOMS</code>	MOH FP	Condoms
<code>PHARM_ANTIBIOTIC</code>	MOH PHARM	Drug class (antibiotic)

#### Supported criteria types:

- `lab_test` — Checks `encounters.tests` JSONB against `lab_test_catalog`
- `fp_method` — Checks prescriptions joined with `stock_items.category`
- `drug_class` — Checks prescriptions for drug classes
- `age_range` — Not implemented (demographics handled by layer 1)

#### Implementation:

- Bound to trigger `trg_process_moh_indicators` on `encounters` table (INSERT + UPDATE OF tests)
- Loops through `moh_indicator_definitions` table
- Requires corresponding definition row to generate tags

## Decision: Demographics Stay Hardcoded

**Date:** 22 Jul 2026

#### Rationale:

1. Demographics are universal — used by MOH 705, 717, and future tools
2. `process_encounter_indicators()` only implements `lab_test` and `fp_method` criteria; `age_range` would be redundant
3. Easier to maintain — add a new demographic code by editing one function
4. Avoids proliferation of definition rows for concepts that never change

## Adding New MOH Tools

#### For demographics:

Edit `tag_encounter_demographics()` in the database.

#### For tool-specific indicators:

1. Add item to `lab_test_catalog` (for lab tests) or `stock_items` (for products)
2. Add a row to `moh_indicator_definitions`
3. OR use the UI: When adding a service or stock item, expand "MOH Indicator" section

## Tables

Table	Purpose
<code>encounter_indicator_tags</code>	Individual tags per encounter
<code>moh_indicator_definitions</code>	Definition of tool-specific indicators
<code>moh_monthly_aggregates</code>	Monthly aggregated counts
<code>lab_test_catalog</code>	Lab test services with categories

<code>stock_items</code>	Pharmaceutical/products with categories
<code>room_indicator_map</code>	Room to indicator mappings

## Refresh Function

---

Run `SELECT refresh_moh_aggregates('YYYY-MM-DD'::date)` to recompute monthly aggregates from all tags.

---

## AegisCare HMS — DHA Compliance Assessment

System	AegisCare HMS (LabTrack v5.5 codebase)
Repository	<code>fmurage6331-dev/confit-core</code>
Branch assessed	<code>arena/01a037b8-confit-core</code>
Date	2026-08-25
Document type	DHA certification readiness / compliance gap assessment
Related docs	<code>docs/certification/DOC-1...8</code> , <code>docs/certification/DOC-7-self-attestation-gap-analysis.md</code> , <code>docs/dha-compliance-task-*.md</code>

**Scope.** This assessment reviews the current AegisCare HMS implementation against Kenya's Digital Health Act, 2023; the Digital Health (Data Exchange Component) Regulations, 2025; the Digital Health (Health Information Management Procedures) Regulations, 2025; the Kenya eClaims FHIR Implementation Guide; the Kenya Health Information Systems Interoperability Framework (KHISIF); the Kenya Data Protection Act, 2019; the Social Health Insurance Act, 2023; and the DHA AfyaLink HIE / SHA claims requirements.

This is NOT a formal DHA certificate. It is a developer gap assessment intended to be used as the basis for a DHA Certification self-attestation (Form HMIS 4 pathway) and for remediation planning.

### Executive Summary

AegisCare HMS has a **substantial, well-structured local clinical and SHA claims layer**: FHIR R4 Encounter/Patient/Condition builders, ICD-11 diagnosis sync, SHA claims tables and auto-generation, FHIR Claim generation, dha\_outbound\_queue dispatch, SHA claim status history, state-machine UI, PHF marker, claims aging view, KMHFL auto-sync, DSAR export, audit triggers, and RLS-based role protection.

However, the system is **not yet DHA-certified or live SHA-submitting**. The major blockers are:

1. **Missing external credentials** — SHA/DHA API keys, JWT/OAuth credentials, AfyaLink test and production onboarding.
2. **Fragile SHA Claim FHIR conformance** — `build_fhir_claim()` now emits correct `sha_claim_items` columns, but the output has **not** been validated against the Kenya eClaims FHIR implementation guide, and no full SHA Claim **Bundle** (Organization + Patient + Coverage + Claim) is assembled for submission.
3. **PHC / PHF zero-total rule not enforced in data** — AegisCare stamps a PHF no-copay note, but a live AfyaLink rule requires PHC claims to have a **zero total amount** and requires `servedPeriod` on claim items. This is a real submit-time rejection risk.
4. **ODPC / DHA administrative readiness** — ODPC registration, DPIA submission, DHA certification application and facility onboarding are not complete.
5. **Security hardening** — MFA, penetration testing, and some access-control baselines are not yet demonstrated.

**Overall readiness: PARTIAL** — not ready for DHA certification submission or live SHA claims until the gaps below are closed.

### Legal & Regulatory Framework

#### Requirement: Digital Health Act, 2023 (No. 15 of 2023)

**Source:** <https://www.health.go.ke/sites/default/files/Digital%20Health%20Bill%20Final.pdf> **Status:** PARTIAL **Implementation:**

AegisCare implements a digital health record with structured clinical data, audit trails, role-based access control, consent records, and FHIR R4 output intended for DHA HIE. **Gap:** No DHA certificate of compliance (valid 2 years per the Health Information Management Procedures Regulations), no ODPC-certified controller/processor evidence submitted, no DHA registration of health data controller notification. **Remediation:**

Complete DHA certification application; register with ODPC; notify DHA of the controller/processor registration; retain and submit evidence of the DPIA and policies.

### Requirement: Digital Health (Data Exchange Component) Regulations, 2025 (Legal Notice 77)

**Source:** <https://new.kenyalaw.org/akn/ke/act/ln/2025/77/eng@2025-04-11> **Status:** ● **GAP Implementation:** Codebase has FHIR edge functions and a `dha_outbound_queue` intended for ESB traffic; app\_settings includes facility identifiers and KMHFL code; KMHFL auto-sync builds facility registry data. **Gap:** Not onboarded onto the Enterprise Service Bus (ESB); no application for onboarding with proof of ODPC registration, DPIA, certification, and onboarding fee; no certificate of compliance. **Remediation:** Submit onboarding application to the Agency (regulation 6); provide DPIA, ODPC certificate, developer/facility particulars, and certificate of compliance; pay applicable fee; obtain ESB user licence.

### Requirement: Digital Health (Health Information Management Procedures) Regulations, 2025 (Legal Notice 76)

**Source:** <https://new.kenyalaw.org/akn/ke/act/ln/2025/76/eng@2025-04-11> **Status:** PARTIAL **Implementation:** Audit triggers on `sha_claims`, `sha_claim_items`, `sha_claim_status_history` and many other tables; audit log UI; consent OTP flow. **Gap:** Audit-log retention policy for the required 20 years is not yet implemented as a documented, enforceable process; no migration of data to national/county health data banks; no archival/death workflow. **Remediation:** Define and document 20-year audit retention; implement secure archival; prepare national/county health data bank transfer mapping; document legacy data migration plan.

### Requirement: Kenya Health Information Systems Interoperability Framework (KHISIF)

**Source:**

[https://www.data4sdgs.org/sites/default/files/services\\_files/Kenya%20Health%20Information%20Systems%20Interoperability%20Frame](https://www.data4sdgs.org/sites/default/files/services_files/Kenya%20Health%20Information%20Systems%20Interoperability%20Frame)

**Status:** PARTIAL **Implementation:** AegisCare uses ICD-11/LOINC/site-specific codes, FHIR R4, standard identifier systems, and interoperable JSON payloads. Existing MOH 204/505/642/705/706/707/717 reports follow MOH tagging architecture. **Gap:** Not demonstrated against the KHISIF common service layer / HIS certification framework; health facility data exchange with KHIS/KHISIF is not implemented. **Remediation:** Perform KHISIF compliance mapping; document interoperability diagrams; integrate with the national HIS certification pathway.

## DHA AfyaLink HIE Compliance

### Requirement: FHIR R4 conformance (Bundle, Encounter, Patient, Condition, Claim, Coverage, Organization)

**Source:** <https://afyalink.dha.go.ke/apidocs/claim-bundle>, <https://igeclaims.intellisoftkenya.com/> **Status:** PARTIAL **Implementation:**

- `generate_fhir_encounter(p_encounter_id UUID)` produces FHIR R4 Encounter with ICD-11 diagnoses.
- Edge functions `fhir-patient`, `fhir-encounter`, `fhir-condition` produce FHIR R4 resources.
- `generate_fhir_condition()` and `generate_fhir_encounter()` are used by `dha_outbound_queue`.
- `build_fhir_claim()` emits a FHIR R4 `Claim` with corrected `sha_claim_items` columns (`intervention_code`, `description`, `amount`), `is_included = true` filter, and `preauth_id` reference.
- `auto_build_fhir_claim()` trigger writes `fhir_bundle` on `sha_claims` insert.

**Gap:**

- No validated DHA/Kenya eClaims profile conformance report (FHIR validator output).
- No submission `Bundle` (`type=message`) that includes Organization + Patient + Coverage + Claim with matching `fullUrl` and `identifier` \$.
- `servedPeriod` (start/end date per item), `billablePeriod`, and `CareTeam` / `Practitioner` references are not built into `build_fhir_claim()`.
- Claim items do not yet use the SHA intervention code value-set consistently with OCL/SHA/PFMS.
- `meta.profile` points to generic `http://hl7.org/fhir/StructureDefinition/Claim` rather than a Kenya eClaims / SHA profile.

**Remediation:**

- Validate all FHIR output against the Kenya eClaims IG and DHA profiles.
- Extend `build_fhir_claim()` to include `servedPeriod`, `billablePeriod`, `CareTeam`, `fullUrl` /Bundle entry structure, and DHA profile meta.
- Build a submission Bundle RPC/edge function that wraps `fhir_bundle` into a `message` Bundle.

4. Add a tests/QA script validating generated bundles with the HL7 FHIR validator.

### Requirement: DHA HIE authentication (JWT/OAuth2 client credentials)

**Source:** <https://afyalink.dha.go.ke/claim-integration>, <https://afyalink.dha.go.ke/sha-portal-api-integration> **Status:** GAP  
**Implementation:** `claims-dispatcher` edge function exists and is designed to send queued items, but it is not live because no credential configuration exists. **Gap:** Access/secret keys, OAuth2 client credentials, callback URL configuration, and IP allow-listing have not been provisioned.  
**Remediation:** Obtain credentials from <https://developer.dha.go.ke> / AfyaLink onboarding; configure secrets; register callback URL; test in sandbox/UAT.

### Requirement: Client Registry (CR ID) and member identity resolution

**Source:** <https://hie-docs.dha.go.ke/registries/client-registry> (per existing DOC-7) **Status:** PARTIAL **Implementation:** `patients.cr_number`, `patients.national_id / national_id_type`, `sha_member_number`, `sha_claims.cr_number_at_claim`, `sha_claims.sha_member_no_at_claim`, and `cr_number_missing / sha_member_missing` flags are implemented and surfaced in the SHA claims state machine as "Missing data" badges. **Gap:** No live CR lookup (credentials; API not connected). Some existing claims may have been created before CR resolution. **Remediation:** Once DHA credentials are available, implement CR lookup at registration/encounter sign and resolve missing flags.

### Requirement: ICD-11 diagnostic coding on claims

**Source:** <https://afyalink.dha.go.ke/claim-integration> **Status:**  COMPLIANT (locally) **Implementation:** `encounter_diagnoses` table plus `trg_sync_encounter_diagnoses` and ICD-11 format/API (`icd11-search`) edge function, MOH ICD-11 mapping for 705). **Gap:** Need to confirm every SHA claim's linked encounter carries at least one structured ICD-11 diagnosis; no live terminology service pre-population for all codes. **Remediation:** Add a claim submission guard requiring ICD-11 diagnosis on included claim items; optionally integrate with DHA terminology/OCL in production.

## SHA Claims Compliance

### Requirement: SHA claim lifecycle (draft → submitted → approved/rejected → resubmission → payment)

**Source:** Kenya SHA / AfyaLink claims guide; DHA SHA Claim Bundle API docs **Status:**  COMPLIANT (local workflow) / PARTIAL (live submission)  
**Implementation:**

- `sha_claims`, `sha_claim_items`, `sha_benefit_packages`, `sha_tariffs`, `sha_claim_packages`.
- `auto_generate_sha_claim()` creates draft claims on encounter sign.
- `auto_build_fhir_claim()` builds FHIR Claim into `fhir_bundle`.
- `sha_claim_status_history` records all admin transitions.
- `resubmission_count`, `payment_reference`, `payment_date`, `last_status_check` on `sha_claims`.
- UI state machine in `src/routes/admin.queue.tsx` (`SHA Claims` tab): Submit, Approve, Reject, Record Payment, Resubmit, with status/age/missing-data/PHF badges.

**Gap:** Live submission and claim-status polling are not connected (credentials). Aging UI uses the `sha_claims_aging` view but production callback handling for `ClaimResponse` states is not built.

**Remediation:** Connect `claims-dispatcher` to the live AfyaLink claim endpoint; implement callback ingestion for `queued/approved/rejected/in-review/clinical-review/payment-completed`; reconcile the SHA state machine with DHA `ClaimResponse` states.

### Requirement: SHIF / PHF / ECCIF fund mapping and PHC zero total amount

**Source:** <https://afyalink.dha.go.ke/claim-integration> ("Any PHC claim must have zero total amount") **Status:**  GAP **Implementation:** `sha_claims.fund_type` (`PHF / SHIF / ECCIF`), `sha_fund_type` on encounters, benefit packages by fund, `set_sha_fund_type()` defaults, and `enforce_phf_zero_claim()` banners PHF claims with "PHF: zero patient copay — SHA covers 100%". **Gap:** The PHF trigger does **not** enforce the AfyaLink submission rule that PHC claims must have a zero total amount and zero item net amounts. Current `total_amount` may carry a non-zero insurance-covered amount and item `amount` values may be non-zero. **Remediation:** Before submission, for PHF/PHC-eligible claims set included item `net / amount` and claim `total_amount` to 0 (or reject non-PHC PHF claims), and add a submitting guard with a clear error message. Re-run the adjusted trigger and validate against the AfyaLink rules.

## Requirement: Pre-authorization (preauth) workflow and 72-hour SLA

**Source:** AfyaLink Preasync / SHA Portal Preauth integration guide **Status:** PARTIAL **Implementation:** `preauth_number`, `preauth_id`, `preauth_status`, `preauth_requested_at`, `preauth_approved_at`, `preauth_submitted_at`, `preauth_notes`; local preauth keywords; preauth missing warning; 72hr SLA timer in the insurance dialog; Eligibility placeholder. **Gap:** No live preauth API call or callback handling; no automatic submission of preauth bundles; no persisted `preauth_submitted_at` update from API until credentials are configured. **Remediation:** Once SHA/DHA credentials are available, implement the preauth endpoint and `servedPeriod` /claim-required services mapping.

## Requirement: Claim items use valid SHA/PFMS intervention codes

**Source:** <https://afyalink.dha.go.ke/claim-integration>, eClaims IG downloads **Status:** PARTIAL **Implementation:** `sha_claim_items.intervention_code`, `invoices/line_items`, `sha_tariffs` (seeded Legal Notice 146/147 tariff codes), and `build_fhir_claim()` uses `intervention_code` falling back to `item_type`. **Gap:** No validation gate ensuring `intervention_code` is present and in the SHA intervention value set for every included item; SHA/PFMS coverage extension for eligible members not implemented. **Remediation:** Add submission-time validation of `intervention_code` against the SHA value-set; populate the coverage extension for PFMS-eligible claims.

## Requirement: Claims aging and payment reconciliation

**Source:** AfyaLink ClaimResponse states (`payment-completed`, `payment-declined`, `sent-for-payment-processing`) **Status:** PARTIAL **Implementation:** `sha_claims_aging` view with `age_days`, `aging_status` (overdue\_submission, overdue\_response, needs\_resubmission); state machine UI shows overdue / resubmitted / missing-data badges; `payment_reference`, `payment_date`, `payment_completed` status. **Gap:** No DHA payment-notice ingestion; no automatic reconciliation of `payment_reference` / `payment_date` from SHA ClaimResponse to the local record. **Remediation:** Implement callback ingestion to populate `payment_reference` / `payment_date` and reconcile approved/paid/payment-completed states.

## ODPC Data Privacy Compliance

### Requirement: ODPC registration as data controller/processor

**Source:** <https://www.clydeco.com/en/insights/2026/03/health-data-protection-in-kenya-strategic-compliance> **Status:**  GAP **Implementation:** Privacy policy, consent dialog, DSAR export, audit logs, break-glass access logging, and role-based access controls exist in the product. **Gap:** No ODPC registration certificate, no DPO appointment on record, no DHA notification of controller registration (within 7 days), no published privacy notice for the live deployment. **Remediation:** Register with ODPC; appoint/certify DPO; publish Privacy Notice; notify DHA.

### Requirement: Data Protection Impact Assessment (DPIA)

**Source:** DHA Certification framework; Digital Health (Data Exchange) Regulations **Status:** PARTIAL **Implementation:** `docs/certification/DOC-2-dpia.md` exists as a draft DPIA. **Gap:** DPIA not yet validated/assessed by a qualified assessor; not submitted with DHA certification application. **Remediation:** Finalize DPIA, obtain assessor validation, submit as part of DHA certification.

### Requirement: Data subject access / DSAR and export

**Source:** Kenya Data Protection Act, 2019; DHA Health Data Rules **Status:**  COMPLIANT (feature) / PARTIAL (governance) **Implementation:** "Export Patient Data (DSAR)" button on patient record (admin-only) exports patient, encounters, admissions, clinical notes, lab orders, radiology orders, prescriptions, invoices, and consents as JSON, and logs `DSAR_EXPORT` to `audit_log`. **Gap:** No automated deletion/rectification workflow; no retention notices wired into an enforceable policy; DSAR acknowledgement SLA (30 days) not tracked as a task. **Remediation:** Add DSAR request tracking and deletion/erasure procedure; set 30-day SLA alerts.

### Requirement: Consent and sensitive personal data safeguards

**Source:** Data Protection Act 2019; DHA SHR consent requirements **Status:** PARTIAL **Implementation:** OTP consent flow for SHA claim consent, consent records in `consent_otps` / `patient_consent`s, `ConsentDialog`, explicit preauth/consent requirements. **Gap:** Not all processing activities have documented lawful-basis assessments; biometric identity verification is not implemented; consent-token SHR flow not connected to live DHA. **Remediation:** Perform lawful-basis assessment for each processing activity; document biometric identity gap in DPIA; implement SHR consent APIs once credentials are available.

### Requirement: Breach notification (ODPC 72 hours / DHA 48 hours)

**Source:** <https://www.clydeco.com/en/insights/2026/03/health-data-protection-in-kenya-strategic-compliance> **Status:** ● GAP

**Implementation:** No automated incident/breach notification workflow exists. **Gap:** No documented breach response plan or notification timers to ODPC and DHA. **Remediation:** Implement incident-response runbook with 72-hour ODPC and 48-hour DHA notification workflows and test it.

### Requirement: 20-year health data retention

**Source:** Digital Health (Health Information Management Procedures) Regulations, 2025; Digital Health Act 2023 s.31 **Status:** PARTIAL

**Implementation:** Audit logs are retained in DB; existing certification docs include backup/recovery policy. **Gap:** No enforceable data-retention matrix for 20-year health data; no archival workflow. **Remediation:** Publish retention matrix, enable archival, retain audit logs 20 years, update backup/recovery policy with tested RPO/RTO.

## Infrastructure & Hosting Compliance

### Requirement: Secure, auditable infrastructure and access

**Status:** PARTIAL **Implementation:** Supabase RLS, role-based permissions, break-glass logging, audit triggers, server-side Supabase auth. **Gap:**

- MFA not implemented.
- Penetration test not performed.
- Server-side password policy / account lockout not demonstrated.
- Secrets configuration not verifiable outside the build env.
- Free-tier Supabase cron/automation pauses can interrupt dispatch or compliance jobs.
- Facility/country-level data residency for Kenya health data not confirmed (hosted outside Kenya).

**Remediation:** Enable MFA; run penetration test; document password/lockout policy; move to a paid Supabase plan; confirm/harden data residency and cross-border transfer safeguards (SCCs/TIA); perform backups and restores with tested RTO/RPO.

### Requirement: Audit trail completeness and tamper protection

**Source:** Digital Health (Health Information Management Procedures) Regs, 2025 (audit logs retained 20 years) **Status:** PARTIAL **Implementation:** `audit_trigger_fn()` attached to many tables, including `sha_claims`, `sha_claim_items`, `sha_claim_status_history`; admin audit-log view.

**Gap:** Audit log immutability/append-only and 20-year retention not fully demonstrated. **Remediation:** Enforce append-only/immutable storage for audit\_log; implement off-site archival with 20-year retention.

## Gap Analysis Summary Table

#	Area	Status
1	DHA certification application / certificate	<span style="color: red;">●</span> GAP
2	DHA ESB onboarding	<span style="color: red;">●</span> GAP
3	DHA/AfyaLink credentials (OAuth/JWT/access keys)	<span style="color: red;">●</span> GAP
4	Kenya eClaims FHIR profile conformance	PARTIAL
5	Full SHA Claim <code>message</code> Bundle assembly	<span style="color: red;">●</span> GAP
6	<code>servedPeriod</code> / <code>billablePeriod</code> / CareTeam on Claim	<span style="color: red;">●</span> GAP
7	PHC/PHF zero total amount	<span style="color: red;">●</span> GAP
8	SHA intervention code validation	PARTIAL
9	Local SHA claims state machine	<span style="color: green;">✓</span> COMPLIANT
10	<code>build_fhir_claim()</code> column fix	<span style="color: green;">✓</span> COMPLIANT
11	<code>sha_claim_status_history</code>	<span style="color: green;">✓</span> COMPLIANT
12	<code>resubmission_count</code> / payment columns	<span style="color: green;">✓</span> COMPLIANT



13	<code>sha_claims_aging</code> view	✅ COMPLIANT
14	KMHFL facility sync	✅ COMPLIANT
15	DSAR export	✅ COMPLIANT (feature) / PARTIAL (governance)
16	Preauth 72hr SLA timer	✅ COMPLIANT (UI) / PARTIAL (live)
17	SHA Eligibility check	❌ GAP (pending credentials)
18	ODPC registration / DPO	❌ GAP
19	DPIA submission	PARTIAL
20	Breach notification workflow	❌ GAP
21	MFA	❌ GAP
22	Penetration test	❌ GAP
23	20-year retention/archival	PARTIAL
24	Kenya data hosting / cross-border controls	PARTIAL
25	Biometric verification	❌ GAP
26	Supabase free-tier cron availability	PARTIAL
27	Audit log immutability	PARTIAL
28	KHIS / MoH reporting integration	PARTIAL
29	Live ClaimResponse / payment reconciliation	❌ GAP

## Compliance Roadmap

### Phase 1 — Now (no external dependency)

- Enforce PHC/PHF zero total amount and `servedPeriod` in `build_fhir_claim()` /submission guard.
- Add a submission Bundle builder and FHIR validator QA script.
- Publish ODPC privacy notice, data-protection policy, 20-year retention matrix, breach runbook, DSAR 30-day SLA process.
- Enable MFA; run penetration test; harden password/lockout.
- Track and close the identified security gaps.

### Phase 2 — Credentials & sandbox (blocked on DHA/SHA onboarding)

- Register with ODPC, obtain certificate, appoint DPO, notify DHA.
- Obtain AfyaLink test credentials; configure secrets and callback URL.
- Complete FHIR IG conformance validation; submit DHA certification application (Form HMIS 4).
- Implement live Client Registry lookup, SHA eligibility check, preauth submit, claim submit.
- Implement ClaimResponse callback ingestion and payment reconciliation.

### Phase 3 — Production readiness

- DHA certificate of compliance (2-year validity).
- ESB onboarding and user licence.
- National/county health data bank transfer plan + 20-year archival.
- Biometric identity verification (if in scope for deployment).
- Move to paid Supabase tier, confirm data residency, implement cross-border transfer controls.
- Complete KHIS and MOH reporting interoperability mapping.
- Annual re-certification and ongoing ODPC conformity.

# AegisCare HMS — Developer Self-Attestation

System	AegisCare HMS (LabTrack v5.5 codebase)
Repository	fmurage6331-dev/confit-core
Branch / commit basis	arena/01a037b8-confit-core (based on 7b59358 )
Date	2026-08-25
Prepared by	Francis Muhoro / AegisCare development team
Document type	Developer self-attestation for DHA certification readiness (Form HMIS 4 pathway)
Related docs	docs/dha-compliance-assessment.md , docs/certification/DOC-1 ... DOC-8

## Developer Declaration

This self-attestation is a **voluntary, honest pre-submission review** of AegisCare HMS against the Digital Health Agency (DHA) certification expectations, the Kenya Digital Health Act 2023, the Digital Health (Data Exchange Component) Regulations 2025, the Digital Health (Health Information Management Procedures) Regulations 2025, the Kenya eClaims FHIR Implementation Guide, the Kenya Health Information Systems Interoperability Framework (KHISIF), the Kenya Data Protection Act, 2019, and the Social Health Insurance Act, 2023.

I/we declare that:

- The information in this attestation reflects the actual state of the code and infrastructure at the date above.
- Known gaps are disclosed below rather than hidden.
- No regulatory certification is claimed on the basis of this document alone.
- DHA certification, ODPC registration, healthcare-facility licensing, and clinical approval remain separate responsibilities for the deploying facility / organization.

## AI Tool Usage Declaration

### Claude (Anthropic) — Primary Development Assistant

Claude was used throughout the AegisCare HMS development for code generation, code review, documentation drafting, SQL review, data-model design, and FHIR/SHA structure drafts. All generated code was reviewed, tested, and adapted by the developer before inclusion.

### Arena AI — Autonomous Agent

An autonomous Arena AI agent was used in this session to implement the SHA claims state-machine UI, KMHFL sync, DSAR export, 72hr SLA timer, Supabase type updates, and draft compliance docs. The agent operated under explicit instructions to:

- Read existing files before changing them.
- Not modify protected files ( `src/routes/moh.*` , `src/lib/moh-reports.ts` , `src/components/moh/*` , `patches/` , and `src/routes/inpatient.tsx` except where explicitly authorized).
- Run ESLint/Prettier and TypeScript checks before committing.
- Commit only to the session branch `arena/01a037b8-confit-core` .
- Give Francis exact SQL rather than running SQL directly.

## What AI Was Used For

AI was used for/assisted with:

1. **Schema & SQL review** — reviewing existing `sha_claims`, `sha_claim_items`, `build_fhir_claim()`, triggers, audit triggers, and generated Supabase types.
2. **SHA claims state machine** — Admin → Claims queue tab listing `sha_claims_aging`, color-coded status badges, Submit / Approve / Reject / Record Payment / Resubmit buttons, `sha_claim_status_history` writes, age/missing-data/resubmission badges, PHF banner.
3. **KMHFL sync** — "Sync from KMHFL" button in facility settings calling the public KMHFR API.
4. **DSAR export** — admin-only patient data export as JSON with audit logging.
5. **Preauth SLA UI** — 72-hour SLA timer and SHA eligibility placeholder in the insurance dialog.
6. **Docs** — DHA compliance assessment and this self-attestation draft.
7. **SQL linting** — ensuring provided SQL is idempotent/re-runnable for Francis to apply.

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## What Developer Did Without AI

The developer (Francis Muhoro / AegisCare team) performed (at minimum):

- Provisioning and configuration of the Supabase project, OAuth, Supabase edge functions, and deployment.
- Data-model decisions and approval of SQL before execution.
- Manual verification/execution of schema changes in the project database.
- Live facility/account and SHA-provider setup where applicable.
- Business, clinical, and facility-level requirements definition.
- Selection and validation of clinical terminology mappings (ICD-11, MOH indicators).
- HIPAA/ODPC/clinical-safety judgment calls where AI output was judged insufficient.
- Any deployment and release approval.

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## AI Output Review Process

For AI-generated code in this session:

1. `npx eslint <file> --fix` was run.
2. `npx eslint <file> --max-warnings=N` was run (0 warnings except `rooms.$id.tsx`, which is allowed 4 existing warnings).
3. `npx tsc --noEmit` was run; the only reported errors are pre-existing errors in auto-generated MCP route files (`src/routes/[.mcp]/*`, `src/routes/mcp.ts`, `src/routes/[.well-known]/oauth-protected-resource.ts`).
4. Changes were reviewed via `git diff` before committing.
5. SQL was not run directly by the agent; exact SQL was provided to Francis and applied by the developer.
6. Dangerous/invasive files were not touched.
7. The session work was committed only to `arena/01a037b8-confit-core`.

Known limitation: `src/integrations/supabase/types.ts` was updated manually because the Supabase CLI had no access token in the sandbox. It should be regenerated from the live project before production to confirm exact generated output.

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## Known Compliance Gaps

These are acknowledged gaps. They are intentionally not hidden.

1. **SHA API credentials pending** — no live SHA eligibility, preauth, claim submission, or ClaimResponse ingestion until credentials are configured.
2. **DHA credentials pending** — no AfyaLink HIE access/JWT/OAuth billing/test onboarding.
3. **Biometric verification not built** — patient identity relies on manual/national-ID fields; DHA's Afya Yangu biometric/self-registration path is not implemented.
4. **Data hosted outside Kenya** — current hosting arrangement is not confirmed to be in-Kenya; cross-border transfer safeguards (SCCs/TIA) are incomplete.
5. **ODPC registration pending** — no confirmed ODPC controller/processor certificate and no DPO appointment.
6. **DHA facility registration pending** — no confirmed DHA certification application, ESB onboarding, or facility/DHA certificate of compliance.

7. **AT SMS Sender ID pending** — the SMS/OTP delivery path is not production-verified with a registered Kenyan messaging Sender ID.
8. **Penetration test not done** — no third-party penetration test report on the deployment.
9. **MFA not implemented** — login currently does not enforce multi-factor authentication.
10. **Supabase cron pauses on free tier** — background automation (billing accrual, dispatcher) may not run reliably on a free-tier Supabase plan.

Additional gaps:

- PHC/PHF zero total amount assertion not enforced in claim data yet.
- Full SHA Claim `message` Bundle (Organization + Patient + Coverage + Claim) not assembled with DHA profile meta and `servicedPeriod`.
- Kenya eClaims FHIR profile conformance not validated.
- 20-year audit/health-data retention not yet enforced as a lifecycle.
- 48-hour DHA / 72-hour ODPC breach notification runbook not implemented.
- DSAR is implemented as an export but no systematic erasure request workflow.

## Security Self-Assessment

Control	Status	Notes
Authentication	⚠ PARTIAL	Supabase Auth implemented; MFA not enforced
Authorization / RBAC	✅ COMPLIANT	roles + <code>PermGuard</code> , admin-only surfaces
Row-level security	✅ COMPLIANT	enabled on clinical and SHA tables
Audit logging	⚠ PARTIAL	<code>audit_trigger_fn</code> , audit log UI; immutability/20-year retention to be proven
Consent / OTP	⚠ PARTIAL	local OTP consent; live SHA consent token flow pending
Access to sensitive data	✅ COMPLIANT	role-gated patient/encounter views, break-glass logging
Data encryption at rest/transit	⚠ PARTIAL	Supabase/TLS assumed; not independently verified
Vulnerability / PEN test	🔴 GAP	not performed
Secrets management	⚠ PARTIAL	env-based secrets; not verified against a dedicated secrets manager
Backup / recovery	⚠ PARTIAL	policy exists; tested restore not demonstrated

## Clinical Safety Declaration

AegisCare HMS is intended to support clinical workflow **but is not a substitute for clinical judgment**. No warranty of clinical safety is made by this self-attestation. Before use:

- Clinical workflows must be validated by a qualified clinical safety officer (or equivalent).
- Any diagnostic or treatment decision supported by the system must be reviewed by a licensed clinician.
- The system must not be used for emergency clinical decisions without appropriate clinical oversight.
- Medicare/SHA reimbursement claims generated by the system must be reviewed by the facility's billing/claims team prior to submission.

## Intellectual Property Declaration

- The AegisCare HMS codebase, including the current branch work, is the property of AegisCare / the repository owner.
- No third-party proprietary code was knowingly copied into the codebase.
- Open-source packages used by the system are listed in `package.json` / `bun.lock` / `package-lock.json` and retain their respective licenses.
- The AI tooling used did not vest copyright in the developer's code.
- Regulatory/documentation content in `docs/` is drafted for the project owner's use.

Formal Attestation Signature

Attestation	I/we hereby attest that the development work, code, SQL, and documentation described in this self-attestation were produced as described, and that known gaps are honestly reflected above.
Name	Francis Muhoro
Role	Developer / System Owner (AegisCare)
Date	2026-08-25
Signature	(electronic signature pending)

This is a developer's self-attestation, not a DHA/ODPC certificate. Final compliance determination rests with the DHA certification framework, ODPC registration/certification, and the deploying facility's governance.